# From Flat to Fair?

## The Effects of a Progressive Tax Reform


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**Abstract**

We study the effects of a progressive tax reform on tax compliance, using a research design that distinguishes between two channels. First, using a quasi-experimental design, we estimate the *direct* effects of the reform—namely, how changes in a household's own tax rate affect its own compliance. Second, leveraging a large-scale natural field experiment, we estimate the *indirect* effects: holding a household's own tax rate constant, we examine how its compliance is influenced by changes in the tax rates of other, poorer or richer, households. We find substantial direct effects: lowering taxes for poor households increases their compliance, while raising taxes for rich households reduces theirs. We also find sizable indirect effects: when poor households learn about the tax hikes on the rich, their stated perceptions of tax fairness and their actual compliance both increase. Among rich households, learning about tax cuts for the poor also improves perceived fairness, but, if anything, reduces compliance. Using an additional reform and follow-up field experiment conducted a year later, we replicate both the quasi-experimental and experimental results. Together, our findings show that tax compliance responds not only to a household's own tax burden but also to its perception of fairness of the broader tax system. Our results also underscore the potential disconnect between stated and revealed preferences for redistribution. Finally, we present a counterfactual analysis that illustrates the implications of the direct and indirect effects for designing progressive tax reforms.



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# 1 Introduction

Progressive tax schedules are widespread throughout the world and play a crucial role in the redistribution of income (Piketty and Saez, 2007; Saez and Zucman, 2019a).[1] There are, however, significant differences in the degree of tax progressivity between countries and even across different taxes within a given country (Fisher-Post and Gethin, 2023). A natural hypothesis regarding the adoption of progressive tax schedules is that individuals care not only about how policies will affect them, but also about how they will impact others, for example, through other-regarding preferences or social comparisons (e.g., see Stantcheva, 2021). For instance, wealthy households may tolerate a reform that increases their tax burden if it advances fairness objectives, such as supporting disadvantaged households. Although it is such a pillar of modern tax systems, empirical evidence on the effects of tax progressivity is scarce. This study addresses this gap by analyzing how taxpayers responded to a progressive tax reform.

Our novel research design allowed us to disentangle two key causal mechanisms, corresponding to the direct and indirect effects of the reform. The *own-rate* channel posits that a taxpayer's tax compliance may respond to a change in their own tax rate, irrespective of what happens to the tax rates of other households. For instance, lowering taxes for a poor household may increase its tax compliance, while increasing taxes for a rich household may lower its tax compliance. The *cross-rate* channel posits that, while keeping its own tax rate constant, the compliance of a taxpayer may also depend on their perception about the tax rates of other households. In the context of a progressive reform, a poor household may adjust its tax compliance upon learning that, in addition to lowering its own tax rate, the government has increased the tax rate on the rich. Likewise, a rich household may change its tax compliance upon learning that the government has reduced tax rates for the poor, beyond the effect of an increase in its own tax rate. Even middle households, who were not directly affected by the reform, may have adjusted their compliance in response to learning about the tax cut for the poor and the tax increase for the rich. Ex ante, cross-rate effects could be either positive or negative, depending on how different taxpayers are affected and their underlying social preferences. For example, some taxpayers may react negatively to increased progressivity if they believe that it would be fairest for everyone to pay the same rate. On the other hand, if taxpayers want more progressive taxes, finding out about a progressive reform may boost their tax morale.

We examined a progressive property tax reform enacted by Tres de Febrero, a major municipality in the Province of Buenos Aires, Argentina, and a suburb of the capital city. The tax is levied monthly on the assessed values of properties, with an average annual-

[1] For example, according to Fisher-Post and Gethin (2023), progressive tax systems reduce inequality by 5% to 15% in Western Europe and the United States. For more evidence on developing countries, see Lustig (2023) and Bachas et al. (2024).

ized tax rate of about 2.6% in 2022. The municipality uses tax revenues to fund basic services such as street lighting and urban sanitation. In January 2023, the municipality implemented a progressive tax reform intended to be revenue-neutral. The properties at the bottom 37% of the value distribution received a tax cut of about 30%. Hereinafter, we refer to these as *poor households* for short.[2] Properties in the top 27% (*rich households*) faced a significant tax increase of about 15%.[3] Tax rates for properties with valuations in the middle of the distribution were not affected by the reform—hereinafter, we refer to these as *middle households* for short.

The setting in which this reform occurred has two advantages in addressing the research question at hand. First, there is substantial inequality, and thus there is ample scope for redistribution through taxation.[4] This inequality is also reflected in home values. For example, a property in the 90th percentile of assessed value is worth 6.9 times as much as a property in the 10th percentile.[5] The second advantage is that, due to limitations in tax enforcement, compliance choices provide insight into an individual's tax morale. For example, in the year prior to the reform, the average probability of making timely payments (i.e., within three months of the due date) was 47.9%. Even in developed countries, there are contexts of low enforcement and compliance—such as the reporting of self-employment income (Cullen et al., 2021) or the church tax in Germany (Dwenger et al., 2016)—in which tax morale plays a similarly central role. Thus, there is scope for tax compliance to increase or decrease in response to the reform.

We identified the reform's own-rate effects by leveraging two sources of identification. First, the reform involved two sharp discontinuities: households with property values below a predetermined threshold (poor households) received a tax cut, while households with property values above a different threshold (rich households) were hit with a tax hike. Properties in the middle group did not experience any tax change. We used a two-cutoff cumulative regression discontinuity design (RDD) (Cattaneo et al., 2016). Moreover, we leveraged the timing of the reform as a second source of causal identification: we compared the evolution of the outcomes right before versus right after the reform took effect.

The RDD estimates revealed significant own-rate effects of the reform on tax compliance, which we can summarize as behavioral elasticities. A 1% reduction in the tax rate

[2] We use "households" for brevity, but the sample also includes some non-residential properties, which represent approximately 16% of the total.

[3] The 30% tax reduction for the poor and 15% surcharge for the rich are simplified summaries of the reform. Section 2.3 details its full implementation.

[4] For instance, the Gini coefficient for the Greater Buenos Aires Metropolitan Area (which includes Tres de Febrero) was 0.404 (CEDLAS, 2024), which is comparable to that in cities such as Miami, Lisbon, and Brussels (OECD, 2018).

[5] Households with more expensive homes are expected to have, on average, higher incomes as well. This is not only a core prediction of standard economic theory, but also a widely documented empirical correlation (Goodman and Kawai, 1982; Jimenez and Keare, 1984; Zheng et al., 2018; Karmali and Weng, 2022).

for the poor increased their compliance by 0.257%, implying an elasticity of $\varepsilon^{own}_{poor}$=0.257. On the other hand, a 1% increase in the tax rate for the rich reduced their compliance by 0.484%, corresponding to an elasticity of $\varepsilon^{own}_{rich}$=0.484.

These RDD estimates are robust to a host of sensitivity checks. Most notably, we provide two key falsification tests: we found null effects when using placebo thresholds and when using a placebo event date for the policy change. Moreover, we leveraged a second progressive tax reform that took effect in January 2024, a year after the first reform. The new reform used the same thresholds to determine which households would experience a tax cut and which would experience a tax hike. We reproduced the RDD analysis exactly as for the first reform. Reassuringly, the results were replicated: we found elasticities that are consistent across the two reforms, both qualitatively and quantitatively.

To identify the cross-rate effects of the reform, we conducted a pre-registered, large-scale, natural field experiment, in which we randomized information about the progressive nature of the tax reform. As part of regular communication with taxpayers, the municipality sent letters to a large sample of taxpayers. We embedded an information provision experiment in the January 2023 mailers. Both types included information on the amounts added to or deducted from recipients bills as a result of the tax reform. We randomly assigned each subject to receive one of two types of letters. The first type included a message on how the reform changed the household's own tax rate. The second type included the same message, but with additional information about how the reform affected the tax rates of other household groups; for example, a poor household would learn that the reform also raised taxes on the rich.

Using administrative records, we estimated the impact of treatment on subsequent tax compliance. In addition, to provide complementary evidence on the causal mechanisms at play, we collected survey data. We invited a subsample of taxpayers with email addresses in file to complete a brief online survey. We received 1,723 responses, corresponding to a response rate of 11.8%. A concern in information-provision experiments is that subjects may ignore the information or that the control group may learn it through other channels. To address these concerns, the survey included a question assessing awareness of the progressive tax reform: subjects saw a list of recent policies and indicated which they had heard of. This allowed us to test whether the treatment increased awareness. Finally, to investigate the tax morale mechanism, the survey included a question on perceived fairness of the tax system, rated on a scale from 0 (very unfair) to 10 (very fair).

We begin by presenting evidence that a significant proportion of subjects opened the letters sent to them. To establish this, we selected a small random sample of households *not* to receive a letter. We found that households that received the control letter, compared to those who did not receive any letter, exhibited a significant increase in tax compliance. Survey data also indicate that households paid attention to the letter content: only a few in the control group were aware of the progressive tax reform, while awareness was significantly higher among those who received the treatment letter.

Next, we used administrative records to compare tax compliance between households with treatment and control letters. We began with the results for poor households and found that the treatment had a positive and statistically significant effect on their tax compliance. Specifically, poor households increased their compliance upon learning about the tax hike on the rich. Compared to poor households who received the control letter, the treatment led to a statistically significant increase in tax compliance of 0.808 percentage points (pp, p-value = 0.006). We estimated a cross-rate elasticity of $\varepsilon_{poor}^{cross}$=0.098 for poor households: a 1% increase in the (perceived) taxes of the rich increased the poor household's own tax compliance by 0.098%. To contextualize this magnitude, we compare it to the effect of the household's own tax rate: a 1% increase in the tax rate of the rich had the same effect on a poor household's compliance as lowering the poor household's own tax rate by 0.38%. While this intention-to-treat effect is already substantial, the true effects are likely larger, as a significant fraction of households may not have read the letters fully or carefully. This cross-rate effect must be driven by changes in tax morale. Consistent with this interpretation, survey data shows that when informed about the tax hike on the rich, poor households reported higher perceptions of the tax system's fairness.

We now turn to the cross-rate effects for rich households. As with poor households, we found that the treatment increased rich households' awareness of the progressive tax reform. Moreover, when the rich found out that the poor were getting a tax cut, they were more likely to state that the tax system is fair. But although their stated perceptions of the tax system's fairness increased by about the same degree as for the poor, we did not find a positive effect on tax compliance for the rich. If anything, the rich became *less* likely to pay their taxes after finding out that the poor experienced a tax cut, although the effect is close to zero in magnitude ($\varepsilon_{rich}^{cross}$=0.019) and not statistically significant. One interpretation of this finding is that, although rich households may *say* they view the tax cut for the poor as fair and claim to be supportive, they may in fact be secretly resentful. We also estimated the cross-rate effects for middle households. Although they did not experience a change in their own tax rates, finding out about the tax hike on the rich and the tax cut on the poor may still have affected their compliance. Within this group, as with the poor and rich households, the treatment had a significant positive effect on awareness of the progressive tax reform and on perceived fairness of the tax system. However, the effect on tax compliance was close to zero and statistically insignificant.

The results of the field experiment are robust to a host of sensitivity checks. Most importantly, an event-study analysis shows that the timing of treatment effects coincides exactly with the timing of the mailing intervention. Moreover, to assess the replicability of our findings, we conducted a second pre-registered field experiment in January 2024, leveraging the implementation of a new progressive tax reform introduced at that time. The results for the 2024 experiment were largely similar to those from the 2023 experiment. Furthermore, to assess whether these results were surprising or predictable, we conducted an expert forecast survey with 39 academics with relevant research experience in public finance and related fields. After receiving a description of the experiment, these experts were asked to forecast the effects of the treatment. The survey revealed a lack of expert consensus, with significant variation in predictions and most experts expressing uncertainty about their forecasts. Only a minority predicted effects similar to our experimental estimates.[6]

We show that our findings have implications for the design of progressive tax reforms. As emphasized in the public finance literature (Saez and Zucman, 2023), policymakers should account for behavioral responses to accurately forecast the effects of tax policy. We note that policymakers should take into account both direct behavioral responses but also indirect ones derived from taxpayers' social preferences. Our counterfactual analysis evaluates a hypothetical tax reform that, absent behavioral responses, would be revenue-neutral and would increase the progressivity of the tax schedule. We adopt a sufficient statistics approach (Chetty, 2009): the counterfactual analysis depends solely on the four behavioral elasticities estimated with the quasi-experimental and experimental designs. Relative to the counterfactual without behavioral responses, the real effects of the reform are significantly different, with an effective tax progressivity that is 48% lower and tax revenues that are 4.4% lower. To the extent that indirect effects drive a sizable portion of these results, they underscore the importance of considering not only the direct impact of a reform but also how taxpayers perceive its broader implications for fairness.

Our paper relates and contributes to multiple strands of literature. First, we contribute to the literature on tax progressivity, which has recently received growing attention. This increased interest is particularly notable in the context of property taxes, which tend to have flat schedules, strongly contrasting with modern progressive income and wealth taxes (Chancel et al., 2022; Dray et al., 2023). Recent studies have documented trends in progressivity and its connection with income inequality (Piketty and Saez, 2007; Bachas et al., 2024), while theoretical work has examined optimal progressivity levels that balance efficiency and equity (Heathcote et al., 2017). A growing body of research examines how property tax design can be improved in developing countries, particularly through changes in tax rates and enforcement (Brockmeyer et al., 2023; Bergeron et al., 2024). There is, however, no direct evidence on the tax compliance effects of making tax schedules more progressive. Our study fills this gap.

Second, our study contributes to the broad literature on social preferences, tax preferences, and redistribution preferences. While there have been some advances in under-

[6]For more details about the design and the results of the forecast survey, see Appendix K.

standing perceptions and support for tax progressivity, they are all based on survey data. Using survey experiments, Tarroux (2019) found that people seem to attach intrinsic value to tax progressivity. Stantcheva (2021) found that educational videos on redistribution increase stated support for progressive taxes, while efficiency-focused videos have no effect. Similarly, Hoy (2025) found that self-reported tax morale rises when individuals are informed about the progressive structure of their tax systems.[7] Although valuable, these studies rely exclusively on stated preferences, which may be subject to social desirability and other biases (Bursztyn et al., 2025). For example, rich households may *say* they are willing to pay more in taxes but behave differently when stakes are real.[8] We complement this literature by examining support for progressive taxation through revealed preferences in a natural, high-stakes context. Indeed, we found significant discrepancies between stated and revealed preferences: while middle and rich households reported higher perceived fairness after learning about the progressive reform, this did not translate into increased compliance.

Our evidence on the own-rate effects of rich households relates to the literature examining the tax evasion responses to wealth taxes. Several papers document how high-income or high-wealth individuals react to increased tax burdens, often using registry data and policy changes. For example, Seim (2017), Brülhart et al. (2022), and Rauh and Shyu (2024) study responses to wealth and income taxes in Sweden, Switzerland, and California, respectively, while Locks (2023) and Londoño-Vélez and Ávila-Mahecha (2024) provide related evidence for Brazil and Colombia. These studies show that high-wealth individuals often respond to increases in their own tax rates by reducing their reported tax bases or shifting assets across borders. Our findings extend this literature by estimating compliance effects not only for the rich but also for the poor, and by separately identifying both direct and indirect effects. Beyond estimating a broader set of effects, we do so within a unified framework that allows for meaningful quantitative comparisons across groups (poor, middle, and rich) and channels (direct and indirect).

Lastly, our findings on the cross-rate effects of a progressive reform contribute to the growing body of evidence on tax morale—for a comprehensive literature review, see Luttmer and Singhal (2014) and Slemrod (2019). To mention a few examples, Dwenger et al. (2016) study the role of intrinsic versus extrinsic motivations. Cullen et al. (2021) show that tax compliance is higher when taxpayers trust the government in power. Giaccobasso et al. (2022) and Kresch et al. (2023) show that tax compliance increases when taxpayers believe they benefit from government spending. Besley et al. (2023) provides evidence that the United Kingdom's regressive poll tax increased tax evasion. And Del Car-

[7] Other related studies are Ballard-Rosa et al. (2017) and Kerschbamer and Müller (2020).

[8] For more discussion of the generalizability of data on social preferences, see Epper et al. (2024) and Fehr et al. (2024). While too numerous to list exhaustively, other studies explore redistributive preferences and fairness norms (e.g., Tyran and Sausgruber, 2006; Kuziemko et al., 2015).

pio (2014), Nathan et al. (2024) and Best et al. (2025) discuss the role of fairness and inequity in tax compliance. We contribute to this literature by providing novel evidence of tax morale and by identifying a new tax morale channel—preferences for progressivity.

The rest of the paper proceeds as follows. Section 2 describes the institutional context and data. Section 3 discusses the own-rate effects, while Section 4 discusses the cross-rate effects. Section 5 presents the counterfactual analysis. The last section concludes.

# 2 Institutional Context and Data

## 2.1 The Property Tax

Our study focuses on Tres de Febrero, a major urban municipality in the Greater Buenos Aires metropolitan area, with approximately 365,000 residents as of 2022. Tres de Febrero levies a local property tax known as *Tasa por Servicios Generales* (TSG), applied to around 96,000 residential properties and 18,000 non-residential properties.[9] Although our sample includes some non-residential properties, the vast majority (84%) are residential, so we refer to taxpayers as "households" for brevity. Tax revenues are used to provide local public services such as street lighting, urban sanitation, and maintenance. This tax represents the standard revenue mechanism in Argentine municipalities, acting as their main funding source.[10] For our analysis, we have access to monthly administrative data from the municipality's tax records.[11]

We illustrate the tax schedule by means of statistics for the pre-reform year, 2022 (see Appendix A.1 for further details). The average property tax liability was AR$ 32,561.1 per year in current Argentine pesos, equivalent to 529 U.S. dollars.[12] On average, in 2022 households paid an annual tax rate of 2.6% of the assessed value of the property. The provincial tax authority's cadastre determines property assessments, but these values systematically underestimate the real market values (for details, see Appendix B), so the municipality uses relatively high statutory rates to compensate. Calculating the tax rates relative to the market values (instead of assessed values) results in an average tax rate of about 0.90%, which is similar to the average property tax rate in the United States, and close to the average rates in Missouri (0.82%) and Maryland (0.96%) (Tax Foundation,

[9]Property owners are legally responsible for paying the property tax.

[10]In Tres de Febrero, the TSG comprised 20% of total resources and 45% of locally generated revenue in 2021. These proportions roughly align with the contribution of real estate property taxes to state and local revenues in the United States (Saez and Zucman, 2019b).

[11]We excluded from the analysis 22,935 properties without property value assessments and 142 households with incomplete tax compliance records.

[12]This conversion is based on the USD PPP exchange rate of AR$ 61.5 for 2022 (World Bank, 2025). Throughout the remainder of the document, unless explicitly stated otherwise, all monetary amounts are expressed in current Argentine pesos.

2024).

This property tax consists of two components: a variable part, calculated as a percentage of the property's assessed value (with tax rates ranging from 0.42% to 3.22% across eight property categories), and a fixed component, earmarked for specific services such as security and health.[13] For reference, for a household with a median-value home, about 62% of the total tax burden comes from the variable component, while the remaining 38% is attributed to the fixed component. Although the variable part is progressive, the flat charge makes the overall tax schedule somewhat regressive. For example, a property in the 25th percentile of the assessed value distribution faced an effective tax rate of 2.47%, compared to 1.99% for a property in the 75th percentile.

## 2.2 Tax Compliance

In Tres de Febrero, property taxes are billed monthly, with each installment due on the 15th of each month.[14] Our primary outcome of interest is a binary variable that indicates whether a household made a tax payment within three months of the due date. While the three-month threshold is somewhat arbitrary, we adopt it to align with definitions used in related studies (Del Carpio, 2014; Castro and Scartascini, 2015; Carrillo et al., 2021; Cruces et al., 2025; Flores et al., 2025). Importantly, our results are robust to alternative definitions. Intuitively, delinquency tends to be persistent: when a payment is not made within the first few months, it is likely to remain unpaid for an extended period—often years.

Compliance in the pre-reform year (2022) was far from perfect, with only 47.9% of monthly bills paid on time. Such low compliance rates are common in developing countries, where tax agencies often face significant challenges in enforcing property taxes due to limitations in human, technical, and legal resources. For example, as in other developed and developing countries, delinquent taxes are subject to financial penalties in the form of interest charges (Perez-Truglia and Troiano, 2018). However, as in those contexts, simply imposing an interest rate is not a magical solution—the same limitations that hinder enforcement of the principal also apply to the interest. The weaker enforcement capacity in developing countries results in higher tax delinquency compared to developed economies. For example, the 40 largest U.S. cities collect nearly 95% of property taxes

[13]Although the tax reform applied uniformly across property types, the tax structure differs between residential and non-residential properties—see Appendix A.1 for more details.

[14]A portion of the property tax is collected in advance through the electricity bill, as a fixed charge. This mechanism increases compliance by ensuring that even non-compliant households contribute at least partially to the tax, since nearly all households pay their electricity billslikely due to the fact that the utility company can cut off power to those who do not. Although households can technically request the removal of this charge from their electricity bill, only a small fraction choose to do so. In 2023, 98% of the households paid at least part of their property tax through this mechanism, highlighting its high compliance rate.

on time (Chirico et al., 2016); in contrast, studies from Argentina, Brazil, Ghana, Haiti, Liberia, Malawi, Mexico, Pakistan and Peru report that less than 50% of households pay their property taxes on time (Giaccobasso et al., 2022).[15] Even in developed countries, there are contexts of low enforcement and compliance—such as self-employment income reporting in the United States (Cullen et al., 2021) or the church local tax in Germany (Dwenger et al., 2016).

The average compliance rate of 47.9% reported above masks substantial heterogeneity between households. At one extreme, 42.3% of taxpayers did not make any on-time payments throughout the year—some households, in fact, go years without paying any property tax bills at all. At the other extreme, 26.7% of households paid on time every month. The remaining households made timely payments in some months but not in others. As a result, there is considerable room for households to adjust their compliance either upward or downward. And given the interest in tax progressivity, it is relevant to examine differences in tax compliance across the income distribution. In 2022, the compliance rate was relatively similar for the three groups: 45.8% for poor households, 48.7% for middle-income households, and 49.8% for rich households. This pattern of persistent non-compliance across income levels suggests that tax compliance decisions are driven by more than just income or wealth effects and may instead reflect other factors, such as tax morale.

## 2.3 The Progressive Tax Reform

Prompted by political and equity considerations, the municipality made the property tax more progressive. The main reform to the tax schedule took effect in January 2023, modifying the variable component of the tax while keeping the fixed component unchanged, with the goal of maintaining revenue neutrality. The government implemented this reform swiftly, limiting advertising beyond the experiment. For instance, the municipal council discussed and passed the 2023 tax reform on December 5, 2022, published it in the annual tax bill on December 13, 2022, and the new rates became effective on January 1, 2023.[16] Our taxpayer survey indicates that, at baseline, most respondents, even among the rich, *stated* that they supported a tax reform that would lower taxes for the poor and raise them for the rich (see Appendix C for more details). This suggests a relatively favorable context for a progressive reform, assuming respondents answer such survey questions honestly.

The reform established four taxpayer groups based on sharp, binding thresholds of assessed property values: properties valued at or below AR$ 750K (in the bottom 37%

[15] See Khan et al. (2015); Kapon et al. (2024); Okunogbe (2021); Brockmeyer et al. (2023); Dzansi et al. (2022); Manwaring and Regan (2024).

[16] Source: link. Likewise, the 2024 reform was also approved shortly before it became effective.

of the value distribution) received a tax cut equal to a 30% reduction in the variable component. Properties valued between AR\$ 750K and AR\$ 1.5M (the middle 36% of the distribution) were not affected by the reform. Properties valued above AR\$ 1.5M (the top 27% of the distribution) experienced a tax hike. More precisely, the properties assessed between AR\$ 1.5M and AR\$ 3M faced a 10% surcharge in the variable component, while the properties assessed above AR\$ 3M experienced a 20% increase in the variable component. These cutoffs were exclusive to this reform and not used in other policies. We should note that when estimating own-rate effects, given the small share of properties above the AR\$ 3M mark (around 6%), we lack sufficient statistical power to analyze the discontinuity at that threshold, so we focus on the AR\$ 1.5M threshold instead.

In January 2024, the municipality further increased the progressivity of the property tax, offering a unique opportunity to assess whether our findings replicate.[17] The 2024 reform maintained the same eligibility thresholds as in 2023, but it strengthened the reform further: in addition to the existing changes to the variable components, they made the fixed component more progressive as well. More precisely, the 2024 reform transformed the previously uniform fixed component into a group-specific charge, making it progressive. Properties valued at or below AR\$ 750K received a reduction in the fixed component of approximately 16%. Properties assessed between AR\$ 1.5M and AR\$ 3M faced a surcharge in the fixed component of approximately 39%, while properties valued above AR\$ 3M experienced a surcharge of approximately 64%. And, like in the 2023 reform, properties valued between AR\$ 750K and AR\$ 1.5M (the middle 36% of the distribution) were not affected by the 2024 reform.

# 3 Own-Rate Effects

## 3.1 Research Design

In this section, we discuss how the changes to taxpayers' own tax rates affected their own tax compliance. We estimated these own-rate effects by combining two sources of identification. First, we took advantage of the sharp discontinuities in tax rates induced by the reform as a function of the property valuation brackets. Second, we exploited the timing of the reform by comparing the outcomes before and after its implementation.[18]

The treatment assignment was based on a two-cutoff cumulative design (Cattaneo et al., 2016), where the running variable is the property value in 2021 ($V$) and the treat-

[17] An earlier reform occurred in 2022, but we do not study it due to its smaller scope, lack of public awareness, and reliance on outdated valuations—for more details see Appendix A.2.

[18] Since we leveraged threshold discontinuities and temporal changes around the implementation date, our research design resembles a difference-in-discontinuity approach.

ment is the tax rate, $\tau$, which is a function of $V$, $\tau = \tau(V)$. Specifically, households below the low cutoff ($c_L$ = AR\$ 750K), households between the two cutoffs and households above the high cutoff ($c_H$ = AR\$ 1.5M) experience different tax schedules. Following Cattaneo et al. (2023), we analyzed these two cutoffs separately, comparing observations below $c_L$ to observations between $c_L$ and $c_H$ around cutoff $c_L$, and observations above $c_H$ to observations between $c_L$ and $c_H$ around cutoff $c_H$. Notice that the treatment $\tau(V)$ is piecewise continuous, with discrete jumps at each cutoff, a setting analyzed by Dong et al. (2023) for the single-cutoff case.

We begin by assessing whether there was a first-stage effect—that is, a sharp change in tax rates around the thresholds. For this analysis, the relevant outcome is the change in tax rates from before to after the reform.[19] In the subsequent reduced-form analysis, we examine how compliance rates changed around those thresholds. More precisely, the outcome of interest is defined as the change in the proportion of bills paid on time (i.e., within three months of the due date) between the six months before and the six months after the reform.[20,21] We estimated the RDD parameters using nonparametric local-linear regression with an optimal mean squared error bandwidth (Calonico et al., 2014).

Our approach of using changes in outcomes rather than levels is key for the replication exercise of the 2024 reform. Since this reform used the same eligibility thresholds as the 2023 reform, by focusing on changes rather than levels, we isolate the incremental effect of each reform rather than the cumulative effects of previous reforms. Focusing on changes rather than levels is also crucial for the falsification test based on a placebo implementation date, which we discuss below.

## 3.2 Main Results

Figure 1 presents our main own-rate results. Panels (a) and (c) focus on poor households. Panel (a) presents the first-stage results, showing how the reform altered tax rates. The x-axis corresponds to the assessed value of the properties, which serves as the running variable for the RDD. The y-axis corresponds to the change in tax rates after the reform. As expected, there is a sharp discontinuity at the AR\$ 750K threshold: poor households (left of the cutoff) experienced a 0.392 pp reduction in their tax rate (p-value $< 0.001$). Panel (c) presents the corresponding reduced-form effect, where the y-axis corresponds to

[19] More specifically, we defined the tax rate as the annual tax liability divided by the property valuation.

[20] For instance, if a property paid four bills in the last semester of 2022, and then paid five bills in the first semester of 2023, her outcome would be 0.167 ($= \frac{5}{6} - \frac{4}{6}$).

[21] For October, November, and December 2022, we restricted our binary variable that indicates whether a household made a tax payment by not allowing payments of a pre-treatment installment to be made in the post-treatment period. In cases where such payments were made after the post-treatment period, their values were set to zero. This correction was made to prevent pre-treatment outcomes from being contaminated by post-treatment choices. In any case, this is a minor detail, as the results remain almost identical without this correction.

the change in tax compliance. The discontinuity at the threshold indicates that the tax cut for the poor (i.e., left of the threshold) increased compliance by 2.718 pp (p-value = 0.002). This suggests the presence of a strongly significant own-rate effect for the poor, as they increased their tax compliance following a reduction in their tax rate.

Panels (b) and (d) of Figure 1 present the corresponding results for rich households. Panel (b) shows that, as expected, the changes in the reform were binding and rich households (those to the right of the AR$ 1.5M threshold) faced a 0.146 pp increase in their tax rate (p-value $< 0.001$). Panel (d) then presents the behavioral response—the change in tax compliance measured in the y-axis. Rich households reduced their compliance by 2.248 pp (p-value = 0.050). The estimates for poor and rich households provide compelling evidence that own-rate effects are substantial and symmetric: lower taxes increase compliance, while higher taxes reduce it.

Our preferred interpretation of the own-rate effects is that they reflect a combination of a price effect and, potentially, a tax morale effect. The price effect captures the impact of changes in tax obligations (e.g., see Brockmeyer et al., 2023; Bergeron et al., 2024). For poor households, lower tax burdens make payment more affordable and thus more likely. For rich households, higher obligations may reduce affordability and, in turn, compliance. At the same time, tax morale may also contribute. For instance, poor households facing a tax cut may increase compliance out of a sense of reciprocity or due to a perceived improvement in tax fairness. Conversely, rich households may reduce compliance after a tax hike, either as a retaliatory response or because the increase undermines their perception of fairness in the system.

Next, we present a series of robustness and falsification tests. We begin with a placebo date test, in which we replicated the main analysis using a fictitious reform date of July 1, 2023, instead of the actual reform date. Since no tax reform actually took place on that date, we should not have observed any effect on tax compliance. If significant effects were observed around this placebo reform date, it would suggest that our main results—based on the true reform date—might be spuriously driven by, for example, unusual time trends or other confounding policy changes.

The results from this placebo exercise are presented in Figure 2. Because tax rates remain constant within a calendar year, the first-stage results for this placebo are exactly zero by construction, and we therefore omit those graphs. The reduced-form estimates are presented in panels (a) and (b), for poor and rich households, respectively. As expected, the estimated discontinuities at the placebo threshold are close to zero and statistically insignificant ($\beta^{own}_{poor} = -0.074$, p-value = 0.907; $\beta^{own}_{rich} = 0.046$, p-value = 0.827). The null results from the placebo test provides reassurance that the effects of the actual reform reflect true causal impacts rather than spurious correlations. It is important to note that, although July 1 may appear arbitrary, it is in fact the only feasible date for this placebo test. Since the outcome is defined as the change from six months before to six months after the reform date, a fake date after July 1 could partially capture the effects of the 2024 reform, while an earlier one could partially pick up the effects of the 2022 reform.

As an additional robustness check, Figure 3 presents the results of a placebo test in which we re-estimate the RDD using arbitrary thresholds. Specifically, we use placebo cutoffs at AR$ 500K and AR$ 2M, rather than the true thresholds of AR$ 750K and AR$ 1.5M. Again, if we observed significant discontinuities at the placebo thresholds, it would suggest that our main results—based on the true thresholds—might be spurious. On the contrary, Figure 3 shows no significant discontinuities at the placebo thresholds, and thus provides reassurance for the validity of our main results. More precisely, panels (a) and (b) show that the estimated discontinuities in tax rates at these points are close to zero and statistically insignificant ($\beta^{own}_{poor} = 0.003$, p-value $= 0.592$; $\beta^{own}_{rich} = -0.002$, p-value $= 0.652$). Similarly, panels (c) and (d) show no significant effects on tax compliance ($\beta^{own}_{poor} = -1.612$, p-value $= 0.274$; $\beta^{own}_{rich} = -1.596$, p-value $= 0.371$). Moreover, while Figure 3 presents results based on two arbitrary placebo thresholds, Figure D.2 demonstrates that the conclusions hold more generally across a wider range of placebo thresholds.

Taken together, these placebo tests provide strong evidence that our findings are not driven by pre-existing trends or confounding factors. Moreover, we show that the results are robust to a wide host of additional robustness checks. Appendix D.1 presents the density of the running variable and shows no evidence of manipulation around either cutoff. This is as expected, given the context—manipulation by households is virtually impossible, since assessed values are calculated and reported by the provincial tax authority. Appendix D.2 presents additional robustness checks, including the inclusion of pre-reform controls, restricting the sample to residential properties only, using alternative definitions of the dependent variable (e.g., payments within six months of the due date instead of three), applying a uniform kernel, and estimating models with polynomials of different degrees, among others.

### 3.3 Own-Rate Elasticities

To facilitate interpretation, we express the magnitude of the own-rate effects in terms of behavioral elasticities. Let $\beta^{own}_j$ represent the reduced-form effect on compliance for group $j$, where $j \in \{poor, rich\}$, obtained from the RDD. Let $\alpha_j$ denote the first-stage effect of the reform on tax rates for group $j$, also obtained from the RDD analysis. Let $C^{own}_j$ and $\tau_j$ denote the baseline compliance levels and tax rates, captured by the levels around the relevant thresholds. For each elasticity shown below, we report 90% confidence intervals in brackets, calculated with 5,000 bootstrap iterations:

$$\varepsilon^{own}_{poor} = \frac{\frac{\beta^{own}_{poor}}{C^{own}_{poor}}}{\frac{\alpha_{poor}}{\tau_{poor}}} = \frac{\frac{-2.718}{60.1}}{\frac{0.392}{2.221}} = \underset{[-0.388,-0.103]}{\text{-0.257}} \tag{1}$$

$$\varepsilon^{own}_{rich} = \frac{\frac{\beta^{own}_{rich}}{C^{own}_{rich}}}{\frac{\alpha_{rich}}{\tau_{rich}}} = \frac{\frac{-2.248}{58.3}}{\frac{0.146}{1.837}} = \underset{[-0.906,-0.023]}{\text{-0.484}} \tag{2}$$

The own-rate elasticity for poor households is -0.257, indicating that for every 1% decrease in their tax rate, compliance increases by 0.257%. In turn, the own-rate elasticity for rich households is -0.484, implying that for every 1% increase in their tax rate, their compliance decreases by 0.484%. Both elasticities are not only statistically significant, but also economically meaningful, reflecting substantial behavioral responses to tax rate changes. The elasticity is larger (in absolute value) for the rich (-0.484) than for the poor (-0.257), suggesting that households may be more likely to reduce compliance after a tax hike than to increase it after a tax cut. However, we must take this result with a grain of salt, as the difference between these two elasticities is statistically insignificant (p-value = 0.402).

### 3.4 Replication: The 2024 Reform

The local government implemented a further progressive reform in 2024, this time targeting the fixed component of the tax rather than the variable one. Thus, we were able to replicate our own-rate analysis of the 2023 reform using data from the 2024 reform. There are two limitations of the 2024 analysis, however. First, the experiment was conducted during a more turbulent political and economic context—explained in more detail in Section 4.6 below, as it is more relevant to the field experiment. Second, for the 2024 analysis, we had only two months of post-treatment data. However, this is not a major limitation, as this time horizon still provides reasonable statistical power.

The own-rate effects from the 2024 reform are presented in Appendix I.2 and summarized below. In terms of sign and statistical significance, the results for the 2024 reform are consistent with those of the 2023 reform. Poor households responded to the tax reduction by increasing their compliance, whereas rich households responded to the tax increase by reducing their compliance. Notably, the own-rate elasticities are nearly identical between the two reforms: for poor households the estimates are -0.257 for 2023 and -0.288 for 2024, while for rich households the estimates are -0.484 for 2023 and -0.471 for 2024. This consistency lends confidence to the accuracy and robustness of the results.

# 4 Cross-Rate Effects

## 4.1 Experimental Design

In this section, we seek to estimate how changes to the tax rates of others affect one's own perceptions of fairness and tax compliance. We designed a natural field experiment to identify these cross-rate effects. In close collaboration with the research team, the municipality sent a personalized letter to taxpayers in January 2023. Our main sample includes approximately 92,000 taxpayers who received a letter.[22,23] For reference, Appendix L.1 displays a sample letter, in this case sent to a recipient from the group of poor households. These letters included personalized information to each account holder, such as the assessed value of the property, a monthly breakdown of the tax amount due, their expiration dates, and payment options. This level of detail ensured that the recipients had a clear understanding of their financial obligations.

Both treatment and control letters included on the front page the exact peso amount by which the recipient's own tax amount changed, under the "progressivity correction" line (for reference, see Appendix L.1). The experimental messages were embedded on the second page. All letters contained a broad statement about the implementation of a tax reform aimed at enhancing equity and specified whether the recipient's *own* tax rate was reduced, increased, or remained unchanged. However, while the control letters did not provide further information about how the reform affected other groups' tax rates, the treatment letters included an infographic and accompanying text that showed how the reform affected tax rates across property valuation brackets (poor, middle, and rich households) and by how much. Panel (a) of Figure 4 shows an example of the message received by poor households in the control group, while panel (b) shows the corresponding message for the treatment group.[24]

Since it was an official government communication (as opposed to a letter from researchers), the municipality had full discretion over how to convey the reform. The municipality prioritized a simple and easy-to-understand message that was both non-deceptive and technically accurate. In the treatment message, the municipality communicated a discount of 30% for poor households and a surcharge of 15% for rich households. This message conveys the essence of the reform and some of this detail, but it is built on two simplifications. First, the figures communicated by the municipality correspond to the ac-

[22]We excluded 3,568 taxpayers who made up-front payments for 2023 before the letters were sent, as their compliance decisions had already been made. Additionally, we excluded 12,689 taxpayers with a registered email address from the main sample, as they constituted a separate survey sample. As discussed in the following section, the results remained consistent when they were included.

[23]This was the only bill of 2023 delivered in paper format. The municipality sent additional communications during that year, but only digitally to the subset of taxpayers with a registered email address.

[24]For sample letters sent to middle and rich households, see Appendix L.1.

tual reform—changes to the variable component of the tax. However, due to the existence of a fixed component, the actual discounts and surcharges on the total tax liability were somewhat smaller than those described in the message.[25] Second, the surcharge for the rich contained two levels. The 15% surcharge for rich households mentioned in the letter was computed as an unweighted average of the change for properties valued between AR$ 1.5M and AR$ 3M (facing a 10% surcharge) and properties valued above AR$ 3M (facing a 20% surcharge).

Because the reform was neither widely discussed publicly nor covered in the media—and was approved and implemented over a relatively short period—most taxpayers were likely unaware of it or its details. This interpretation is supported by the survey data presented in the results section, which show that only a minority of control group respondents reported being aware of the reform. Given this context, we believed the treatment had a meaningful opportunity to raise awareness and salience of the reform.

Each subject selected to receive a letter was randomly assigned, with equal probability, to either a treatment or a control letter.[26] Because subjects in different groups (poor, middle, and rich) were impacted differently by the reform, we wanted to study the treatment effects separately for each group. For that reason, we stratified the randomization across the three property valuation groups. Appendix E shows that, consistent with random assignment, the treatment arms are balanced in observable characteristics.

Lastly, a minor design feature is that a small group of taxpayers (5,913) was randomized to receive no letter at all. The purpose of this secondary treatment arm was to assess whether recipients were attentive to the letters. As shown in prior research on tax reminders (Cruces et al., 2025), letters that include useful payment information and serve as reminders can increase complianceprovided they are opened. Thus, comparing the control letter group with the no-letter group offers a straightforward way to validate whether taxpayers were reading the correspondence.

## 4.2 Complementary Taxpayer Survey

The main outcome of interest of the experimental design is the taxpayer's tax compliance choices, sourced from administrative records. Additionally, we conducted a post-treatment survey of taxpayers to complement the results and to shed light on the potential underlying mechanisms. Of the original population of 92,000 in our experimental sample, we

[25] For example, properties in the bottom 37% of the value distribution (poor households) saw a median tax reduction of about 13.2%, while those in the top 27% (rich households) experienced a median tax surcharge of approximately 9.9%—see Appendix A.1 for more details.

[26] For the experimental analysis, we excluded from the sample individuals who had already made payments in 2023 before the letters were sent, as they had already made their choices and could not be influenced by the letter.

set aside a group of 14,646 households with email addresses registered in the municipality database—called the *email sample*. As part of the main experiment, households in this sample were assigned to treatment and control groups and received the respective informational letters in January 2023. Additionally, only for this group, we emailed an invitation to complete a survey about municipal taxes and services 45 days after the original mailing.[27] We thus sent 14,646 emails inviting taxpayers to participate, providing a short description of the study and a direct survey link. Because taxpayers invited to the survey received additional information (both in the body of the invitation email and in the survey) and questions related to the reform (as described below), as well as further reminders by email, we excluded them from the main experimental sample in our analysis.[28]

This survey was conducted between February and April 2023. Potential respondents received a first email on February 23, 2023, and a reminder on March 10, 2023. The response rate was 11.8%, which is high compared to benchmarks in similar survey-based tax compliance studies.[29] After excluding a small number of households that did not have property value assessments, we ended up with a final sample of 1,723 survey responses. Table E.5 indicates that treatment and control groups are balanced in their observable characteristics even within the sample of survey respondents. In addition, there are no significant differences in the response rate to the survey between treated and control households. One caveat, however, is that the email sample is not fully representative of the broader taxpayer population. Most notably, pre-treatment compliance rates are significantly higher in the email sample than in the main sample.[30] Despite this difference, most other baseline characteristics are fairly similar between the email sample and the rest of the taxpayers (see Appendix G). In any case, the survey was primarily designed to provide qualitative insights into the underlying mechanisms, so full representativeness is not essential for its purpose.

Appendix M.2 presents the complete survey instrument, which we describe below. One challenge was that the survey would take place significantly later than the letter: the median respondent completed the survey 52 days after they received the letter. Drawing on existing literature that documents the decay of information treatment effects over time (e.g. Cavallo et al., 2017; Bottan and Perez-Truglia, 2022), we were concerned that the treatment effects could have largely dissipated by then. For that reason, we reinforced

[27]Due to concerns about potential contamination from the survey content, our baseline results excluded the email sample from the analysis of the treatment effects on compliance. Nonetheless, Appendix D.2 shows that the results remain consistent when this group is included.

[28]We have also excluded them in our control versus no letter estimates in Appendix F.2.

[29]For example, Bergolo et al. (2020) reported a response rate of 8.9% in an online survey of taxpayers in Uruguay who received email invitations similar to ours.

[30]The average compliance rate in the control group is 74.1% in the email sample, compared to 47.9% in the main sample.

the message by showing it to subjects in both the email invitation and on the second page of the survey.[31]

The first goal of the survey was to assess whether the treatment message had the intended effect on awareness of the reform. Participants were asked about their awareness of various recently implemented municipal policies. In particular, we asked whether they were aware of a measure introducing “discounts for low-valuation properties and additional charges for high-valuation properties” in the local property tax. This information appeared in the treatment letter but not in the control letter, so we would expect different responses across the two groups. Awareness of other municipal policies—mentioned in both the treatment and control letters—can serve as a placebo outcomes.

The second goal of the survey was to measure the effect of the treatment on perceived tax fairness. Participants were asked to rate the fairness of the property tax schedule: “How fair do you think the current distribution of municipal tax rates is between people with more and fewer resources—that is, how much each of these groups pays?” Responses were recorded on a 0–10 scale, ranging from “Very Unfair” to “Very Fair.” A positive treatment effect would suggest that households viewed the progressive reform as making the schedule fairer, while a negative effect would indicate the opposite. We also included an adapted version of a General Social Survey question on the government’s role in reducing the gap between rich and poor. As discussed below, we used this question as a placebo outcome: because it was designed to capture deeply held general preferences rather than views on a specific policy, we did not expect it to be affected by the information on the property tax reform.

## 4.3 Effects on Reform Awareness and Perceptions

The validity of mailing experiments like ours relies on households opening and engaging with the letters. In the extreme, if taxpayers did not open the letters—or did not turn to the back of the page—they would not have been exposed to the information treatment at all. We begin by presenting evidence that a significant share of subjects opened the letters and paid attention to their content. We first compared tax compliance between households that received a control letter and a small fraction that was randomly assigned to receive no letter at all. The results, presented in Appendix F.2 and summarized here, show that taxpayers with control letters consistently and substantially increased their tax compliance relative to households that received no letter at all. Overall, the results from this auxiliary experiment suggest that a significant share of taxpayers opened the letters we sent.[32] Importantly, this finding is consistent across poor, middle-income, and rich

[31] For a sample of an email invitation of a household in the treatment group, see Appendix M.1.

[32] The average difference of about 5 pp aligns with findings from a recent systematic review of similar experiments (Antinyan and Asatryan, 2024). It is worth noting that experiments of this type often find

households, meaning that all three groups were equally likely to open the correspondence.

Next, we investigated whether the treatment message, compared to the control message, increased awareness of the reform as intended. In the survey, we asked respondents whether they knew about the reform. As depicted in panel (a) of Figure 5, only about 15.3% of the respondents in the control group indicated that they were aware of it, with relatively similar levels of knowledge among taxpayers in the three property valuation groups. This aligns with our prior expectation that taxpayer knowledge of the reform was low given the lack of public discussion about it. Panel (b) shows a consistent and significant treatment effect across all valuation groups (in gray) as well as separately for poor households (in blue), medium households (in green), and rich households (in red). Our information treatment significantly increased awareness of the reform in the three groups by about 8 pp on average.

Next, we provide results from a falsification check. The back of the letter also included information about three additional policies the municipality had implemented—business credits, free and express commercial permits, and reduced paperwork for new businesses. We also asked respondents about their awareness of these policies in the survey. In contrast to the information about the reform's progressive nature (present only in treatment letters), both control and treatment letters included information about these three policies. So, we should not expect differences in awareness of these policies across treatment and control groups. Panels (c) and (d) of Figure 5 replicate panels (a) and (b), but for the first of the placebo outcomes, awareness of the business credits—results for the other two placebos are presented in Appendix H. As expected, the treatment effect on this placebo outcome is close to zero and statistically insignificant. The overall null effects on the placebo outcomes are reassuring, suggesting that the observed effects on perceived fairness are neither spurious nor driven by mechanical factors such as anchoring.

Having established that taxpayers read and incorporated the information in our treatments, we were able to turn to a series of outcomes related to the taxpayers' perceived tax fairness. These results are presented in Figure 6. More precisely, panel (a) shows relatively high perceived levels of fairness reported by taxpayers in the control group, with an average stated perception of approximately 5.5 points (on a 1–10 scale), and roughly similar across poor, middle, and rich households. Most importantly, panel (b) of Figure 6 shows that the treatment increased perceived fairness across all groups, indicating that, on average, households viewed the more progressive tax schedule as fairer.[33] In the full sample, the average treatment effect is positive, statistically significant and economically

large effects because they reflect a substantial reminder effect of receiving a letter, beyond the content of the letter itself. In contrast, our experiment produced smaller effects because it compared subtle variations between two letter types, effectively netting out the basic reminder effect.

[33]This finding is consistent with prior evidence that some taxpayers have appetite for progressive tax schedules (e.g., Durante et al., 2014; Stantcheva, 2021; Hoy, 2025; Nathan et al., 2024).

significant too (0.57 points on a 0–10 scale). Moreover, this effect is similar across all three groups—poor, middle, and rich households.

We also provide a falsification check using a placebo outcome. We tested whether our treatment had any effect on the respondents' general views on the role of government in reducing the gap between the rich and the poor, which reflects deeply rooted general preferences. Reassuringly, the results in panel (d) of Figure 6 show that our information treatment did not influence this placebo outcome. The absence of placebo effects strengthens the credibility of our findings, indicating that the treatment's impact on perceived fairness is unlikely to be an artifact of cognitive biases, such as anchoring, or other mechanical survey effects.

## 4.4 Effects on Tax Compliance

We now turn to the core outcome of our experiment: the cross-rate effects on actual tax compliance, as measured using administrative records. Panel (a) of Figure 7 presents an event-study analysis of treatment effects on tax compliance (i.e., comparing tax payments between treatment and control households) at the most granular level, monthly payments. We estimated the following regression separately for each pre-treatment and post-treatment month:

$$\begin{aligned} Y_i = \theta_1 \cdot \text{Treat}_i \cdot \text{Poor}_i + \theta_2 \cdot \text{Treat}_i \cdot \text{Middle}_i + \theta_3 \cdot \text{Treat}_i \cdot \text{Rich}_i \\ + \theta_4 \cdot \text{Poor}_i + \theta_5 \cdot \text{Middle}_i + \theta_6 \cdot \text{Rich}_i + X_i\theta_7 + \varepsilon_i \end{aligned} \tag{3}$$

The outcome $Y_i$ is an indicator variable for whether that month's tax bill was paid within three months of the due date. The key coefficients of interest are the interactions between the treatment indicator ($\text{Treat}_i$) and the group indicators ($\text{Poor}_i, \text{Middle}_i, \text{Rich}_i$),[34] allowing us to estimate group-specific average treatment effects. $X_i$ is a vector of additional control variables: 12 months of pre-treatment outcomes,[35] indicator variables for whether the property was an "always payer" or "never payer" in the previous year, property valuation (in logs), an indicator for residential properties, and an indicator for properties that made an annual payment in the previous year.[36] $\varepsilon_i$ is the usual error term. The standard errors are clustered at the individual level.

Panel (a) of Figure 7 shows the results from the event-study analysis. The x-axis

[34]An omitted category is not needed because the regression does not include a constant term.

[35]For the post-treatment periods, these are the compliance indicators during each of the 12 months of 2022. For the pre-treatment periods, these are the compliance indicators during each of the previous 12 months.

[36]Some control variables have a few missing values. To avoid dropping these observations, for each of these variables, we imputed missing values with zeros and included an additional variable indicating whether the value was imputed.

indicates the number of months before or after the letters were mailed, while the y-axis corresponds to the treatment effects on tax compliance. The pre-treatment coefficients, which are close to zero and statistically insignificant, demonstrate successful random assignment: i.e., prior to the mailing intervention, tax compliance was balanced between treatment and control groups. This is true within each of the three groups—poor, middle, and rich households.

We begin by describing the effects for poor households. In the post-treatment period, there is a clear, positive, and significant impact of the treatment on compliance—the six post-treatment coefficients are mostly statistically significant and range from 0.5 to 1.0 percentage points. This indicates that, relative to poor households who did not receive the information, those who learned about the tax hike on the rich increased their tax compliance. In terms of the evolution of the effects, panel (a) of Figure 7 suggests that the impact may gradually dissipate over time. This pattern is consistent with evidence of waning effects in other correspondence experiments, presumably because the information fades from taxpayers' immediate memory (e.g., Bergolo et al., 2023).[37]

In contrast to the significant positive effects observed for poor households, the treatment effect for rich households was negative, though less precisely estimated and mostly statistically insignificant. Relative to rich households who did not receive the information, those who learned that the poor received a tax cut—if anything—reduced their compliance. This finding stands in sharp contrast to the survey results, which show that rich households responded to the same information by reporting higher perceptions of fairness. For middle households, we find a similar inconsistency between their survey responses and their actual behavior. After learning about the effects of the reform on other groups—namely, the tax cut for the poor and the tax hike for the rich—the survey data indicate that middle households reported increased perceptions of fairness. However, this information did not alter their compliance behavior, suggesting they did not put their money where their mouths were.

One limitation of the event-study analysis is that it measures effects at the most granular level (monthly), which is inefficient in terms of statistical power. To provide a more precise estimate of the treatment effects, we present results at a more aggregated level (quarterly). Specifically, we estimate the following regression separately for the three months before the mailing (the pre-treatment period) and the three months immediately after (post-treatment period):

$$Y_{i,t} = \theta_0 + \theta_1 \cdot \text{Treat}_i + X_i\theta_2 + \omega_t + \varepsilon_{i,t} \tag{4}$$

[37] Also consistent with this interpretation, the gradual decline in treatment effects aligns with the results of the comparison between control letters and no letters.

The outcome $Y_it$ is an indicator for whether month $t$'s bill was paid within three months of the due date. $\text{Treat}_i$ and $X_i$ are defined as in equation (3). One difference with equation (3), however, is that this specification estimates the regression separately for each group (poor, middle, and rich). $\omega_t$ is a vector of month fixed effects and $\varepsilon_i$ is the usual error term. The standard errors are clustered at the individual level.

Panel (b) of Figure 7 presents the results from this alternative specification. For poor households, we find a positive treatment effect of 0.808 pp (p-value=0.006). As discussed in more detail in Section 4.5 below, this effect is not only statistically significant but also economically meaningful. For rich households, we find a negative treatment effect of -0.313 pp, but even though it is more precisely estimated, it is still statistically insignificant at conventional levels (p-value=0.393). For middle households, we find an effect close to zero (0.152 pp) and statistically insignificant (p-value=0.619).

We show that these results are robust to several checks. Appendix F.1 demonstrates that the treatment effects described above are also evident in the raw time series data on tax compliance. Appendix D.2 shows that the results hold under alternative sets of control variables, different sample restrictions (e.g., limiting to residential properties), and alternative definitions of compliance (e.g., payments made within six months of the due date instead of three), among others.

The cross-rate effect for poor households reported above likely understates the true size due to substantial attenuation bias.[38] While we provided evidence that a significant share of households opened the letters, this does not necessarily imply that all recipients read them fully or carefully. The control and treatment messages were on the second page of the letter, and it is likely that a significant share of households either did not flip to the second page or did not pay attention to the letter at all. Thus, the treatment effect reported above (0.808 pp for poor households) should be interpreted as an intention-to-treat (ITT) effect. To obtain the Average Treatment Effect on the Treated (ATET), we need to scale up the ITT by an adjustment factor that accounts for non-compliance. The recent literature on correspondence experiments discusses adjustment factors between 2 and 7.5 (Perez-Truglia and Cruces, 2017; Gerber et al., 2020; Nathan et al., 2020; Bottan and Perez-Truglia, 2025). For example Perez-Truglia and Cruces (2017) note that, according to the Environmental Protection Agency, nearly 50% of unsolicited mail is discarded unopened, determining a conservative non-uptake adjustment factor of 2.[39] This would result in a scaled-up cross-rate effect for poor households of 1.616 pp ($= 0.808 \cdot 2$).

[38] It could be argued that the own-rate effects (Section 3) could be affected by *some* attenuation bias as well, because not every taxpayer was necessarily aware of the change in their own tax liability. However, this attenuation bias is probably minor, and adjusting for it would be more challenging.

[39] An alternative approach using survey results yields a much higher adjustment factor—see Appendix F.3 for details.

## 4.5 Cross-Rate Elasticities

We can assess the magnitude of the cross-rate effects for poor and rich households by expressing them as behavioral elasticities.[40] Let $\beta_j^{cross}$ represent the treatment effect on compliance for group $j$ as reported in the experiment, where $j \in \{poor, rich\}$. Let $C_j^{cross}$ denote the baseline compliance of each control group in 2023, and let $\frac{\Delta\tau_{-j}}{\tau_{-j}}$ represent the tax rate change for the opposite group. That is, for a poor household, the relevant change in the opposite group's tax rate corresponds to the tax hike on the rich; for a rich household, it corresponds to the tax cut for the poor. For simplicity, we characterize the tax changes as a 30% decrease in the tax rate for poor households and a 15% increase for rich households, matching the information conveyed in the letters regarding the variable component of the tax.[41] The results are presented below, with 90% confidence intervals in brackets, calculated using 5,000 bootstrap iterations:

Our cross-rate elasticities are computed using the variable component in the denominator to maintain consistency and simplicity, as this was how the tax change was communicated in the treatment messages. On the other hand, the own-rate elasticities are based on the total tax.

$$\varepsilon_{poor}^{cross} = \frac{\frac{\beta_{poor}^{cross}}{C_{poor}^{cross}}}{\frac{\Delta\tau_{rich}}{\tau_{rich}}} = \frac{\frac{0.808}{55.2}}{0.15} = \underset{[0.040,0.158]}{0.098} \tag{5}$$

$$\varepsilon_{rich}^{cross} = \frac{\frac{\beta_{rich}^{cross}}{C_{rich}^{cross}}}{\frac{\Delta\tau_{poor}}{\tau_{poor}}} = \frac{\frac{-0.313}{54.8}}{-0.3} = \underset{[-0.018,0.055]}{0.019} \tag{6}$$

On the one hand, the cross-rate elasticity for rich households (0.019) is statistically insignificant, which is unsurprising given that the treatment effect for this group was also insignificant. However, we can reject the null hypothesis that the cross-rate elasticity for the rich equals that for the poor (p-value = 0.059), highlighting a meaningful difference in responses between the two groups. On the other hand, the cross-rate elasticity for poor households (0.098) is statistically significant. This is not surprising given that the treatment effect for the poor was significant, but this elasticity helps illustrate the economic magnitude of the response: for each 1% increase in the tax rate of rich households, poor

[40] We do not compute a cross-rate elasticity for middle households because we lack sufficient information. The treatment for this group provided information about both a tax cut for the poor and a tax hike for the rich. As a result, even if their compliance responded to the treatment, we would not be able to attribute that response to one change or the other.

[41] Intuitively—and for simplicity—the calculations in this section assume that households interpreted the information about changes to other groups tax rates as applying to the entire tax, rather than just the variable component. This is likely a reasonable assumption given the limited-attention context and the fact that details about the variable component appeared in fine print. In any case, this simplified approach is the more conservative one, as the alternative would result in smaller denominators and thus larger elasticities.

households increase their compliance rate by 0.098%. To put this in perspective, we can compare it to the own-rate elasticity for poor households (-0.257): a 1% increase in the tax rate of the rich has the same effect on poor households' compliance as lowering the poor households' own tax rate by 0.38% ($= -\frac{0.098}{-0.257}$). Moreover, because of substantial attenuation bias from imperfect treatment uptake, the true cross-rate elasticity is likely at least twice as large. Under a conservative adjustment factor of 2, a 1% increase in the tax rate of the rich would have the same effect as lowering the poor households own tax rate by 0.76%. In sum, this evidence suggests that, beyond the direct effect of ones own tax rate, perceptions of the broader tax schedule can play an important role in shaping tax morale.

### 4.6 Replication: The 2024 Reform

Taking advantage of the new progressive reform in 2024, we conducted a second large-scale field experiment to assess whether the results of the 2023 experiment replicated. In addition to the changes to the variable components from the 2023 reform, the 2024 reform made the fixed component more progressive. The 2024 treatment message was similar to that of 2023, emphasizing that the new reform further reduced the tax for poor households, maintained it for middle households, and increased it for rich households. We re-randomized households into treatment and control letters within each of the property value groups.[42]

In addition to replicating our results, we used the 2024 experiment as an opportunity to refine the design by making the distinction between the treatment and control messages even more subtle. In the 2023 letters, both treatment and control versions displayed the magnitude of the progressivity discount for the recipient on the front page. However, a limitation of the 2023 control letter was that it did not include an infographic and therefore did not reiterate that information about the size of the discount in the message on the second page. As shown in the screenshots provided in Appendix L.2, the 2024 letters further minimized the differences between treatment and control messages. In contrast to 2023, when only the treatment group received information in the form of an infographic, in 2024 both groups received one. The treatment group's infographic showed the tax changes for all three groups—poor, middle, and rich—while the control group's infographic featured a single bar representing the change for the households own group only.[43]

[42]For details on the 2024 experimental design see Appendix I.1.

[43]For example, for poor households, the control message included an infographic with a single bar indicating the discount for poor households, while the treatment message featured an infographic with three bars: the discount for poor households, the surcharge for rich households, and no change for middle households.

While the reform and letters were similar in 2023 and 2024, two key contextual differences could have attenuated the treatment effects in 2024. First, the 2024 reform and its letters were introduced during a turbulent economic and political period, right after a heated presidential election, which may have made the reform seem less significant relative to broader national events. Second, the similarity between the reforms (and messages) may have led some taxpayers in 2024 to misinterpret our communications as referring to the previous year's reform, potentially weakening their response.

With these caveats in mind, we replicated our analysis of the 2023 experiment using data from the 2024 experiment. The complete set of results is presented in Appendices I.3 and I.4 and is summarized below. Consistent with the 2023 results, we again observe in 2024 a significant difference in compliance between subjects who were assigned to the control letter and those who were assigned to no letter. This suggests that, despite the volatile environment, a substantial fraction of the subjects opened and paid attention to the letters. Most importantly, the difference in compliance between treatment and control letters in 2024 mirrors the results of the 2023 experiment. For poor households, we again found a significant treatment effect of 0.721 pp (p-value = 0.040), which is similar to, and statistically indistinguishable from, the corresponding effect of 0.808 pp estimated for the 2023 experiment. As in 2023, in the 2024 experiment we found a negative but statistically insignificant effect on rich households and an effect for middle households that is close to zero and statistically insignificant. The fact that the two experiments yielded consistent findings, despite notable differences in context, highlights the robustness of our results.

# 5 Counterfactual Analysis

## 5.1 Framework

In this section, we conduct a counterfactual analysis to illustrate the policy implications of our findings. We consider the effects of a hypothetical progressive reform that, similar to the actual reform implemented in January 2023, reduces the tax rate on poor households by $\%\Delta\tau_{poor}$ and increases the tax rate on rich households by $\%\Delta\tau_{rich}$. These parameters are calibrated so that, in a world with fixed tax compliance, the reform is revenue-neutral and achieves a high progressivity target: rich households pay an effective tax rate 0.5 pp higher than that of poor households. This outcome is achieved by changing the tax rates by $\%\Delta\tau_{poor} = -45.5\%$ and $\%\Delta\tau_{rich} = +17.32\%$. We aim to understand the effects of this reform on effective tax progressivity and tax revenues under behavioral responses, and contrast it to a counterfactual scenario that ignores behavioral responses.[44]

[44] Throughout this section, we use *effective* tax progressivity, which is different from *statutory* tax progressivity. The latter refers to the progression of tax rates as determined by law—essentially, how tax

Our hypothetical economy was calibrated to reflect many features of the real-world context, but we simplified certain aspects for the sake of brevity. We assume households are homogeneous within each property value bracket, allowing us to focus on average effects.[45] Let subscript $j \in \{poor, middle, rich\}$ denote poor, middle, and rich households. Let $\gamma_j$ be the share of households in each of these three groups. Let superscript $s$ denote the scenarios: $s = 0$ represents the status quo (i.e., before the reform), while $s = A$ is the *actual* post-reform scenario (with behavioral responses) and $s = C$ is the *counterfactual* post-reform scenario without behavioral responses. Let $L_j^s > 0$ denote the average tax liability. For example, $L_{poor}^0$ is the average liability of poor households before the reform. Let $C_j^s \in (0, 1)$ denote the average tax compliance. For example, $C_{poor}^0$ is the average probability that a poor household pays its taxes in time before the reform. Lastly, let $T_j^s = L_j^s \cdot C_j^s$ be the amount of taxes paid on time. For this counterfactual analysis, we fix the values $\{\gamma_j, L_j^0, C_j^0\} \forall j$ to match the corresponding averages during the 12 months preceding the 2023 reform, from January through December 2022.[46]

The progressive reform affects taxes paid through two channels. The first is the mechanical channel: lowering tax rates for the poor reduces their tax liabilities, whereas increasing tax rates for the rich increases theirs. In the counterfactual scenario with no behavioral responses, this is the only channel at play. When behavioral responses are included, there is an additional channel operating through compliance: i.e., the amount of taxes paid ($T_j^s = L_j^s \cdot C_j^s$) varies due to changes in tax compliance ($C_j^s$). To compare the outcomes with and without behavioral responses, we adopted a sufficient statistic approach (Chetty, 2009). The behavioral responses can be approximated as a linear function of the four behavioral elasticities estimated above ($\varepsilon_{poor}^{own}$, $\varepsilon_{rich}^{own}$, $\varepsilon_{poor}^{cross}$ and $\varepsilon_{rich}^{cross}$):

$$C_{poor}^A = C_{poor}^0 \cdot \left(1 + \varepsilon_{poor}^{own} \cdot \%\Delta\tau_{poor} + \varepsilon_{poor}^{cross} \cdot \%\Delta\tau_{rich}\right) \tag{7}$$

$$C_{rich}^A = C_{rich}^0 \cdot (1 + \varepsilon_{rich}^{own} \cdot \%\Delta\tau_{rich} + \varepsilon_{rich}^{cross} \cdot \%\Delta\tau_{poor}) \tag{8}$$

Following the conservative approach outlined in Section 4, we doubled the cross-rate elasticities to account for attenuation bias. Since each of the four elasticities is subject to

*liabilities* change with property values. Effective tax progressivity, on the contrary, reflects how tax *payments* change with property values, thus incorporating tax compliance. This distinction is critical to understanding the implications of tax reforms, as statutory progressivity does not necessarily translate into effective progressivity.

[45] In the real reform, the impact of the reform on effective tax rates was heterogeneous because it applied only to the variable component—thus, in proportional terms, it affected some households more than others.

[46] In the real-world context of Tres de Febrero, tax compliance is effectively split into two components: nearly all households pay the fixed portion of the tax, largely due to its inclusion in the electricity bill (in 2023, the electricity bill mechanism covered nearly 90% of the fixed charge), while approximately half of households ultimately pay the variable component. However, for the sake of simplifying the counterfactual analysis, we assume a binary framework in which taxpayers pay the full amount or nothing at all.

sampling error, we used bootstrap for inference.[47]

There are some implicit assumptions in this exercise that are worth noting. First, to simplify the exposition of the results, we assume that the behavioral responses of middle households are exactly zero.[48] Second, there is an implicit extrapolation assumption for the own-rate elasticities. These elasticities were estimated by means of a regression discontinuity design and thus correspond to the local average treatment effects for households near the corresponding thresholds. We applied these estimates to all taxpayers within each group, implicitly assuming that the effects estimated near the thresholds are reasonable approximations of the group averages.[49]

## 5.2 Results

The key results are summarized in Figure 8. Panel (a) shows the results corresponding to the first outcome of interest: the effective tax progressivity. The x-axis displays various scenarios: "Pre-Reform" (baseline), "Post-Reform without BR" (assuming no behavioral responses), "Post-Reform BR from poor" (behavioral responses only from poor households), "Post-Reform BR from rich" (behavioral responses only from rich households) and "Post-Reform BR from poor and rich" (responses from both groups).

The key takeaway from panel (a) of Figure 8 is that the reform ends up being less progressive than intended due to the behavioral responses. The first bar from panel (a) shows a slightly regressive tax system at baseline: before the reform, rich taxpayers faced, on average, an effective tax rate that was 0.31 pp *lower* than that of poor taxpayers. More precisely, the rich paid an effective rate of 1.07% while the poor paid an effective rate of 1.38%. The second bar shows that, ignoring behavioral responses, the reform would have made taxes significantly more progressive, with rich taxpayers paying an effective tax rate 0.5 pp *higher* than that of poor households: rich households pay an effective rate of 1.25% versus 0.75% for poor households. The third bar incorporates the behavioral responses of poor households only.[50] Since taxpayers in this group react by increasing their compliance, they end up paying more in taxes, narrowing the gap between the rich and the poor. More precisely, the behavioral responses of the poor *reduce* the effective tax progressivity by 0.113 pp compared to the counterfactual without the behavioral responses. This difference is precisely estimated and highly statistically significant (p-

[47]Within each bootstrap sample, we estimated the four elasticities of interest and used those values to compute $C_j^A$ following equations (7) and (8). We then used the distribution of the results across the bootstrap samples to calculate the 90% confidence intervals and p-values.

[48]For the own-rate channel, this is true by construction since the tax rates for this group were not affected by the reform. And for the cross-rate channel, there is support for this assumption insofar as the treatment effects for middle households were close to zero and statistically insignificant.

[49]A similar extrapolation assumption applies to the cross-rate elasticity too.

[50]In practice, this operates by simply taking equation (7) and setting the behavioral elasticities corresponding to rich taxpayers to zero.

value<0.001). The fourth bar from panel (b) shows our results on tax progressivity when we incorporate the behavioral responses from the rich only. Since taxpayers in this group actually reduce their compliance, the total taxes paid by this group fall, thus lowering effective tax progressivity. In magnitude, the effect of the behavioral responses from the rich is slightly larger than the corresponding effect from the poor, reducing the effective tax progressivity by 0.127 pp (p-value = 0.048). The fifth and final bar incorporates the behavioral responses from rich and poor groups simultaneously. Since both effects reduce progressivity, the net result is also a reduction in effective tax progressivity of 0.240 pp (p-value = 0.002). In other words, the effective progressivity is 48% lower than it would have been without behavioral responses.

The key takeaway from panel (b) of Figure 8 is that the reform—though intended to be revenue-neutral—ultimately reduced tax revenues due to behavioral responses. Panel (b) mirrors panel (a) but focuses on tax revenues instead of effective tax progressivity. The first bar from panel (b) shows that before the reform, the per capita revenue was AR$ 12,843. The second bar, corresponding to the post-reform without behavioral responses, shows the same per capita revenue as in the pre-reform scenario. This outcome is by design, as the reform was calibrated to be revenue-neutral in the absence of behavioral responses. The third bar shows the per capita revenues after accounting for behavioral responses from the poor only. Since poor households increase compliance, this channel generates AR$ 201 in additional per capita revenue, an effect that is highly statistically significant (p-value < 0.001). In contrast, the fourth bar incorporates the behavioral responses from rich taxpayers only. Since this group lowers its compliance, their responses reduce tax revenue by AR$ 763 (p-value = 0.048). The fifth and final bar shows the results when allowing for behavioral responses from taxpayers in both groups. Since the two behavioral responses move in opposite directions, the net impact depends on which effect dominates. Here, the negative effect prevails, meaning that, despite the reform being designed to be revenue-neutral, behavioral responses reduce tax revenue. However, the direction and magnitude of this result should be interpreted cautiously, as the net effect is imprecisely estimated and statistically insignificant (p-value = 0.145).

Appendix J provides more details and discussion about the results of the counterfactual analysis. In particular, while the above analysis breaks down the behavioral responses between rich and poor households, Appendix J.2 further decomposes those effects into own-rate versus cross-rate effects. We show that, while not a majority, a sizable portion of the counterfactual results are driven by the indirect channel.

# 6 Conclusions

Tax progressivity is a cornerstone of modern fiscal policy, widely adopted to reduce inequality and foster social equity. Although the redistributive effects of progressive taxation are well-documented, empirical evidence on its behavioral impacts—particularly regarding tax compliance—remains limited. This paper fills this gap by examining how progressive taxation shapes compliance behavior, combining quasi-experimental and experimental evidence in the context of a municipal property tax reform in Argentina.

Our analysis revealed significant *direct* effects of the progressive reform: reductions in tax rates for poor households significantly increased their compliance, while increases for rich households resulted in decreased compliance. Additionally, we documented significant *indirect* effects: when informed about tax increases on rich households, poor households increased their perceived fairness of the tax schedule and they also increased their tax compliance. These findings demonstrate that behavioral responses to tax reform are shaped not only by the direct pecuniary incentives but also by perceptions of the fairness of the broader tax system. In contrast, while middle and rich households also stated improved perceptions of fairness due to the progressive reform, their tax compliance did not increase. The contrast between the stated and behavioral effects for rich and middle households serves as a cautionary tale of how stated and revealed preferences can diverge. Individuals may express support for more progressive taxes, but talk is cheap—and they do not always put their money where their mouth is. One interpretation of this finding is that, although middle and rich households may *say* they view the tax cut for the poor as fair and claim to be supportive, they may in fact be secretly resentful.

To apply our findings to other contexts, one must account for institutional and cultural differences. In low-enforcement settings, tax reforms have a greater potential to influence compliance, whereas high-enforcement contexts limit such effects. For example, the 40 largest U.S. cities collect nearly 95% of property taxes on time (Chirico et al., 2016), leaving minimal room for a progressive property tax reform to impact compliance. However, even in developed countries, tax enforcement can be imperfect in other contexts. For instance, the Internal Revenue Service estimates that individuals underreport 63% of income subject to minimal third-party reporting, such as self-employment earnings (Cullen et al., 2021). This noncompliance rate is comparable to that observed in the Argentine setting studied in this paper. In such cases, progressive tax reforms could significantly affect tax evasion, even in developed economies. In addition, cultural or political factors can shape these behavioral responses. For example, the magnitude of the effects documented in this paper is likely to differ in other settings, depending on contextual factors such as the prevailing ideological orientation of the society (e.g., left-leaning vs. right-leaning).

Lastly, our findings have implications for the design of progressive tax reforms. Indeed, property taxes represent one of the oldest forms of taxation and their design has not changed much over time (Chancel et al., 2022; Dray et al., 2023). In most countries, property taxes are either proportional or even regressive, which stands in sharp contrast to the progressive nature of modern income and wealth taxation. However, the recent surge in research on wealth inequality has renewed interest in making property taxes more progressive. Our counterfactual analysis showed that accounting for behavioral responses—direct and indirect—significantly reduces effective progressivity and total revenue compared to projections that ignore these responses. This highlights that policymakers, particularly in settings with limited enforcement, must account for varying compliance behaviors to achieve intended distributional and revenue objectives.

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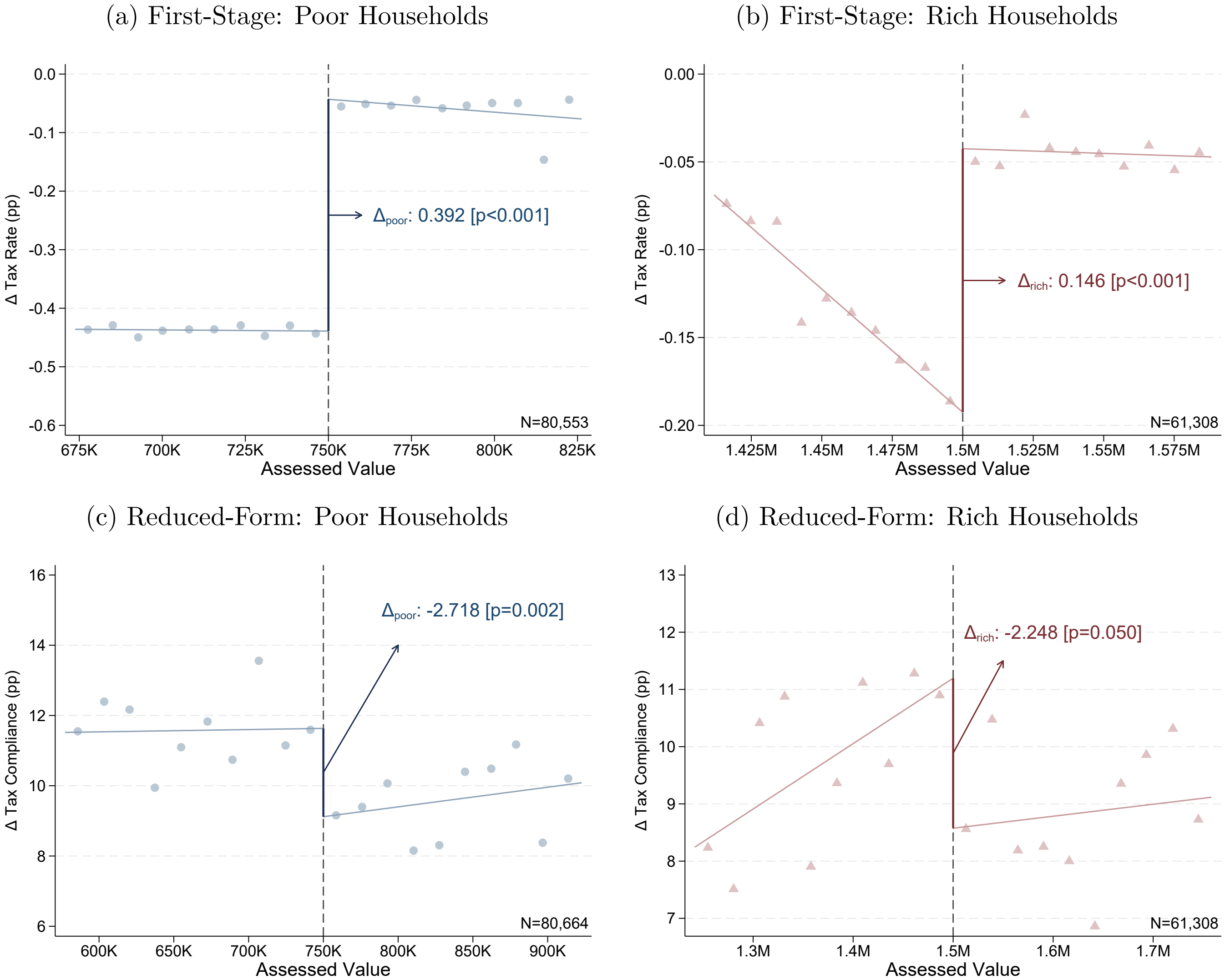
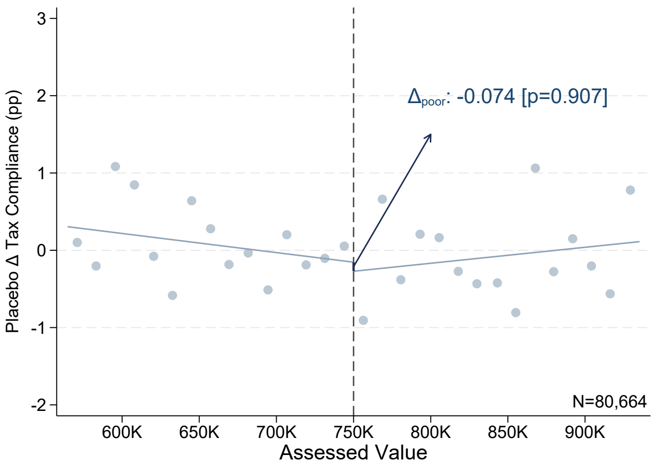
Figure 1: Own-Rate RDD Effects on Tax Compliance for 2023



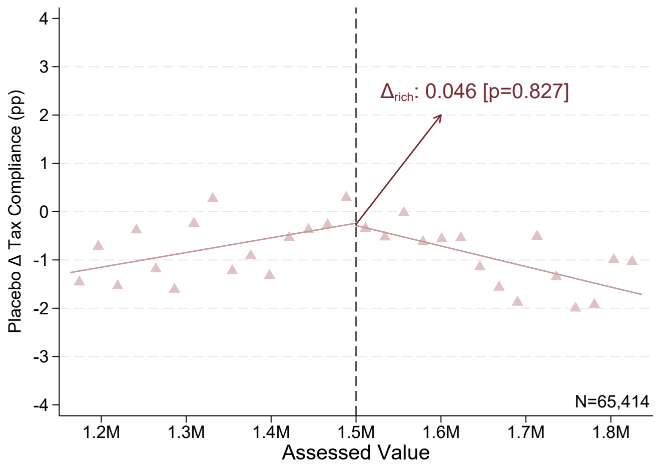

*Note*: This figure calculates the own-rate effects of the progressive property tax reform in 2023, using a two-cutoff RDD. The left figures (in blue) show the results for the poor at the cutoff of AR$ 750K. The right figures (in red) show the results for the rich at the cutoff of AR$ 1.5M. The running variable corresponds to the properties' cadastral values from 2021. The top panels present the first-stage change in tax rates around these thresholds. The outcome is the household-level change in the tax rate between the last semester of 2022 and the first semester of 2023. The bottom panels present the reduced-form effect on tax compliance: how household-level payment rates changed around those thresholds. The outcome is the change in the proportion of monthly bills paid timely between the pre-reform and the post-reform semesters. We define timely payments as bills paid within three months after the due date. Each figure indicates the estimated effect around the cut-off point, where the p-value, taken from a robust bias-corrected inference, is indicated in brackets. So, the RDD point estimates correspond to a change in a time-differenced outcome, reminiscent of a difference-in-discontinuity approach.

Figure 2: RDD Falsification Test using a Placebo Reform Date

(a) Reduced-Form: Poor Households

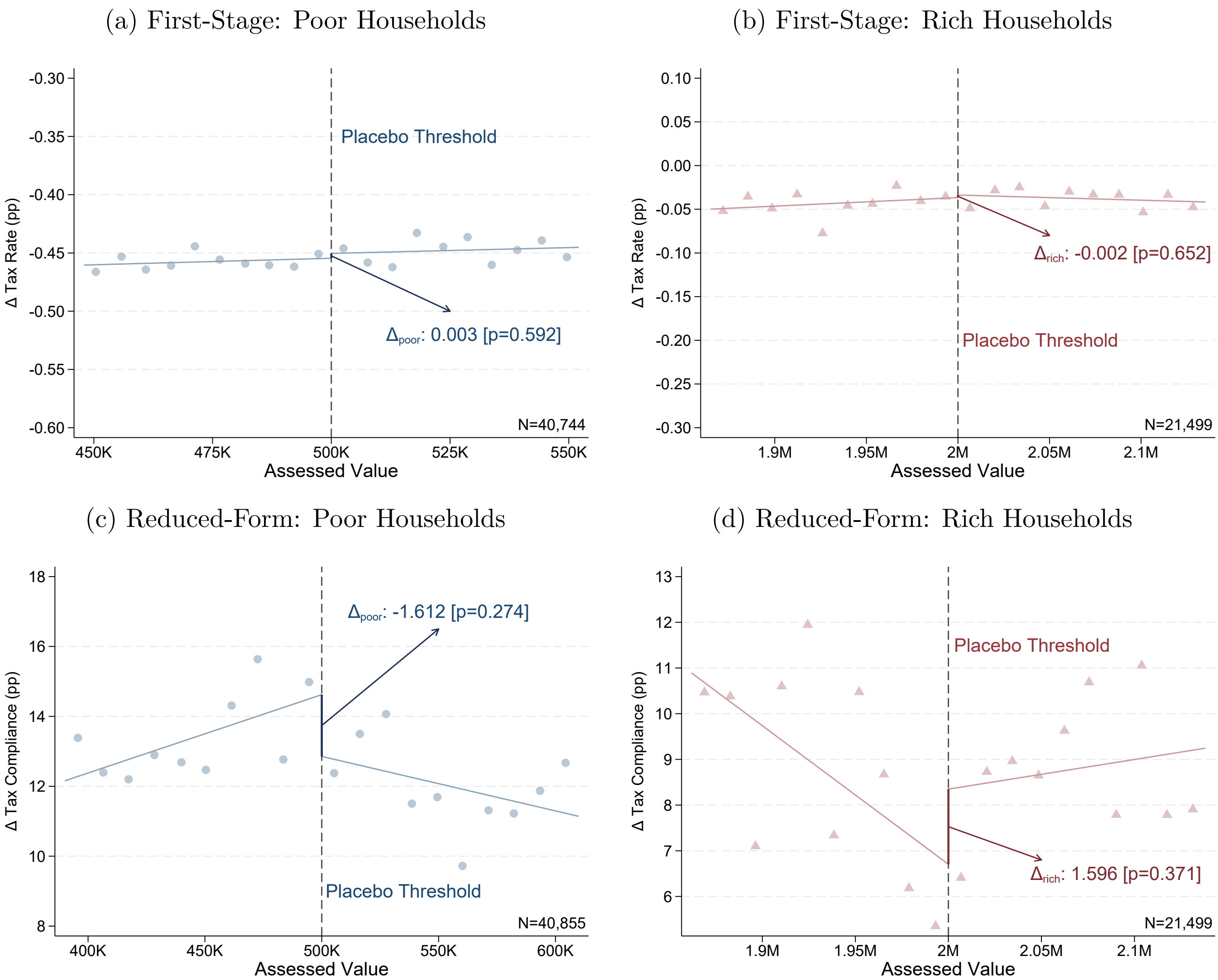



(b) Reduced-Form: Rich Households




*Note*: This figure calculates the own-rate effects using a placebo reform date in the middle of 2023. Panels (a) and (b) present the reduced-form RDD for poor and rich, in which we evaluate the change in tax compliance for both groups for placebo dates between the first semester of 2023 and the last semester of 2023. The x-axis corresponds to the cadastral valuation of the properties from 2021. Each figure indicates the estimated effect around the cut-off point, where the p-value, taken from a robust bias-corrected inference, is indicated in brackets.

Figure 3: RDD Falsification Test using Placebo Thresholds for 2023




*Note*: This figure calculates the own-rate effects using placebo tax thresholds at AR$ 500K and AR$ 2M. Panels (a) and (b) represent the first-stage RDD for poor and rich, respectively. These two figures evaluate the increase in the amounts owed between the last semester of 2022 and the first semester of 2023, in relation to the value of the property. Panels (c) and (d) represent the reduced-form RDD for poor and rich, respectively. They evaluate the change in tax compliance for both groups between the last semester of 2022 and the first semester of 2023. The x-axis corresponds to the properties' cadastral values from 2021. Each figure indicates the estimated effect around the cut-off point, where the p-value, taken from a robust bias-corrected inference, is indicated in brackets.

Figure 4: Sample of Control and Treatment Messages: Poor Households

(a) Control Letter Message

## FAIRER AND MORE EQUITABLE TSG

The TSG **increased below inflation** as in recent years and is also now **fairer and more equitable**.

**Based on your tax valuation, your TSG decreased in relation to the rest.**

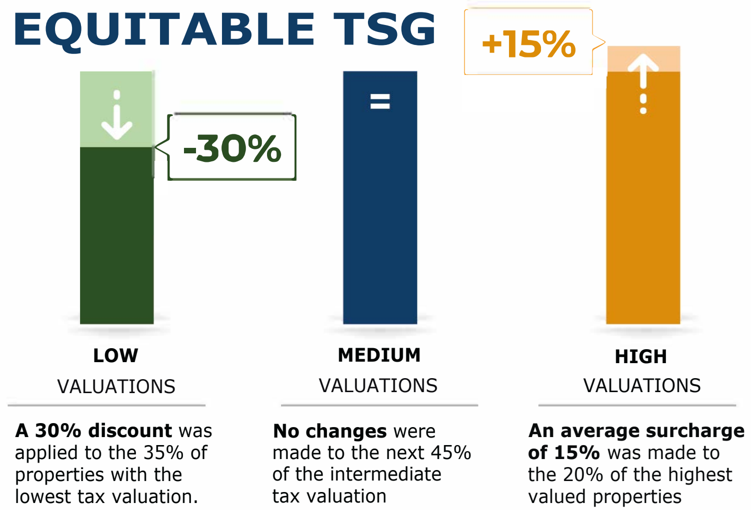
(b) Treatment Letter Message

## FAIRER AND MORE EQUITABLE TSG

The TSG **increased below inflation** as in recent years and also now it is **fairer and more equitable**.

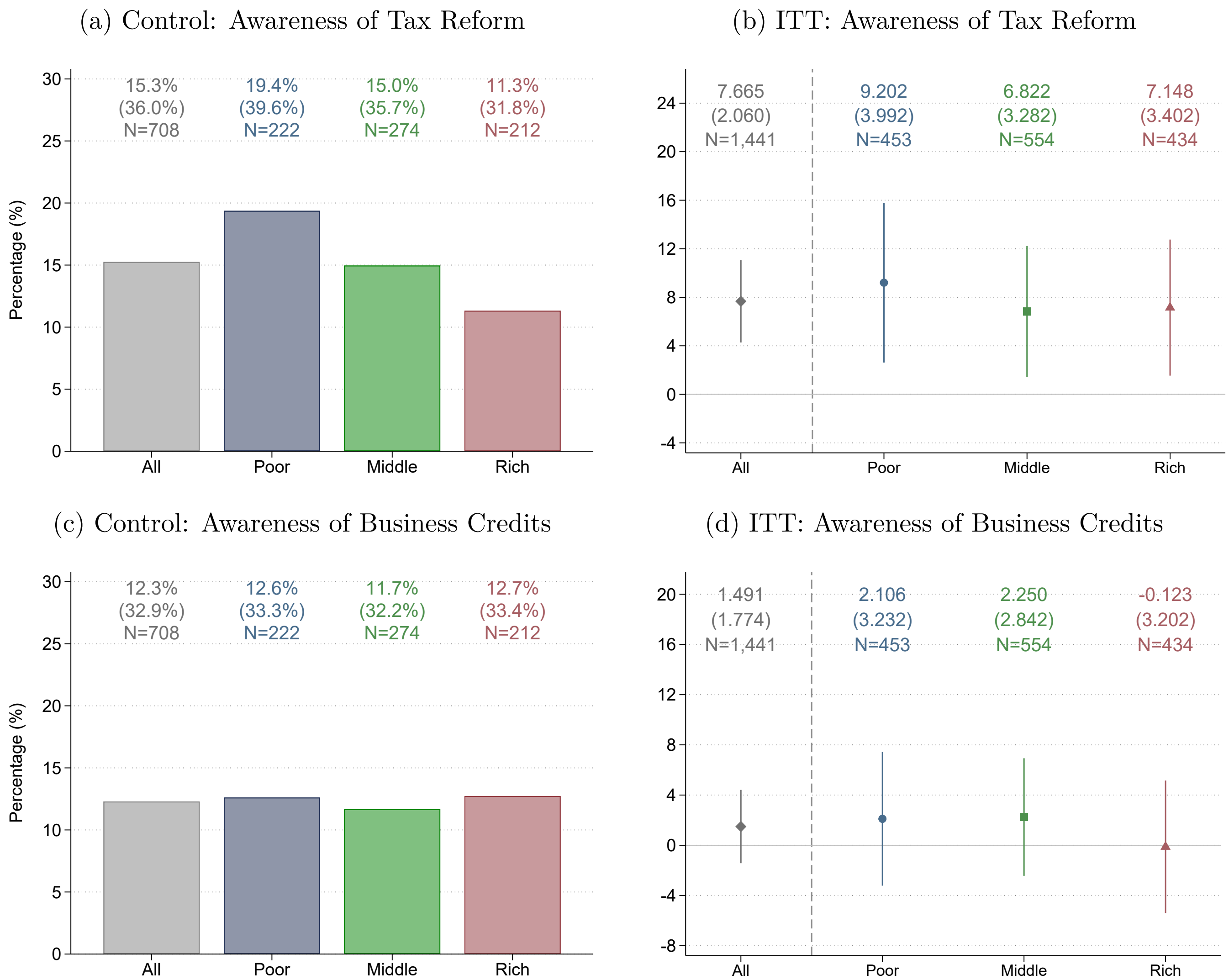
We applied a 30% discount on the variable component of the TSG for properties with the lowest tax valuation, and an average increase of 15% for those with the highest value.

**Based on your tax valuation, your TSG decreased in relation to the rest.**




*Note*: English translation (from Spanish) of the main pieces of information from the mailers. This text was contained on the second page of the mailer (for a full sample of the mailer, see Appendix L.1). TSG refers to *Tasa por Servicios Generales*, the local tax levied by the municipality.

Figure 5: Taxpayers' Awareness of the Reform: Survey Experiment



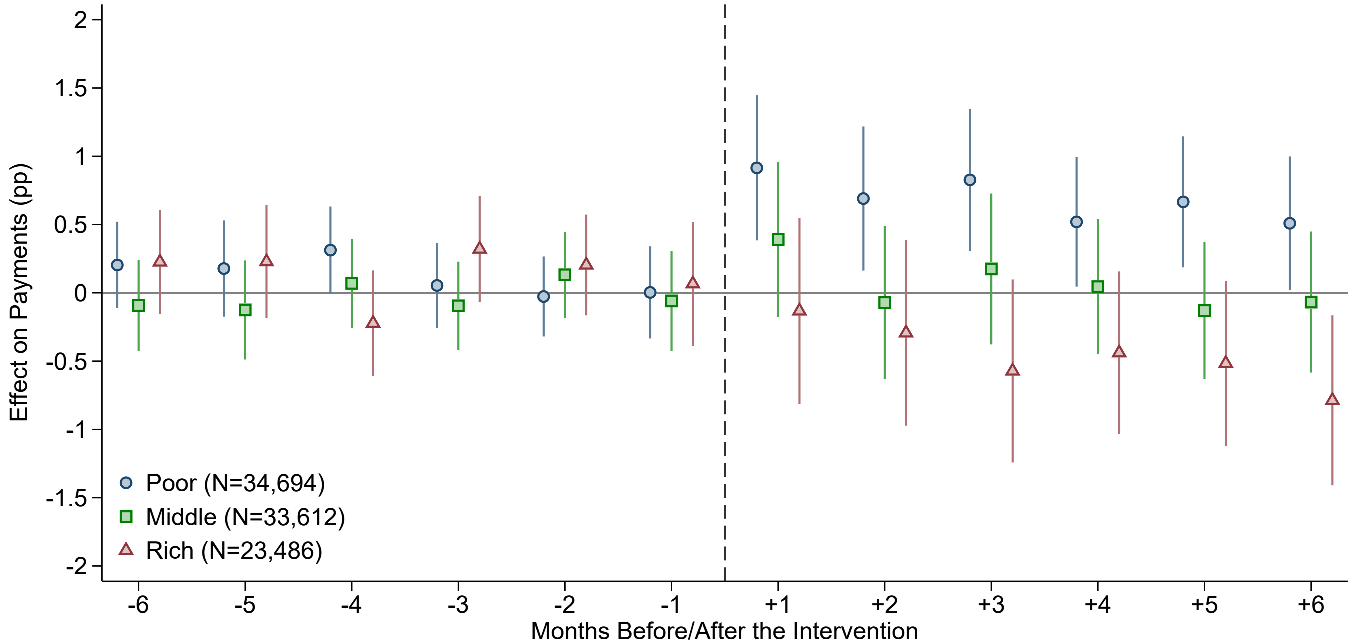

*Note*: This figure uses data from our online survey on taxpayers to assess awareness of the progressive tax reform. We invited a small subsample of subjects with email addresses on file to participate. The respondents received a list of recent policies and were asked to indicate which ones they had heard of. The left panels show the average knowledge of two different policies for households receiving the control letter. The right panels show the effects of the treatment letter on the awareness of the progressive reform and on the knowledge of the Tres de Febrero business credits. The vertical spikes denote 90% confidence intervals. The figure also presents the means (left panels) and treatment effects and their standard errors (right panels).

Figure 6: The Effect on Fairness Perceptions: Survey Experiment

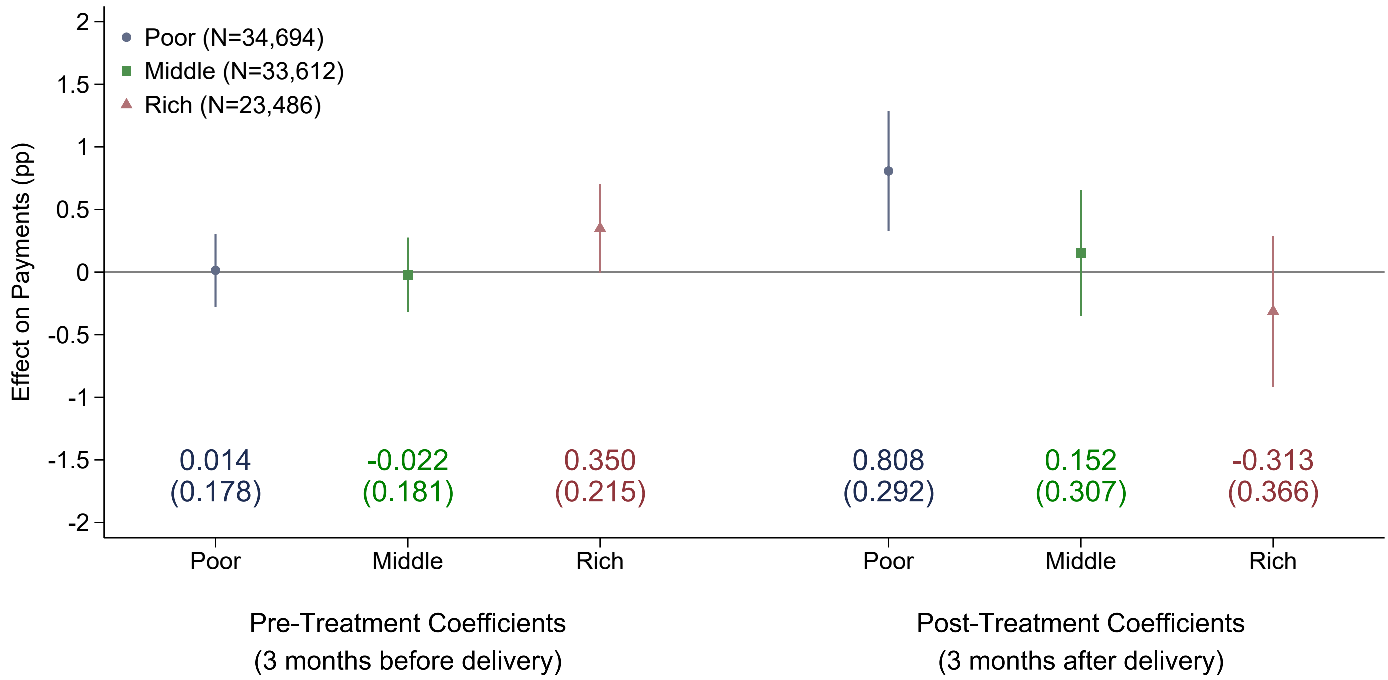

*Note*: This figure uses data from our online survey on taxpayers to estimate the impact of our progressivity information experiment on taxpayers' fairness perceptions. We asked respondents how fair they considered the distribution of the municipal tax rates between rich and poor households, on a scale from 0 to 10 (0 = very unfair, 10 = very fair) in the top panels, and their perception regarding the government's role in addressing inequality (0 = not be concerned, 10 = do everything to reduce it) in the bottom panels. The left panels show the average levels for households receiving the control letter. The right panels show the effects of the treatment letter on these survey questions comparing treated and control respondents. The vertical spikes denote 90% confidence intervals. The figure also presents the means (left panels) and treatment effects and their standard errors (right panels).

Figure 7: Cross-rate Effects of a Progressive Tax Reform on Own Tax Compliance

(a) Event-Study Analysis (Monthly Level)




(b) Pre- and Post-Treatment Effects (Quarterly level)

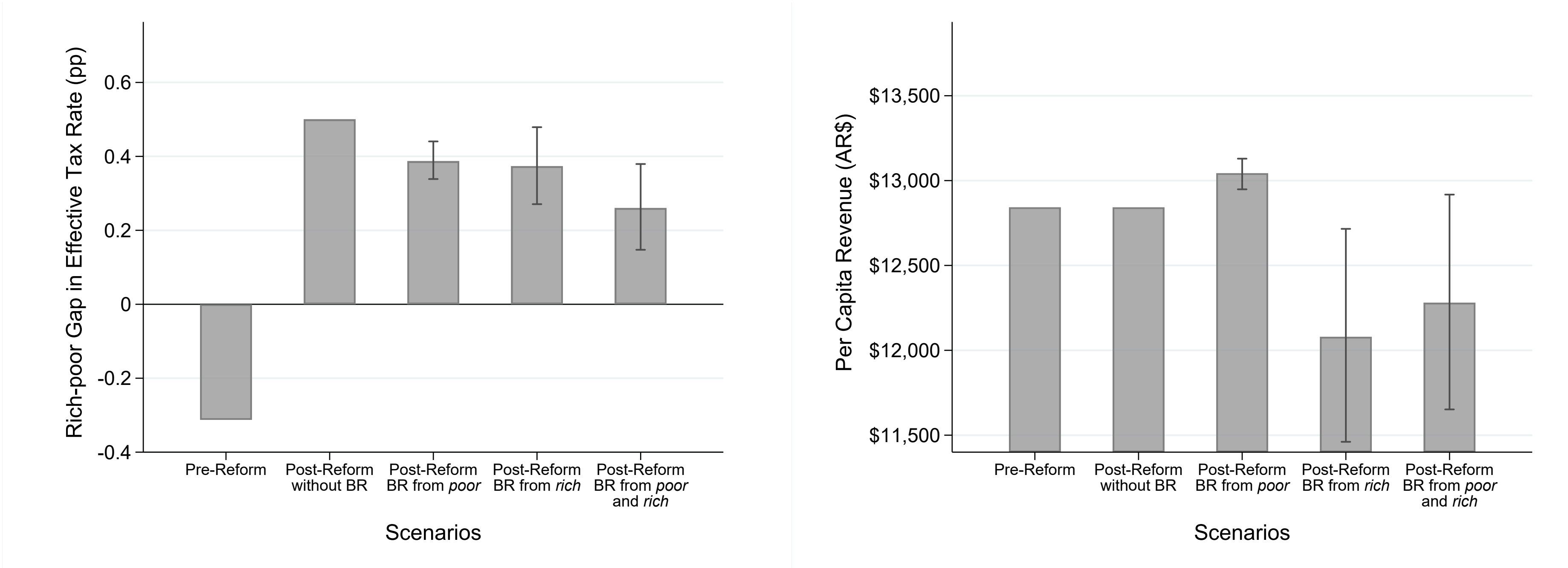



*Note*: This figure compares the likelihood of paying the monthly property tax bill between treatment and control. The control group received a letter with information solely about how the reform changed the household's own tax rate. The treatment group letter included additional information about how the reform affected the tax rates of other households (i.e., informing the progressive nature of the reform). See Figure 4 for an example. Panel (a) shows the dynamic effect of the treatment, while panel (b) shows the three-month pooled effects of the treatment (pre- and post-treatment). 'Poor' (in blue) denotes households with properties valued at AR$ 750K or less, 'Middle' (in green) denotes properties valued between AR$ 750K and AR$ 1.5M, and 'Rich' (in red) denotes properties valued more than AR$ 1.5M. We estimated the coefficient for each monthly bill in separate regressions, including the 90% confidence interval for each group. We reported clustered standard errors for the pooled estimates in parentheses. The sample includes residential and non-residential properties and excludes units that made payments for 2023 before we sent the letters. We controlled for the past 12 months payments in each pre-treatment regression, and for the 2022 monthly payments in each post-treatment regression. Additionally, we included controls for property valuation (in logs), a residential indicator, indicators for whether the property was an "always payer" or "never payer" in the previous year, and a dummy for properties that made an annual payment in the previous year. We included dummies equal to one for missing controls to retain all observations in the regressions. Standard errors clustered at the individual taxpayer level.

Figure 8: Counterfactual Analysis: Tax Progressivity and Revenue with and without Behavioral Responses

(a) Effective Tax Progressivity

(b) Tax Revenue (Per Capita)




*Note*: This figure uses our key point estimates to analyze the effects of a revenue-neutral progressive reform on effective tax progressivity and tax revenues in the presence of behavioral responses (BR), and we contrast it to a counterfactual scenario where behavioral responses are muted. Panel (a) shows the rich-poor effective tax rate gap, assuming different scenarios of behavioral responses (BR). Panel (b) shows the per capita revenue effects, assuming once again different scenarios of behavioral responses. The vertical spikes denote 90% confidence intervals obtained from a 5,000 repetitions bootstrap. Once behavioral responses are factored in, tax progressivity may not increase as much as intended, and a reform that was supposed to be revenue-neutral may not be so.

# Online Appendix (For Online Publication Only)

**"From Flat to Fair? The Effects of a Progressive Tax Reform"**

**Ajzenman, Cruces, Perez-Truglia, Tortarolo and Vazquez-Bare**

**June 12, 2025**

# A Details about the Institutional Context

## A.1 Local Property Tax

Tres de Febrero is a major urban municipality in the Greater Buenos Aires metropolitan area, Argentina.[51] Our study focuses on a local property tax known as *Tasa por Servicios Generales* (TSG), which funds public services such as public spaces, street lighting, urban sanitation, security, and the fire department, among others.[52] This type of tax is common across all Argentine municipalities and serves as the primary source of local revenue. Tres de Febrero is no exception, with the TSG accounting for 20% of its total resources and 45% of its own-source revenue in 2021.

TSG bills are issued monthly and are due during the first weeks of each billing period.[53] Alternatively, residents have the option of making an advance annual payment in the first months of the year instead of monthly installments.

The municipal property tax consists of variable and fixed components. Before the 2022-24 progressive reforms, the tax liability for a residential taxpayer $i$ was calculated as follows:

$$\begin{aligned} \text{Monthly tax}_i = & \Big[\text{Cadastral value}_i \times \text{Tax rate}/12 + \text{Fixed charge}\Big] \\ & \times \text{Module factor} \end{aligned} \tag{A.1}$$

The variable component is calculated by applying a tax rate to each property's assessed value. Tax rates vary across eight property-use categories defined by the municipality: residential, industrial, commercial, wholesale establishments, mixed-use residential with commerce or factory, empty lots, civil entities, and religious entities. In 2022, the statutory tax rates ranged from 0.42% to 3.22% (Table A.1), with commercial properties bearing a higher tax burden through tax rates that generally exceed those applied to residential properties. Property assessments are based on the cadastre maintained by the Revenue

[51] As of 2022, Tres de Febrero had approximately 365,000 residents and 115,000 households, representing 1.2% of Argentina's total population.

[52] When we asked in our survey experiment about satisfaction with the provision of services by the municipality, on a scale from 0 to 10, we found that the degree of satisfaction is around 5.8 units, so it is not excessively high or low.

[53] Since 2023, the municipality has introduced mid-year adjustments to keep up with growing inflation.

Agency of the Province of Buenos Aires (ARBA).

The fixed components of the tax correspond to specific services, including security, health, fire departments, and maintenance of public spaces. These components are measured in 'modules,' with each module valued at AR$ 10 in 2022 (see Table A.2). While cadastral values are seldom updated, both the variable and fixed charges are adjusted annually through a module factor.[54]

Figure A.1 illustrates that for properties in the bottom decile, fixed charges comprise 73% of the tax bill and variable charges 27%. In contrast, for the top decile, fixed charges account for only 14% of the bill, with variable charges making up 86%.

## A.2 The Progressive Tax Reform

For political and equity reasons, the municipality implemented progressive reforms to the municipal property tax during 2022-24. The 2022 and 2023 reforms modified the *variable* component of equation (A.1), while the 2024 reform transformed the previously uniform *fixed* component into a group-specific charge. Each reform used assessed property value thresholds to determine eligibility.[55]

Our analysis focuses on the 2023 and 2024 reforms, as they were more substantive and enabled real-time randomized communication campaigns. However, for completeness and transparency, we describe all three reforms below.

The initial 2022 reform increased the variable component of equation (A.1) by 10% for properties valued above AR$ 1.5M. However, this adjustment was based on outdated assessments and received no public communication or media coverage.

The 2023 reform introduced a four-tier structure. It applied a 30% discount on the variable component of equation (A.1) for properties assessed at AR$ 750K or below, kept it unchanged for properties between AR$ 750K and AR$ 1.5M, retained the 10% surcharge for properties valued between 1.5M and AR$ 3M (now based on an updated cadastre), and imposed a 20% surcharge for properties above AR$ 3M. For a *residential* taxpayer $i$ in cadastral value bracket $b$, the monthly tax became:

[54] Additionally, one of the four fixed components—the security module—varies by property type: for residential properties, it remains strictly fixed, whereas for commercial properties, it starts at the same minimum but can increase based on a percentage of the total tax owed. If this percentage yields an amount lower than the minimum, the business is charged the minimum amount.

[55] The thresholds defining tax brackets remained constant across all three years. Due to amortization, assessed property values decreased by roughly 4% between 2022-23 and by 1% between 2023-24.

$$\begin{aligned}\text{Monthly tax}_i = \Big[&\text{Cadastral Value}_i \times \text{Tax rate}/12 \times \text{ProgreV}_{b(i)} \\ &+ \text{Fixed charge}\Big] \times \text{Module factor}\end{aligned} \tag{A.2}$$

where $\text{ProgreV}_{b(i)}$ represents the progressivity factor applied to the *variable* component, which varies by cadastral value bracket $b(i)$ as follows:

$$\text{ProgreV}_{b(i)} = \begin{cases} 0.7 & \text{Cadastral Value}_i \leq \text{AR\$ 750K} \\ 1 & \text{AR\$ 750K} < \text{Cadastral Value}_i \leq \text{AR\$ 1.5M} \\ 1.1 & \text{AR\$ 1.5M} < \text{Cadastral Value}_i \leq \text{AR\$ 3M} \\ 1.2 & \text{Cadastral Value}_i > \text{AR\$ 3M} \end{cases} \tag{A.3}$$

Lastly, the 2024 reform transformed the previously uniform *fixed* component of equation (A.2) into differentiated charges by property value (see Table A.2): approximately a 16% reduction for properties at or below AR\$ 750K; no change for properties between AR\$ 750K and AR\$ 1.5M; approximately a 39% increase for properties between AR\$ 1.5M and AR\$ 3M; and approximately a 64% increase for properties above AR\$ 3M. Thus, for a residential taxpayer $i$ in cadastral value bracket $b$, the monthly tax is:

$$\begin{aligned}\text{Monthly tax}_i = \Big[&\text{Cadastral Value}_i \times \text{Tax rate}/12 \times \text{ProgreV}_{b(i)} \\ &+ \text{Fixed charge} + \text{ProgreF}_{b(i)}\Big] \times \text{Module factor}\end{aligned} \tag{A.4}$$

where $\text{ProgreF}_{b(i)}$ is the progressivity factor applied to the *fixed* part (Table A.2), which varies by cadastral value bracket $b(i)$ as follows:

$$\text{ProgreF}_{b(i)} = \begin{cases} -100 & \text{Cadastral Value}_i \leq \text{AR\$ 750K} \\ 0 & \text{AR\$ 750K} < \text{Cadastral Value}_i \leq \text{AR\$ 1.5M} \\ 240 & \text{AR\$ 1.5M} < \text{Cadastral Value}_i \leq \text{AR\$ 3M} \\ 390 & \text{Cadastral Value}_i > \text{AR\$ 3M} \end{cases} \tag{A.5}$$

These reforms introduced exogenous variation in effective tax rates at the discontinuities, enabling us to estimate the own-rate effects of tax changes (see Section 3). Additionally, we leveraged the timing of these reforms to conduct an information RCT, varying how different aspects of the reform were communicated to households across brackets. This experimental design allowed us to assess the cross-rate effects of tax changes, as detailed in Section 4.

Table A.1: Property Type and Statutory Tax Rates (STR)

| Category | N Properties | | Statutory Tax Rates (%) | | | | |
|---|---|---|---|---|---|---|---|
| | N | % | 2018 | 2019 | 2020 | 2021 | 2022-2024 |
| Residential | 109,729 | 82.8 | 0.30 | 0.50 | 0.69 | 0.93 | 1.21 |
| Residential with commerce/factory | 10,073 | 7.6 | 0.45 | 0.75 | 1.04 | 1.4 | 1.82 |
| Commercial | 7,425 | 5.6 | 0.70 | 1.18 | 1.68 | 2.27 | 2.95 |
| Empty lot | 1,166 | 0.9 | 0.70 | 1.27 | 1.84 | 2.48 | 3.22 |
| Industrial | 3,699 | 2.8 | 0.70 | 1.22 | 1.77 | 2.39 | 3.11 |
| Civil entities | 220 | 0.2 | 0.10 | 0.17 | 0.24 | 0.32 | 0.42 |
| Religious entities | 170 | 0.1 | 0.10 | 0.17 | 0.24 | 0.32 | 0.42 |
| Wholesale establishments | 1 | 0.0 | 0.45 | 0.76 | 1.10 | 1.49 | 1.94 |
| Assessed values ARBA (year) | | | 2018 | 2018 | 2018 | 2018 | 2021 |

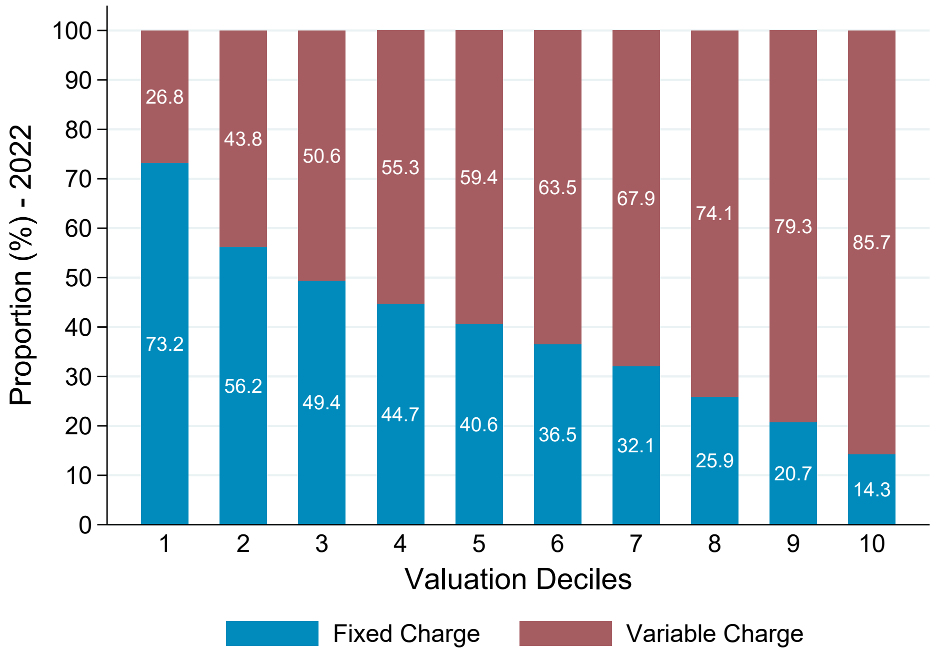
*Note:* This table reports statutory tax rates by category of property as defined in the tax code (*Ordenanza Impositiva*) every year. The first two columns display the number of properties in 2018. The tax base used to calculate the monthly fee corresponds to the assessed value in 2018 provided by the provincial tax authority (ARBA). Despite an annual inflation of 47% in 2018, 54% in 2019, and 36% in 2020, this base remained unadjusted. To compensate, the municipality increased the tax rates.

Table A.2: Fixed Charges (measured in modules)

| | 2022 | 2023 | 2024 | | | |
|---|---|---|---|---|---|---|
| | | | Cadastral Values | | | |
| | | | ≤ AR$ 750K | AR$ 750K-1.5M | AR$ 1.5M-3M | > AR$ 3M |
| *Unit modules (AR$ 10 each):* | | | | | | |
| Security | 25 | 25 | 25 | 25 | 25 | 25 |
| Public spaces | 14.5 | 14.5 | 9.5 | 14.5 | 26.5 | 33.5 |
| Health | 20 | 20 | 15 | 20 | 32 | 40 |
| Firefighters | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |
| Total modules | 61 | 61 | 51 | 61 | 85 | 100 |
| *Tax change (fixed charge)* | | *Unchanged* | -16% | *Unchanged* | 39% | 64% |
| Module factor | 1 | 1.6 | 3.6–9 | 3.6–9 | 3.6–9 | 3.6–9 |

*Note:* The municipality adjusts the tax bill annually for inflation. Since 2024, they have adjusted the bill every other month to keep up with the rising inflation. The module factor is 3.6 in January, 5.1 in March, 7 in May, 8.4 in July, and 9 in September.

Figure A.1: Fixed and Variable Charges of the Property Tax Bill, by Deciles




*Note*: This figure shows the relative importance of variable and fixed components of the property tax bill by deciles of assessed values, for the year 2022.

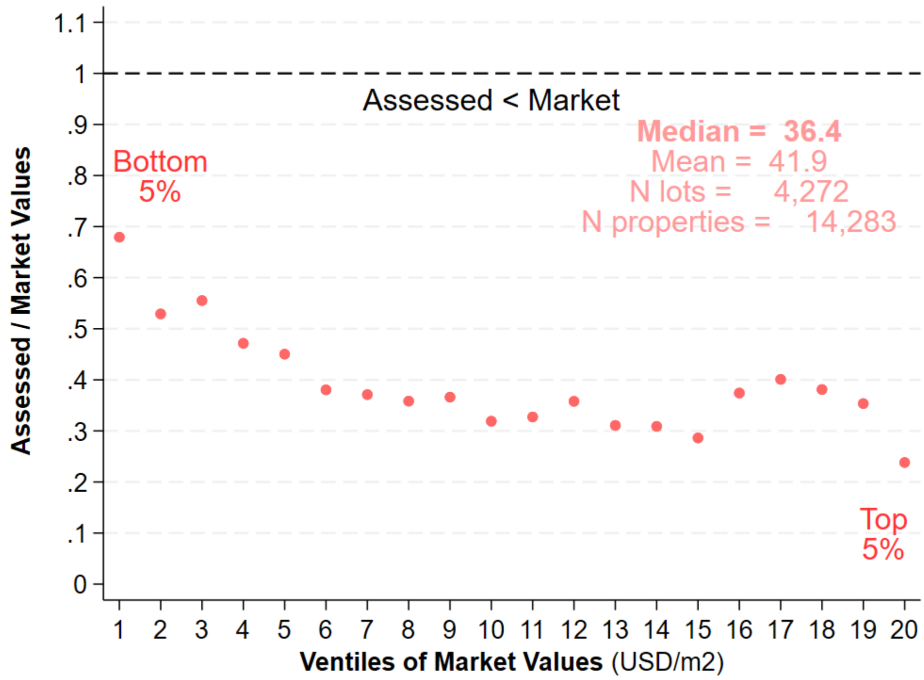
# B Market vs. Assessed Property Values

In this section, we used proprietary data on real estate prices to document that assessed values are systematically below market values. Moreover, the degree of under-assessment is slightly weaker for properties at the bottom of the distribution, which constitutes an additional source of tax regressivity.

We matched cadastral records from the municipal registry with real estate listings from *Properati*, a major online platform in Argentina. We began with a universe of 54,371 listings and applied a series of filters to ensure data quality and comparability. First, we retained only listings for sale in U.S. dollars (excluding rentals and listings in pesos), removed observations with missing or implausibly low prices, and excluded entries without valid geolocation. We then performed a geographic match between listings and cadastral parcels using spatial proximity, and focused on the strictest criterion—where the listing's coordinates fell exactly within the parcel polygon (distance = 0 meters). We restricted the analysis to residential properties or those with mixed residential and commercial use, yielding a final matched sample of 14,172 observations (26.07% of the original listings).

For each matched property, we computed both assessed values (AV) and market values (MV) in USD per square meter, enabling direct comparisons across units. Figure B.1 shows the distribution of AVs and MVs for the matched sample. The figure reveals a substantial and systematic gap: assessed values are consistently and significantly lower than market values. This finding supports anecdotal accounts of widespread fiscal under-assessment in this context, likely driven by infrequent updates to cadastral valuations by the provincial tax authority. The under-assessment is not uniform across the value distribution; it is less pronounced in the bottom two deciles, introducing an additional source of effective tax regressiveness.

Figure B.1: Comparison of Assessed Values vs. Market Values



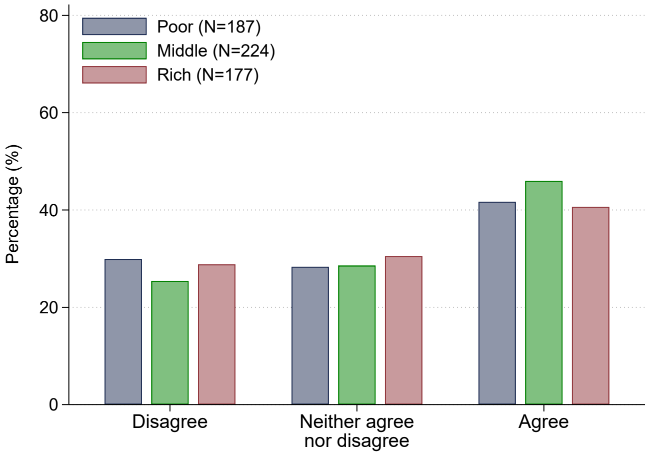

*Note*: This figure displays the ratio of assessed to market property values across ventiles of the market value distribution. Assessed values correspond to 2018 valuations from the provincial cadastral agency (ARBA). Market values are taken from Properati listings posted between 2020 and 2023, where prices are reported directly in USD on the website. Both values are converted to USD per square meter, to make them directly comparable. The sample is restricted to properties with exact or near-exact geospatial matches (within zero meters of distance) to cadastral parcels.

# C Support for Redistribution among Survey Respondents

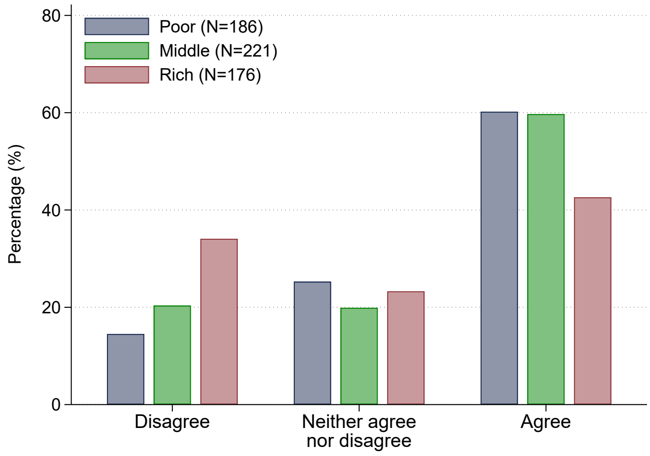
Figure C.1 presents some descriptive statistics from the survey, specifically about the taxpayers' preferences for redistribution. Panel (a) shows support for reducing taxes on low-valuation properties, with respondents evaluating whether lower taxes for the poorest households would be justified despite reducing available funds for public services, while panel (b) examines support for increasing taxes on high-valuation properties, asking whether the richest should contribute more to increase public revenue. The results indicate that while a substantial share of respondents agrees with lower taxes for the poor, support is even stronger for raising taxes on the rich. Panel (c) presents responses to a separate question, where participants were asked whether they support a simultaneous tax cut for the poorest and a tax increase for the richest, keeping overall public revenue unchanged, showing that the majority of taxpayers favor a more progressive tax system overall.

Figure C.1: Support for Different Changes in the Tax

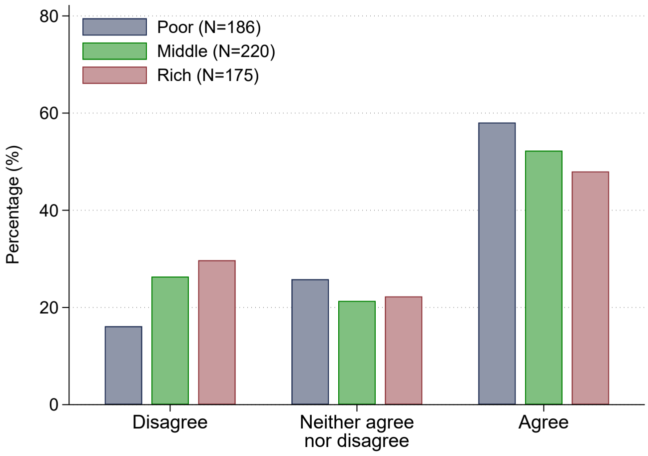
(a) Lower Tax for Poor




(b) Higher Tax for Rich




(c) Lower Tax for Poor, Higher Tax for Rich




*Note*: Variables constructed from respondents' agreement with statements regarding the fairness of potential TSG reforms, based on the 2023 survey experiment (see Appendix M.2 for the full survey instrument). The original variable ranged from 0 to 10. Respondents who rated between 0 and 4 were classified as "Disagree", those who rated 5 as "Neither agree nor disagree", and those who rated between 6 and 10 as "Agree". Only respondents in the control group are included.

# D Robustness Checks

## D.1 Manipulation Test

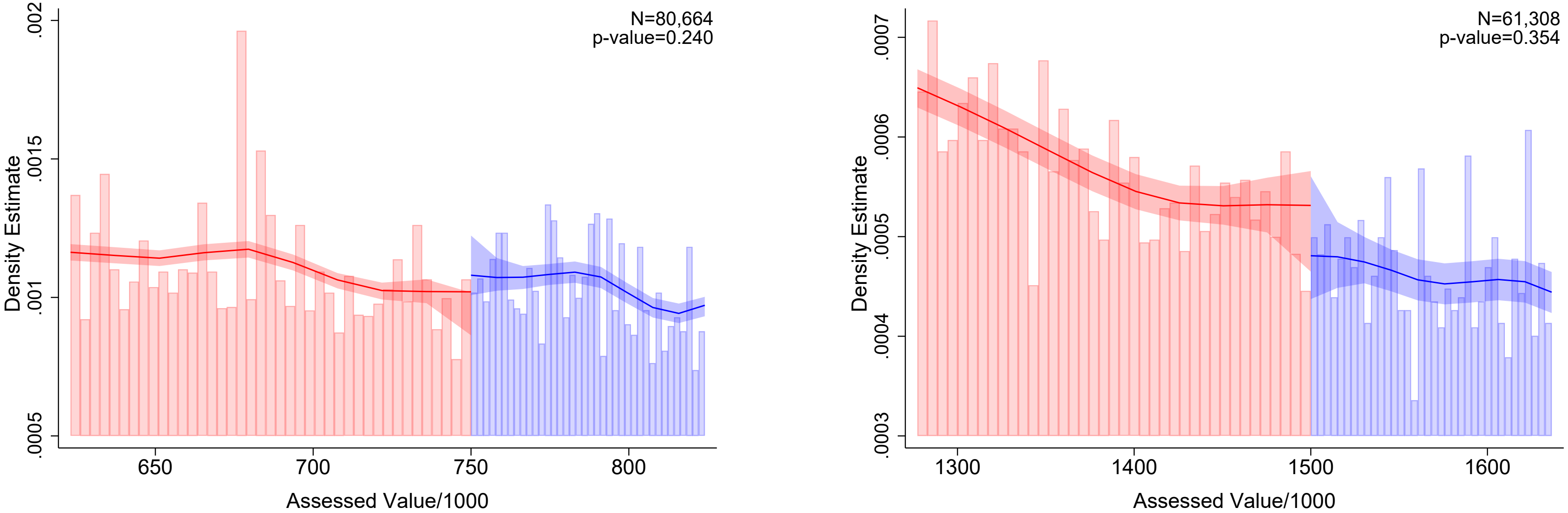
We conducted the standard McCrary (2008) manipulation test to assess the continuity of the running variable's density function based on the methods proposed by Cattaneo et al. (2018, 2020). Intuitively, taxpayers might attempt to influence their own-rate treatment status by adjusting their property's assessed values to fall just above the AR$ 750K threshold or below the AR$ 1.5M threshold—zones where the tax rate remained unchanged. In practice, however, such manipulation is virtually impossible, as assessed values are calculated and reported by the provincial tax authority (ARBA).

The McCrary density test and accompanying figures revealed no evidence of unusual bunching around the thresholds that determine treatment eligibility. While we detected a slight discontinuity for poor properties, the institutional context suggests this is unlikely to reflect deliberate manipulation. Property values are based on fiscal valuations, which are typically established by governmental authorities and cannot be directly altered by individual property owners. We, therefore, consider this result spurious and conclude that it does not compromise the validity of the design. Overall, this evidence reinforces the validity of the regression discontinuity design used to estimate own-rate effects in Section 3.

Figure D.1: Manipulation Test of Assessed Values around the Discontinuities: 2023

(a) Poor Cutoff - 2023

(b) Rich Cutoff - 2023




*Note*: Panels (a) and (b) show the McCrary (2008) manipulation test (with their corresponding p-value) of the property assessed values at the low-value threshold (poor) and the high-value threshold (rich) in 2023. The shaded areas represent the 90% confidence intervals. The x-axis represents assessed property value in AR$ thousands.

## D.2 Alternative Specifications

We conducted a series of robustness checks on the own-rate and cross-rate effects presented in Sections 3 and 4. The results indicate that our findings are highly robust.

Tables D.1 and D.2 present multiple robustness checks for the RDD analysis used to estimate own-rate effects. These checks incorporate: (i) additional pre-treatment controls, (ii) including just households in the sample, (iii) alternative definitions of the outcome variable, (iv) including in the sample only those households which received a control or treatment letter, (v) the use of an uniform kernel (rather than a triangular kernel), (vi) the use of an epanechnikov kernel, (vii) the use of a zero degree polynomial and (viii) the use of a second degree polynomial. The estimates remain stable across these specifications, supporting the validity of the main findings.

Table D.3 focuses on robustness checks for the experimental analysis used to estimate cross-rate effects. These tests examine the sensitivity of the results to: (i) including only pre-treatment payment controls, (ii) incorporating the email sample into the regressions, (iii) including just households in the sample, (iv) using an alternative definition of the outcome, and (v) including the observations of those taxpayers who made up-front payments for 2023 before the letters were sent. The findings remain stable across all specifications.

Finally, Figure D.2 displays the placebo exercise done in Figure 3, but using different thresholds for our placebo. Our results are generally robust to the selection of the threshold.

The consistency of results across these robustness checks strengthens confidence in the reliability of the main conclusions.

Table D.1: Robustness Tests: Own-Rate Effects I

| | Robustness Checks for RDD Results | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Effect of Reform on Tax Compliance:* | | | | | | |
| Low-Valuation HHs | −2.718*** | −1.165* | −2.515*** | −2.597*** | −3.931*** | −1.358 |
| | [0.002] | [0.082] | [0.005] | [0.003] | [0.001] | [0.240] |
| | | | | | [pval diff.=0.142] | |
| High-Valuation HHs | −2.248* | −1.364 | −2.800** | −2.502** | −1.444 | −3.308* |
| | [0.050] | [0.123] | [0.023] | [0.024] | [0.569] | [0.064] |
| | | | | | [pval diff.=0.320] | |
| *Effect of Reform on Tax Rates:* | | | | | | |
| Low-Valuation HHs | 0.392*** | 0.390*** | 0.366*** | 0.392*** | 0.400*** | 0.384*** |
| | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] |
| | | | | | [pval diff.=0.246] | |
| High-Valuation HHs | 0.146*** | 0.145*** | 0.126*** | 0.146*** | 0.144*** | 0.148*** |
| | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] |
| | | | | | [pval diff.=0.706] | |
| *Specification* | | | | | | |
| Additional Controls | | ✓ | | | | |
| Just Households | | | ✓ | | | |
| Alternative outcome (6 months of late payment) | | | | ✓ | | |
| Just Control Sample | | | | | ✓ | |
| Just Treatment Sample | | | | | | ✓ |
| *Observations (Compliance)* | | | | | | |
| Low-Valuation HHs | 29,180 | 31,365 | 28,208 | 33,393 | 30,821 | 30,487 |
| High-Valuation HHs | 61,308 | 61,308 | 54,648 | 61,308 | 15,009 | 16,243 |
| *Observations (Rates)* | | | | | | |
| Low-Valuation HHs | 13,017 | 12,756 | 15,934 | 13,017 | 30,821 | 30,487 |
| High-Valuation HHs | 61,308 | 61,308 | 54,648 | 61,308 | 7,067 | 7,411 |
| *Average Tax Compliance (Baseline)* | | | | | | |
| Low-Valuation HHs | 60.4% | 60.1% | 60.9% | 60.2% | 57.2% | 56.7% |
| High-Valuation HHs | 57.4% | 57.4% | 58.3% | 57.4% | 60.5% | 60.0% |
| *Average Tax Rate (Baseline)* | | | | | | |
| Low-Valuation HHs | 2.2% | 2.2% | 2.1% | 2.2% | 1.8% | 1.9% |
| High-Valuation HHs | 1.8% | 1.8% | 1.7% | 1.8% | 2.2% | 2.2% |

*Note*: This table summarizes Figure 1 using different specifications for the estimates. Column (1) shows the main estimates, column (2) shows the estimates including additional controls such as the payment in previous periods of the corresponding bill, column (3) includes just households in the sample, column (4) uses a alternative outcome that includes the possibility of paying 6 months after the due date instead of only 3 months later. Finally, columns (5) and (6) focus exclusively on the control and treatment samples, respectively, while also reporting the p-value from the test of differences between them estimated through a 5,000 repetitions bootstrap. Each column indicates the estimated effect around the cut-off point, where the p-value, taken from a robust bias-corrected inference, is indicated in brackets.

Table D.2: Robustness Tests: Own-Rate Effects II

| | Robustness Checks for RDD Results | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| *Effect of Reform on Tax Compliance:* | | | | | |
| Low-Valuation HHs | −2.718*** | −3.313*** | −2.860*** | −2.412*** | −3.011*** |
| | [0.002] | [0.001] | [0.001] | [0.004] | [0.003] |
| High-Valuation HHs | −2.248* | −2.071* | −2.181* | −1.878* | −2.538* |
| | [0.050] | [0.069] | [0.057] | [0.077] | [0.070] |
| *Effect of Reform on Tax Rates:* | | | | | |
| Low-Valuation HHs | 0.392*** | 0.389*** | 0.392*** | 0.383*** | 0.395*** |
| | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] |
| High-Valuation HHs | 0.146*** | 0.147*** | 0.147*** | 0.133*** | 0.152*** |
| | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] |
| *Specification* | | | | | |
| Uniform Kernel | | ✓ | | | |
| Epanechnikov Kernel | | | ✓ | | |
| Zero Degree Polynomial | | | | ✓ | |
| Second Degree Polynomial | | | | | ✓ |
| *Observations (Compliance)* | | | | | |
| Low-Valuation HHs | 29,180 | 22,837 | 27,215 | 11,663 | 44,615 |
| High-Valuation HHs | 61,308 | 61,308 | 61,308 | 61,308 | 61,308 |
| *Observations (Rates)* | | | | | |
| Low-Valuation HHs | 13,017 | 18,087 | 12,589 | 13,718 | 24,529 |
| High-Valuation HHs | 61,308 | 61,308 | 61,308 | 61,308 | 61,308 |
| *Average Tax Compliance (Baseline)* | | | | | |
| Low-Valuation HHs | 60.4% | 60.8% | 60.5% | 60.6% | 60.1% |
| High-Valuation HHs | 57.4% | 57.3% | 57.3% | 57.5% | 57.5% |
| *Average Tax Rate (Baseline)* | | | | | |
| Low-Valuation HHs | 2.2% | 2.2% | 2.2% | 2.2% | 2.2% |
| High-Valuation HHs | 1.8% | 1.8% | 1.8% | 1.9% | 1.9% |

*Note*: This table summarizes Figure 1 using different specifications for the estimates. Column (1) shows the main estimates, column (2) shows the estimates using an uniform kernel rather than a triangular one, column (3) uses an epanechnikov kernel rather than a triangular one, column (4) uses a zero degree polynomial and column (5) uses a second degree polynomial. Each column indicates the estimated effect around the cut-off point, where the p-value, taken from a robust bias-corrected inference, is indicated in brackets.

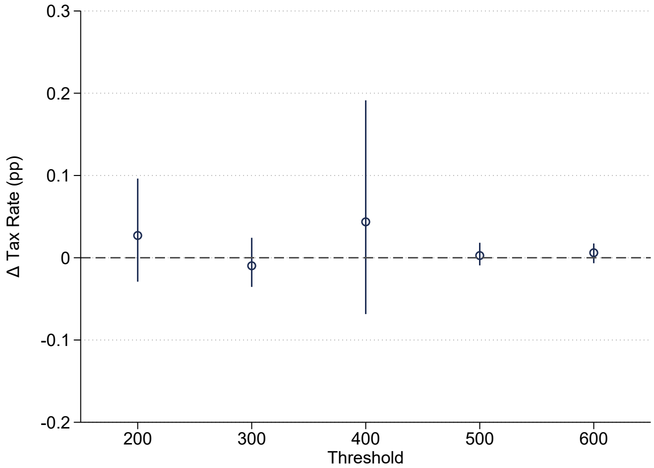
Table D.3: Robustness Tests: Cross-Rate Effects

| | Robustness Checks for Experimental Results | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Effect of Reform on Tax Compliance:* | | | | | | |
| Low-Valuation HHs | 0.808*** | 0.795*** | 0.698** | 0.756** | 0.570** | 0.741*** |
| | (0.292) | (0.295) | (0.278) | (0.304) | (0.290) | (0.279) |
| Medium-Valuation HHs | 0.152 | 0.170 | 0.132 | 0.247 | 0.166 | 0.157 |
| | (0.307) | (0.309) | (0.286) | (0.320) | (0.304) | (0.298) |
| High-Valuation HHs | −0.313 | −0.421 | −0.245 | −0.177 | −0.254 | −0.288 |
| | (0.366) | (0.369) | (0.341) | (0.418) | (0.367) | (0.360) |
| *Specification* | | | | | | |
| Just pre-treat payment controls | | ✓ | | | | |
| Email Sample Included | | | ✓ | | | |
| Just Households | | | | ✓ | | |
| Alternative outcome (6 months of late payment) | | | | | ✓ | |
| Up-Front Payers Included | | | | | | ✓ |
| *Observations* | | | | | | |
| Low-Valuation HHs | 34,694 | 34,694 | 38,678 | 31,999 | 34,694 | 36,519 |
| Medium-Valuation HHs | 33,612 | 33,612 | 38,325 | 30,945 | 33,612 | 34,819 |
| High-Valuation HHs | 23,486 | 23,486 | 26,716 | 17,945 | 23,486 | 24,022 |
| *Average Tax Compliance (Baseline)* | | | | | | |
| Low-Valuation HHs | 54.5% | 54.5% | 57.3% | 54.5% | 54.5% | 56.8% |
| Medium-Valuation HHs | 56.0% | 56.0% | 59.1% | 56.5% | 5.06% | 57.5% |
| High-Valuation HHs | 56.1% | 56.1% | 59.0% | 57.1% | 56.1% | 57.0% |

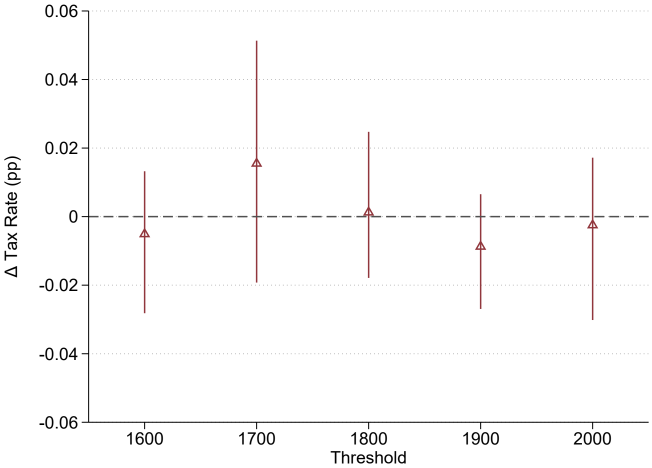
*Note*: This table summarizes Figure 7 using different specifications for the estimates. Column (1) shows the main estimates, column (2) excludes every other control not related to pre-treatment payments, column (3) includes the e-mail survey sample, column (4) includes only residential properties in the sample (excludes non-residential properties), column (5) uses an alternative outcome that includes the possibility of paying 6 months after the due date instead of only 3 months later, and column (6) includes the observations of those taxpayers who made up-front payments for 2023 before the letters were sent. Standard errors clustered at the individual level in parenthesis.

Figure D.2: RDD Falsification Test: Full Distribution of Placebo Thresholds for 2023

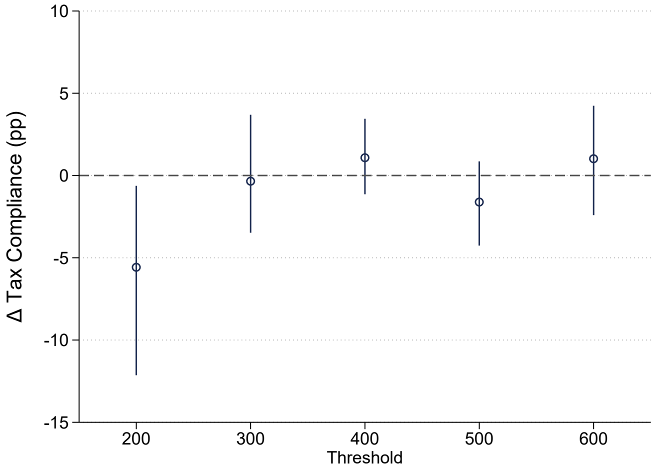
(a) First-Stage: Poor Households



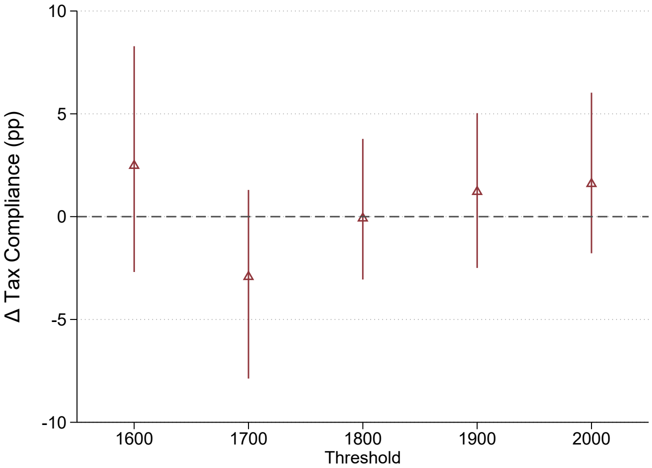

(b) First-Stage: Rich Households




(c) Reduced-Form: Poor Households




(d) Reduced-Form: Rich Households




*Note*: This figure calculates the own-rate effects using different placebo thresholds for poor and rich households. The x-axis shows the threshold used in each estimation. Point estimates and their confidence intervals (at a 90% confidence level) are displayed. Panels (a) and (b) represent the first-stage RDD for poor and rich, respectively. These two figures evaluate the increase in the amounts owed between the last semester of 2022 and the first semester of 2023, in relation to the value of the property. Panels (c) and (d) represent the reduced-form RDD for poor and rich, respectively. They evaluate the change in tax compliance for both groups between the last semester of 2022 and the first semester of 2023.

# E Balance Tables

The balance tables compare baseline characteristics across treatment and control groups (Tables E.1 and E.3), as well as between control and no-letter groups (Tables E.2 and E.4), for the experiments conducted in 2023 and 2024. At the request of the authorities, the no-letter pool in the 2023 experiment consists exclusively of businesses, restricting the control-versus-no-letter comparison to this group. We also compared baseline characteristics across treatment and control groups for our complementary survey (Table E.5).

Table E.1 presents the results for the 2023 experiment, highlighting that key variables such as payment shares for 2021 and 2022, assessed property values, and tax rates are statistically balanced between treatment and control groups. The p-values across all variables confirm no significant differences, ensuring group comparability. Table E.3 replicates this analysis for the 2024 experiment, yielding similar results and further reinforcing the reliability of the randomization process.

For the control letter versus no letter comparison, Table E.2 evaluates the 2023 experiment, showing no significant imbalances in pre-treatment characteristics like payment shares and assessed property values, as indicated by the p-values. Table E.4 extends this comparison to the 2024 experiment, and while the difference is statistically significant for 5 of the pre-treatment comparisons, this significance is driven by the large sample size, while the actual difference remains small in magnitude. Moreover, Table E.4 reports 22 tests, implying that, by chance, 2 or 3 are likely to be statistically significant at the 10% level. Nevertheless, we include these variables as controls in the baseline specification to mitigate potential biases.

Finally, Table E.5 also compares our treatment groups across different valuations for our survey experiment. Overall, we see few significant differences between groups.

Table E.1: Balance Table: 2023 Experiment

| | | All | | | Poor | | | Middle | | | Rich | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | All | Control | Treat | P-value | Control | Treat | P-value | Control | Treat | P-value | Control | Treat | P-value |
| Share Household (%) | 88.1 | 88.3 | 88.0 | 0.168 | 92.3 | 92.1 | 0.491 | 92.0 | 92.1 | 0.855 | 76.8 | 76.1 | 0.199 |
| | (32.4) | (32.2) | (32.5) | | (26.6) | (26.9) | | (27.1) | (27.0) | | (42.2) | (42.7) | |
| Share of Payments (2021) (%) | 44.0 | 44.2 | 43.9 | 0.294 | 42.4 | 41.9 | 0.306 | 44.7 | 45.0 | 0.530 | 46.1 | 45.1 | 0.116 |
| | (45.8) | (45.8) | (45.8) | | (45.4) | (45.5) | | (45.7) | (45.8) | | (46.4) | (46.1) | |
| Share of Payments (2022) (%) | 44.0 | 44.2 | 43.8 | 0.236 | 42.1 | 41.6 | 0.299 | 44.7 | 44.9 | 0.733 | 46.4 | 45.6 | 0.134 |
| | (45.3) | (45.3) | (45.4) | | (44.9) | (44.9) | | (45.3) | (45.4) | | (45.8) | (45.8) | |
| 2023 Assessed Value (million $s) | 129.8 | 127.8 | 131.8 | 0.296 | 47.5 | 47.4 | 0.814 | 105.5 | 105.8 | 0.169 | 279.4 | 292.5 | 0.381 |
| | (587.5) | (391.8) | (733.7) | | (18.9) | (18.8) | | (21.1) | (21.3) | | (754.9) | (1,431.9) | |
| 2022 Tax Rate (%) | 2.80 | 2.80 | 2.80 | 0.87 | 4.00 | 4.01 | 0.96 | 2.08 | 2.08 | 0.73 | 2.06 | 2.07 | 0.52 |
| | (4.34) | (4.20) | (4.49) | | (6.61) | (7.09) | | (0.59) | (0.50) | | (0.79) | (0.81) | |
| Share Poor (%) | 37.8 | 37.7 | 37.8 | 0.751 | - | - | - | - | - | - | - | - | - |
| | (48.5) | (48.5) | (48.5) | | | | | | | | | | |
| Share Rich (%) | 25.6 | 25.4 | 25.8 | 0.201 | - | - | - | - | - | - | - | - | - |
| | (43.6) | (43.5) | (43.7) | | | | | | | | | | |
| Observations | 91,792 | 46,082 | 45,710 | | 17,394 | 17,300 | | 16,982 | 16,630 | | 11,706 | 11,780 | |

*Note*: This balance table presents the difference between control and treatment groups across different categories (All, Poor, Middle, and Rich). Standard deviation in parenthesis. The p-value columns reflect the results of difference tests between control and treatment groups. Results are omitted in categories where comparisons between groups are not applicable.

Table E.2: Balance Table Letter vs No Letter (Businesses Only): 2023 Experiment

| | | All | | | Poor | | | Middle | | | Rich | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | All | No Letter Business | Control Business | P-value | No Letter Business | Control Business | P-value | No Letter Business | Control Business | P-value | No Letter Business | Control Business | P-value |
| Share of Payments (2021) (%) | 42.1 | 41.9 | 42.3 | 0.657 | 42.6 | 42.6 | 0.982 | 36.6 | 38.3 | 0.322 | 44.2 | 44.1 | 0.895 |
| | (45.8) | (45.9) | (45.8) | | (45.8) | (45.8) | | (45.0) | (44.9) | | (46.1) | (46.2) | |
| Share of Payments (2022) (%) | 43.6 | 43.8 | 43.4 | 0.707 | 45.0 | 43.7 | 0.479 | 38.9 | 39.6 | 0.652 | 45.7 | 45.2 | 0.687 |
| | (45.9) | (46.0) | (45.7) | | (45.9) | (45.4) | | (45.6) | (45.4) | | (46.2) | (46.0) | |
| 2023 Assessed Value (million AR$) | 238.6 | 235.3 | 241.9 | 0.713 | 42.9 | 42.9 | 0.993 | 111.0 | 111.2 | 0.857 | 394.2 | 404.5 | 0.773 |
| | (936.4) | (836.4) | (1027.1) | | (18.5) | (18.7) | | (21.7) | (21.4) | | (1163.1) | (1429.1) | |
| 2022 Tax Rate (%) | 3.68 | 3.68 | 3.68 | 0.94 | 5.10 | 5.04 | 0.64 | 3.22 | 3.26 | 0.43 | 3.21 | 3.22 | 0.55 |
| | (2.08) | (2.37) | (1.74) | | (4.18) | (2.34) | | (0.92) | (1.63) | | (0.98) | (0.92) | |
| Share Poor (%) | 24.7 | 24.7 | 24.7 | 0.994 | - | - | - | - | - | - | - | - | - |
| | (43.1) | (43.1) | (43.1) | | | | | | | | | | |
| Share Rich (%) | 50.1 | 49.8 | 50.3 | 0.597 | - | - | - | - | - | - | - | - | - |
| | (50.0) | (50.0) | (50.0) | | | | | | | | | | |
| Observations | 10,831 | 5,425 | 5,406 | | 1,339 | 1,334 | | 1,384 | 1,352 | | 2,702 | 2,720 | |

*Note*: This balance table presents the difference between control and no letter groups across different categories (All, Poor, Middle, and Rich). Table shows only Businesses, since every household received a letter in the 2023 experiment. Standard deviation in parenthesis. The p-value columns reflect the results of difference tests between control and treatment groups. Results are omitted in categories where comparisons between groups are not applicable.

Table E.3: Balance Table: 2024 Experiment

| | | All | | | Poor | | | Middle | | | Rich | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | All | Control | Treat | P-value | Control | Treat | P-value | Control | Treat | P-value | Control | Treat | P-value |
| Share Household (%) | 84.7 | 84.8 | 84.7 | 0.879 | 89.5 | 89.6 | 0.713 | 89.3 | 89.0 | 0.396 | 70.2 | 70.3 | 0.891 |
| | (36.0) | (35.9) | (36.0) | | (30.6) | (30.5) | | (30.9) | (31.2) | | (45.7) | (45.7) | |
| Share of Payments (2022) (%) | 41.5 | 41.7 | 41.3 | 0.295 | 39.4 | 39.3 | 0.859 | 42.3 | 41.9 | 0.437 | 44.4 | 43.9 | 0.365 |
| | (45.1) | (45.1) | (45.1) | | (44.6) | (44.6) | | (45.2) | (45.1) | | (45.8) | (45.8) | |
| Share of Payments (2023) (%) | 51.0 | 51.0 | 51.0 | 0.937 | 50.7 | 50.6 | 0.840 | 51.3 | 51.3 | 0.991 | 51.0 | 51.0 | 0.908 |
| | (46.4) | (46.4) | (46.4) | | (46.5) | (46.5) | | (46.1) | (46.3) | | (46.4) | (46.4) | |
| 2024 Assessed Value (million AR$) | 119.6 | 119.1 | 120.1 | 0.548 | 47.1 | 47.0 | 0.522 | 105.2 | 105.5 | 0.334 | 257.3 | 261.4 | 0.495 |
| | (238.0) | (110.4) | (317.5) | | (19.1) | (18.9) | | (21.1) | (21.2) | | (145.1) | (620.0) | |
| 2023 Tax Rate (%) | 2.66 | 2.65 | 2.68 | 0.31 | 3.50 | 3.58 | 0.32 | 2.06 | 2.07 | 0.21 | 2.15 | 2.13 | 0.06 |
| | (4.16) | (3.79) | (4.50) | | (5.92) | (7.05) | | (0.44) | (0.46) | | (0.87) | (0.85) | |
| Share Poor (%) | 39.8 | 39.8 | 39.9 | 0.657 | - | - | - | - | - | - | - | - | - |
| | (49.0) | (48.9) | (49.0) | | | | | | | | | | |
| Share Rich (%) | 24.3 | 24.3 | 24.3 | 0.996 | - | - | - | - | - | - | - | - | - |
| | (42.9) | (42.9) | (42.9) | | | | | | | | | | |
| Observations | 92,616 | 46,179 | 46,437 | | 18,366 | 18,535 | | 16,585 | 16,612 | | 11,228 | 11,290 | |

*Note*: This balance table presents the difference between control and treatment groups across different categories (All, Poor, Middle, and Rich). Standard deviation in parenthesis. The p-value columns reflect the results of difference tests between control and treatment groups. Results are omitted in categories where comparisons between groups are not applicable.

Table E.4: Balance Table Letter vs No Letter: 2024 Experiment

| | | All | | | Poor | | | Middle | | | Rich | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | All | No Letter | Control | P-value | No Letter | Control | P-value | No Letter | Control | P-value | No Letter | Control | P-value |
| Share Household (%) | 84.7 | 83.9 | 84.8 | 0.105 | 89.3 | 89.5 | 0.759 | 88.1 | 89.3 | 0.101 | 69.3 | 70.2 | 0.496 |
| | (36.0) | (36.8) | (35.9) | | (30.9) | (30.6) | | (32.4) | (30.9) | | (46.1) | (45.7) | |
| Share of Payments (2022) (%) | 41.6 | 40.9 | 41.7 | 0.241 | 39.3 | 39.4 | 0.926 | 42.6 | 42.3 | 0.797 | 40.8 | 44.4 | 0.010 |
| | (45.1) | (45.1) | (45.1) | | (44.6) | (44.6) | | (45.3) | (45.2) | | (45.7) | (45.8) | |
| Share of Payments (2023) (%) | 50.9 | 49.8 | 51.0 | 0.087 | 49.8 | 50.7 | 0.416 | 50.7 | 51.2 | 0.604 | 48.4 | 51.0 | 0.073 |
| | (46.4) | (46.4) | (46.3) | | (46.5) | (46.5) | | (46.1) | (46.1) | | (46.7) | (46.4) | |
| 2024 Assessed Value (million AR$) | 119.1 | 119.0 | 119.1 | 0.932 | 46.3 | 47.2 | 0.073 | 105.3 | 105.2 | 0.854 | 252.8 | 257.3 | 0.302 |
| | (109.8) | (103.0) | (110.4) | | (19.2) | (19.4) | | (21.3) | (21.2) | | (122.7) | (145.1) | |
| 2023 Tax Rate (%) | 2.65 | 2.68 | 2.65 | 0.60 | 3.57 | 3.50 | 0.64 | 2.08 | 2.07 | 0.17 | 2.20 | 2.16 | 0.15 |
| | (3.84) | (4.23) | (3.79) | | (6.69) | (5.92) | | (0.46) | (0.44) | | (0.96) | (0.94) | |
| Share Poor (%) | 39.7 | 38.9 | 39.8 | 0.240 | - | - | - | - | - | - | - | - | - |
| | (48.9) | (48.8) | (48.9) | | | | | | | | | | |
| Share Rich (%) | 24.4 | 24.8 | 24.3 | 0.442 | - | - | - | - | - | - | - | - | - |
| | (42.9) | (43.2) | (42.9) | | | | | | | | | | |
| Observations | 51,070 | 4,870 | 46,200 | | 1,894 | 18,368 | | 1,768 | 16,602 | | 1,208 | 11,230 | |

*Note*: This balance table presents the difference between control and no letter groups across different categories (All, Poor, Middle, and Rich). Standard deviation in parenthesis. The p-value columns reflect the results of difference tests between control and treatment groups. Results are omitted in categories where comparisons between groups are not applicable.

Table E.5: Balance Table: Complementary Survey

| | | All | | | Poor | | | Middle | | | Rich | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | All | Control | Treat | P-value | Control | Treat | P-value | Control | Treat | P-value | Control | Treat | P-value |
| Share Household (%) | 93.0 | 91.9 | 94.1 | 0.081 | 95.7 | 97.1 | 0.381 | 96.7 | 97.0 | 0.874 | 81.7 | 87.6 | 0.060 |
| | (25.5) | (27.2) | (23.6) | | (20.4) | (16.8) | | (17.8) | (17.2) | | (38.7) | (33.0) | |
| Share of Payments (2021) (%) | 81.5 | 80.3 | 82.7 | 0.124 | 77.9 | 82.0 | 0.176 | 81.4 | 83.0 | 0.528 | 81.2 | 83.2 | 0.483 |
| | (32.6) | (33.8) | (31.4) | | (35.8) | (33.0) | | (32.5) | (30.5) | | (33.4) | (30.9) | |
| Share of Payments (2022) (%) | 84.1 | 83.7 | 84.5 | 0.536 | 82.3 | 87.7 | 0.023 | 84.7 | 82.3 | 0.285 | 83.7 | 84.0 | 0.898 |
| | (28.8) | (28.8) | (28.8) | | (29.6) | (25.2) | | (27.6) | (30.6) | | (29.6) | (29.8) | |
| 2023 Assessed Value (million $s) | 135.0 | 132.1 | 137.7 | 0.268 | 51.5 | 52.8 | 0.346 | 107.1 | 105.7 | 0.399 | 246.9 | 261.0 | 0.162 |
| | (105.6) | (95.4) | (114.4) | | (15.9) | (15.5) | | (21.5) | (21.1) | | (95.2) | (132.1) | |
| 2022 Tax Rate (%) | 2.279 | 2.315 | 2.246 | 0.183 | 3.109 | 2.927 | 0.193 | 1.997 | 2.001 | 0.782 | 1.941 | 1.858 | 0.131 |
| | (1.079) | (1.182) | (0.969) | | (1.803) | (1.383) | | (0.236) | (0.210) | | (0.651) | (0.603) | |
| Share Poor (%) | 30.7 | 30.1 | 31.3 | 0.592 | - | - | - | - | - | - | - | - | - |
| | (46.1) | (45.9) | (46.4) | | | | | | | | | | |
| Share Rich (%) | 30.6 | 29.9 | 31.3 | 0.521 | - | - | - | - | - | - | - | - | - |
| | (46.1) | (45.8) | (46.4) | | | | | | | | | | |
| Observations | 1,723 | 844 | 879 | | 254 | 275 | | 338 | 329 | | 252 | 275 | |

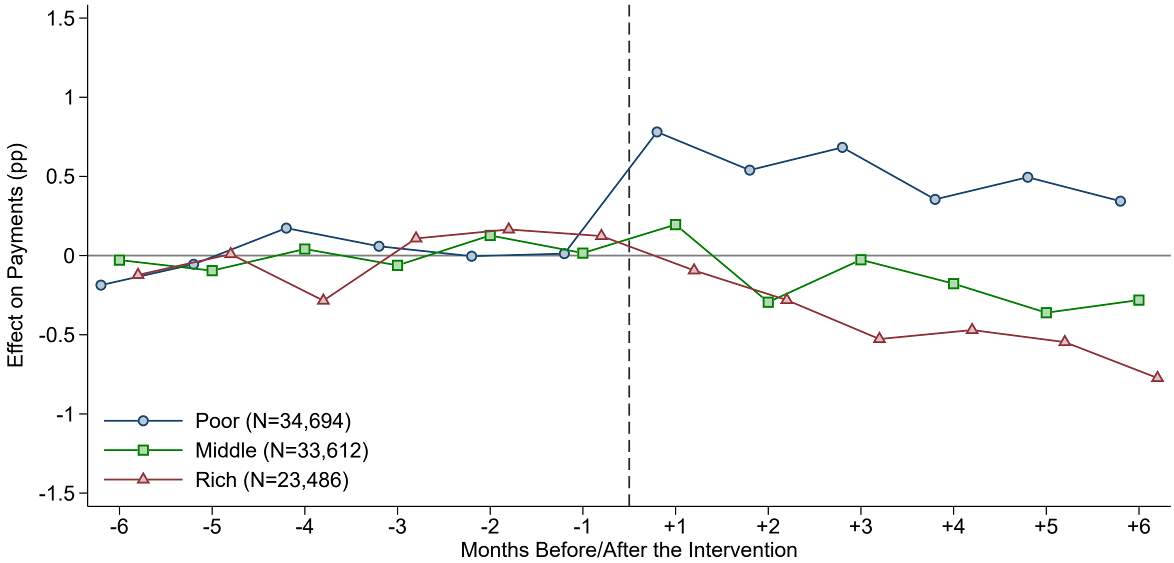
*Note*: This balance table presents the difference between control and treatment groups across different categories (All, Poor, Middle, and Rich) for our survey experiment. Standard deviation in parenthesis. The p-value columns reflect the results of difference tests between control and treatment groups. Results are omitted in categories where comparisons between groups are not applicable.

# F Additional Experimental Results

## F.1 Differences in Payment Rates

This section examines the additional experimental results comparing the treatment and control groups in terms of tax payment behavior. As shown in Figure F.1, six months prior to the intervention, the payment differences between the treatment and control groups were minimal, indicating no significant pre-existing disparities in compliance behavior. However, a notable shift was observed post-treatment, especially among lower-valuation taxpayers. Specifically, an increase in the number of payments was recorded, with lower-value properties in the treatment group showing substantially higher compliance rates compared to the control group. In contrast, taxpayers in the high-value bracket experienced a slight decrease in compliance, while those in the middle bracket displayed no significant behavioral changes.

Figure F.1: Difference in Payment Rates between Control and Treatment Letters in 2023




*Note*: This figure compares the likelihood of paying the monthly bill between treatment and control. The blue line denotes poor households with assessed values below AR$ 750K. The green line denotes middle households with assessed values between AR$ 750K and AR$ 1.5M. The red line denotes rich households with assessed values above AR$ 1.5M. We normalized each series by the average pre-treatment difference. We estimated the coefficient for each monthly bill in separate regressions. The sample includes residential and non-residential properties and excludes units that made payments for 2023 before we sent the letters.

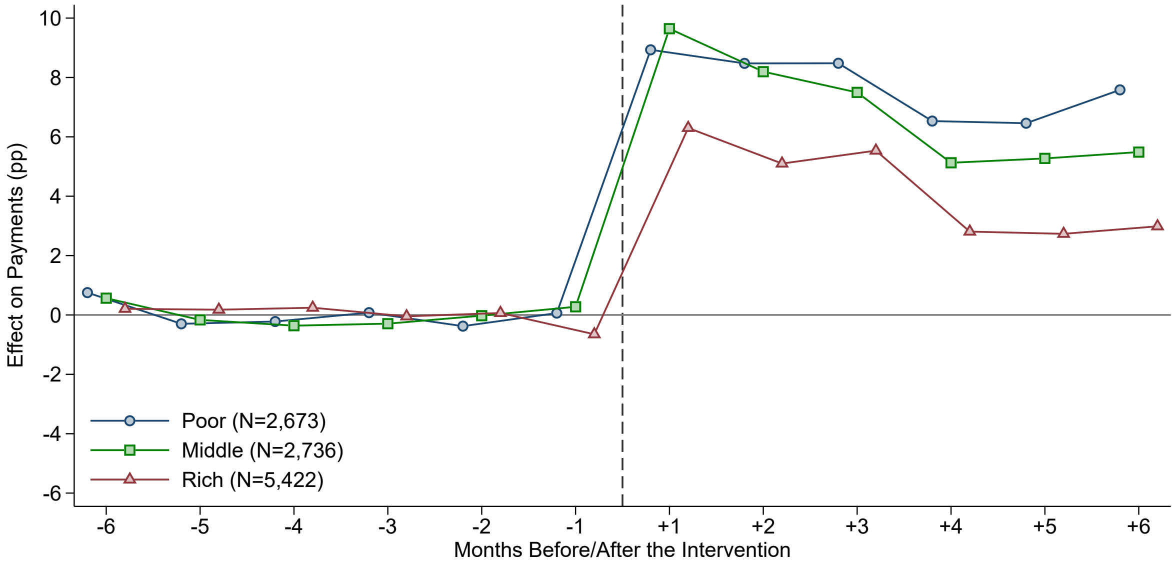
## F.2 Control Letter vs No Letter

This section compares the effects of receiving a control letter versus receiving no letter on tax compliance. Two figures illustrate the findings, focusing on changes in pre- and post-treatment payment behaviors across different valuation brackets.

Figure F.2 shows that before the intervention, payment patterns were similar between groups that received a letter and those that did not, confirming the lack of significant differences before treatment. However, after treatment, properties that received a control letter showed higher compliance rates than those that received no letter, suggesting the mere receipt of a letter influences compliance behavior.

Figure F.3 presents a comprehensive analysis of treatment effects on tax compliance behavior. Panel (a) examines compliance rates before and after the intervention across the two groups. The findings show a noticeable increase in payments for those receiving a control letter during the first two months of payments, with comparable patterns observed across properties of lower, middle, and high value. This increase corresponds to a rise of approximately 4 to 6 percentage points in the payment share for these months. Panel (b) pools data into two-month pre- and post-treatment periods to enhance statistical power. The analysis revealed no significant differences in compliance rates before treatment, but post-treatment, the groups receiving control letters consistently exhibit higher compliance rates across all valuation brackets.

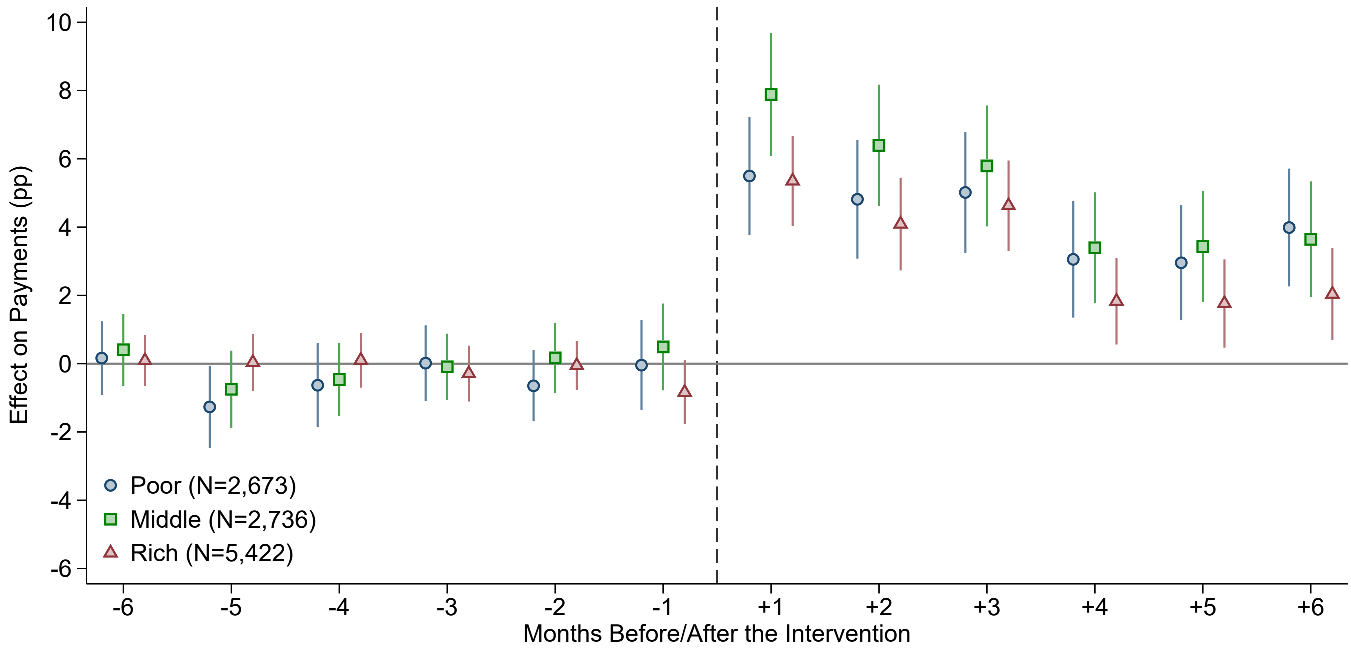
Figure F.2: Difference in Payment Rates between Control and No-Letter Groups in 2023



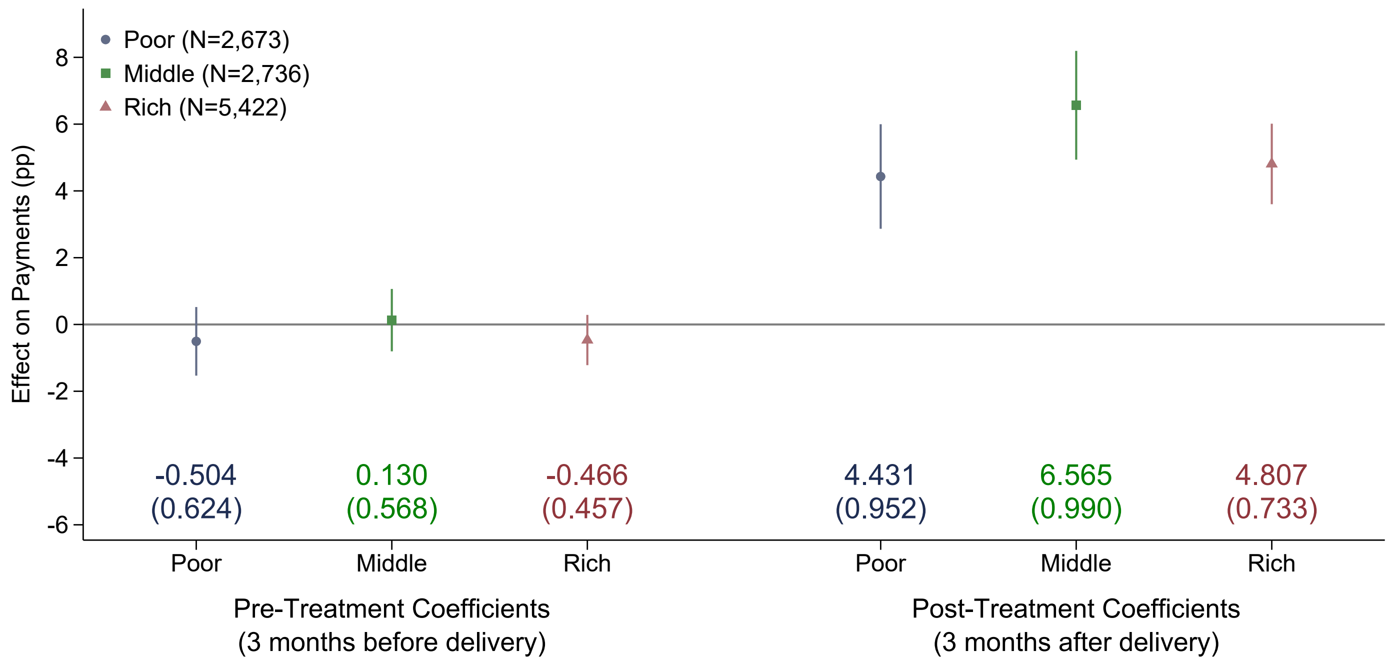

*Note*: This figure compares the likelihood of paying the monthly bill between treatment and control, normalized by the average pre-treatment difference. We estimated the coefficient for each monthly bill in separate regressions. The sample includes residential and non-residential properties and excludes units that made payments for 2023 before we sent the letters.

Figure F.3: The Effect of Receiving a Letter on Own Tax Compliance

(a) Event-Study Analysis (Monthly Level)




(b) Pre- and Post-Treatment Effects (Quarterly level)




*Note*: This figure compares the likelihood of paying the monthly property tax bill between the control and the no-letter groups. The control group received a letter with information solely about how the reform changed the household's own tax rate, while the no-letter group received none. Panel (a) shows the dynamic effect of the letters, while panel (b) shows the two months pooled effects of the letter (pre and post-treatment). 'Poor' (in blue) denotes households with properties valued at AR$ 750K or less, 'Middle' (in green) denotes properties valued between AR$ 750K and AR$ 1.5M, and 'Rich' (in red) denotes properties valued more than AR$ 1.5M. We estimated the coefficient for each monthly bill in separate regressions, including the 90% confidence interval for each group. Clustered standard errors for the pooled estimates in parenthesis. The sample includes only shops (the only ones targeted to receive no letter) and excludes units that made payments for 2023 before we sent the letters. We controlled for the past 12 months payments in each pre-treatment regression, and for the 2022 monthly payments in each post-treatment regression. Additionally, we included controls for property valuation (in logs), a residential dummy, dummies indicating whether the property was an "always payer" or "never payer" in the previous year, and a dummy for properties that made an annual payment in the previous year. We included dummies equal to one for missing controls to retain all observations in the regressions. Standard errors clustered at the individual taxpayer level.

## F.3 Experimental Uptake: Alternative Estimates of Reading Rates

The discussion of the results in Section 4.4 indicates that we may adjust our ITT estimated by a letter reading rate to obtain the Average Treatment Effect on the Treated, since it is likely that a significant share of households did not pay attention at all to the second page of the letter. The main body of the paper uses a conservative adjustment factor to accounts for treatment non-uptake of 2, in line to the existing literature (Perez-Truglia and Cruces, 2017; Bottan and Perez-Truglia, 2025; Gerber et al., 2020; Nathan et al., 2020) and with estimates from the Environmental Protection Agency.

In our context, we can use the evidence on awareness of the reform from the survey sample. Specifically, Figure 5 (panels (a) and (b)) present the treatment effect on the probability of knowing about the reform, and we can use this treatment effect as a proxy for experimental uptake. If 19.4% of poor households in the control group were aware of the reform and 100% in the treatment group were exposed to the information, we would expect awareness to increase by 80.6 pp due to our treatment (from 19.4% to 100%). However, we identified an effect of approximately 9.2 pp (panel (b)), implying a reading rate or uptake of 11.4% ($\frac{9.2}{80.6}$), or a non-uptake adjustment factor of around 8.8.

This adjustment factor suggests that the treatment ATET on the poor was 7.1 pp. This high estimate is probably an upper bound we must interpret cautiously, but it complements the lower bound used in the main body of the paper. The reading rates we estimated from the survey have several limitations, including the fact that the specific question used to measure awareness included different options and reforms, potentially confusing some respondents. Moreover, some taxpayers might have known about the progressive reform, but did not pay attention to the survey.

# G Comparison of Sample Subgroups

Table G.1 presents summary statistics for key taxpayer attributes across different subgroups. Column (1) describes the full sample of taxpayers with registered property valuations, while column (2) focuses on the main sample used in the experimental analysis. Column (3) reports statistics for taxpayers with registered email addresses, who were eligible for the survey and excluded from the experimental analysis, as discussed in Section 4.2. Finally, columns (4) and (5) compare survey respondents and non-respondents. This table shows that that while survey participants exhibit higher compliance levels, their observable characteristics remain largely comparable to those of the broader sample. Taxpayers in the email sample (column (3)) show significantly higher compliance rates in 2022 compared to the universe (column (1)), with payment shares of approximately 47.9% in the full sample versus 74.1% in the email sample. Survey respondents (column (5)) demonstrate even higher compliance, with a payment share of 84.1%. In contrast, there are smaller differences in other observable characteristics such as assessed property values and tax rates. For example, taxpayers in the universe have a somewhat higher tax rate (2.8%) than those in the email sample (2.37%), shown respectively in columns (1) versus (3).

Table G.1: Summary Statistics by Sample Group

| | (1) Universe | (2) Main | (3) All Email | (4) Responded Survey? No | (5) Responded Survey? Yes |
|---|---|---|---|---|---|
| Share Household (%) | 84.3 | 88.7 | 89.9 | 89.4 | 93.0 |
| | (36.4) | (31.7) | (30.1) | (30.7) | (25.5) |
| Share of Payments (%) | 47.9 | 47.4 | 74.1 | 72.6 | 84.1 |
| | (45.4) | (45.4) | (35.6) | (36.2) | (28.8) |
| Assessed Value (AR$) | 137.2 | 133.0 | 133.1 | 132.4 | 137.9 |
| | (589.3) | (583.2) | (182.6) | (191.7) | (104.8) |
| Tax Rate (%) | 2.80 | 2.75 | 2.37 | 2.38 | 2.28 |
| | (4.04) | (4.12) | (1.39) | (1.43) | (1.08) |
| Share Poor (%) | 37.1 | 37.3 | 33.6 | 34.1 | 30.7 |
| | (48.3) | (48.4) | (47.2) | (47.4) | (46.1) |
| Share Rich (%) | 26.6 | 25.8 | 27.5 | 27.0 | 30.6 |
| | (44.2) | (43.7) | (44.6) | (44.4) | (46.1) |
| Observations | 113,956 | 91,786 | 13,145 | 11,422 | 1,723 |

*Note*: This table presents summary statistics for the different sample subgroups. Columns (1) and (2) show the full universe and the main sample, respectively. Column (3) includes taxpayers with registered email addresses, who were eligible for the survey. Columns (4) and (5) compare non-respondents and respondents. Differences in compliance rates are the most pronounced, while other characteristics remain relatively stable across groups. Standard deviations are reported in parentheses. All statistics reflect pre-treatment characteristics from December 2022. However, nine observations in the survey sample are missing from this table, likely due to taxpayers deregistering from the system in 2022.

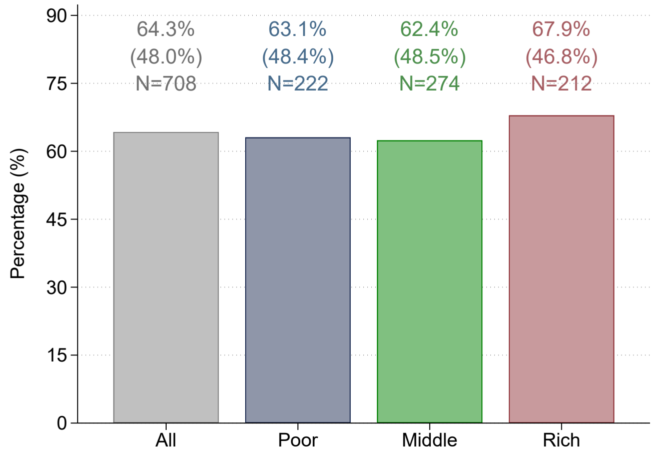
# H Survey Questionnaire Additional Placebos

In this appendix, we present additional results from the survey experiment, focusing on placebo outcomes designed to test the specificity of the treatment effects. In particular, we examined self-reported awareness of two additional municipal policies that were unrelated to the progressive tax reform. These policies were mentioned in both control and treatment letters, and thus serve as placebo outcomes.

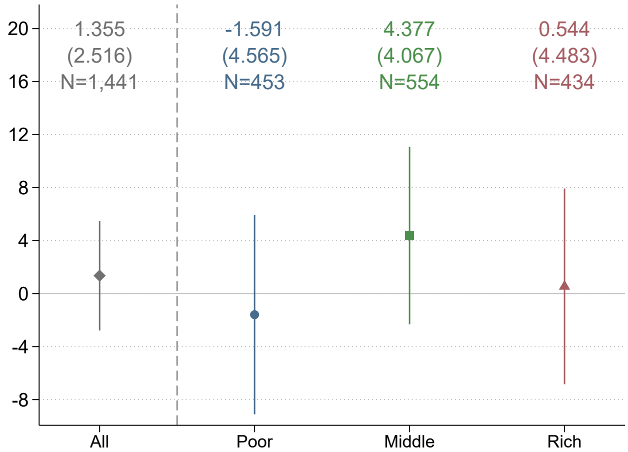
The results are presented in Figure H.1. The two left panels correspond to the free and express commercial permits, while the two right panels correspond to the simplified paperwork for new businesses. Panels (a) and (c) of Figure H.1 show the proportion of respondents in the control group who reported being aware of each of these policies. Awareness is relatively high, which may stem from organic familiarity with the policies or simply from the fact that all letters—including the control letters—publicized them. Most importantly, panels (b) and (d) present the treatment effects on these two placebo outcomes. In panel (b), we find that—consistent with the placebo outcome results reported in Figure 5—the coefficients are close to zero and statistically insignificant. In panel (d), the results are somewhat more mixed, as one of the coefficients—the leftmost one, corresponding to the full sample—is statistically significant. However, our preferred interpretation is that this is a spurious finding: given the large number of placebo analyses reported in the paper, it is expected that a few may appear statistically significant purely by chance.

Figure H.1: Taxpayers' Awareness of Placebo Reforms: Survey Experiment

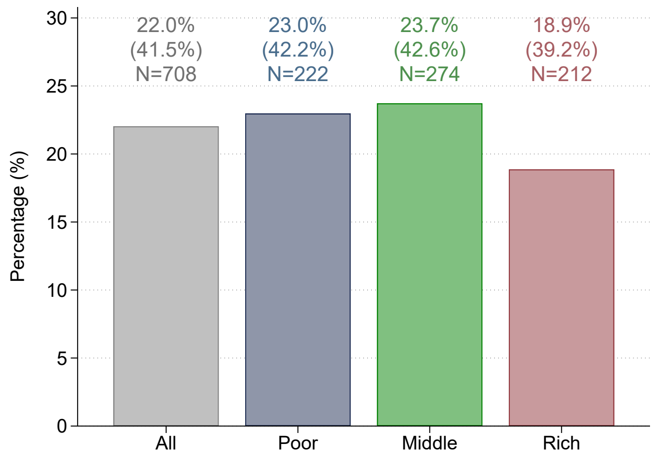
(a) Control: Awareness of New Commercial Permits



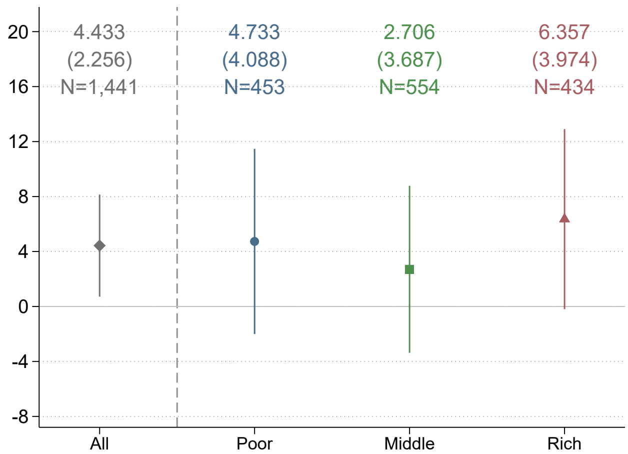

(b) ITT: Awareness of New Commercial Permits




(c) Control: Awareness of Reduced Paperwork




(d) ITT: Awareness of Reduced Paperwork




*Note*: This figure uses data from our online survey on taxpayers to assess awareness of two placebo reforms, implemented by the municipality and shown to both control and treatment letters. We invited a small subsample of subjects with email addresses on file to participate. The respondents received a list of recent policies and were asked to indicate which ones they had heard of. The left panels show the average knowledge of two different policies for households receiving the control letter. The right panels show the effects of the treatment letter on the awareness of the Tres de Febrero free and express commercial permits and on the knowledge of the Tres de Febrero reduced paperwork for new businesses. The vertical spikes denote 90% confidence intervals.

# I Replication of the 2023 Findings in 2024

## I.1 Experimental Design in the 2024 Reform

The municipality implemented a second progressive tax reform in 2024. We re-randomized households into treatment and control letters within each of the property value groups—that is, we did not keep the treatment status of 2023 in 2024.[56] A minor difference for the 2024 experiment with respect to 2023 is that we excluded from the treatment and control letters the "empty lot", "civil entities", "religious entities" and "wholesale establishments" categories (around 1% of the sample). In the 2024 design, we also included within each of the poor, middle, and rich households groups a small subsample of households that did not receive any letter (3.6% of the total) to gauge the pure effect of receiving a letter (see results from this exercise in Appendix I.4)—in 2024 we included no letter households across the whole sample, whereas the analysis of no letter in 2023 in Appendix F.2 was limited to commercial properties only. Finally, we excluded from the mailing experiment sample the 17.35% of the sample that had a registered email with the Municipality, and we reserved this sample for a survey about perception of taxes and rates which we did not want to contaminate with the information from the treatment and control letters.

This new reform built upon the 2023 setup, maintaining the core principles of increasing progressivity while refining the communication strategy to taxpayers. As explained in Section 4.6 above, we took this opportunity to refine our experimental design by adjusting the 2024 control letter to make the difference between the treatment and control letters even more subtle.

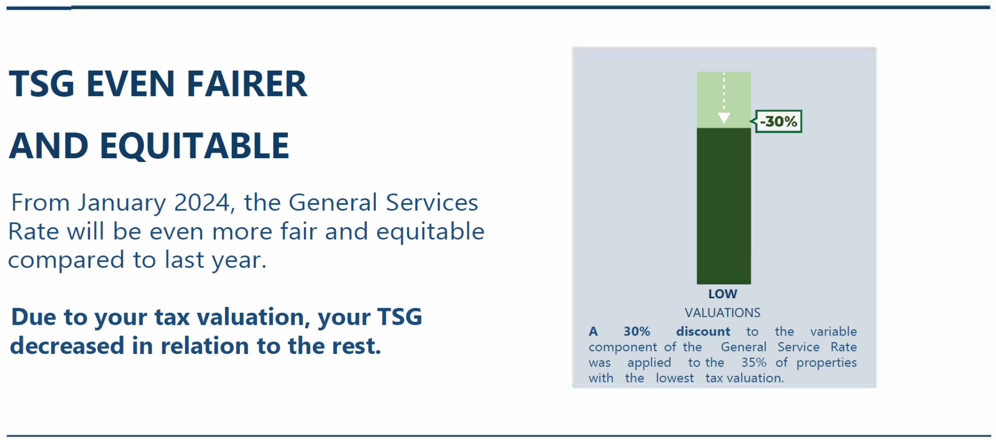
An example of the treatment and control infographic for poor households is shown in Figure I.1, while Appendix L.2 provides a full sample letter along with the treatment and control infographics for middle and rich households. One important detail is the information about the discounts and surcharges. In the 2023 reform, the tax liability adjustments were expressed as a 30% reduction for lower-valuation households and a 15% increase for higher-valuation households in the variable component. The 2024 reform further adjusted the tax system by modifying the fixed component, resulting in an additional approximate 16% reduction for lower-valuation households and an additional increase for higher-valuation households that ranges from approximately 39% to 64%. However, rather than communicating these specific changes, the municipality opted to frame the 2024 policy change as a continuation of the 2023 reform. To ensure continuity and minimize confusion, the municipality framed the reform as a cumulative change in tax liability from 2022 to 2024, rather than focusing on the very technical detail of the

[56] The results from the 2024 experiment discussed below remain virtually unchanged if we control for the household's treatment status in 2023.

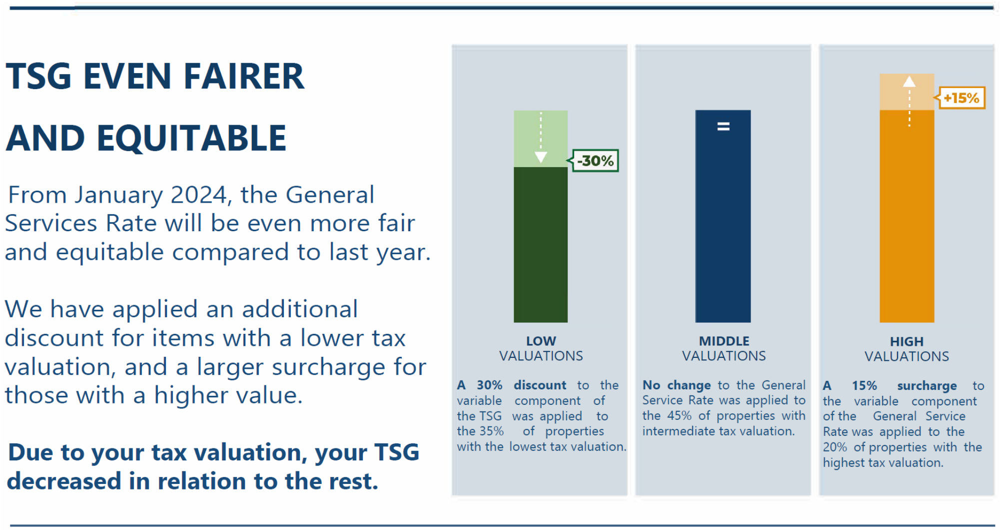
fixed component adjustment introduced in 2024. The justification for this approach was that when considering the total tax liability from 2022 to 2024, the cumulative effect of the reforms translated into an approximate 22.3% reduction for poor households and a 17.7% increase for rich households. Since these numbers closely aligned with the 30% reduction and 15% increase communicated in 2023, the municipality chose to maintain this language to avoid taxpayer confusion. However, while the municipality aimed for clear communication, an unfortunate error occurred during the final letter distribution at the page setup stage. The "small print" below the infographic mistakenly read *variable component* (as in the 2023 letters) instead of *tax liability*. Importantly, this typo was consistent across both treatment and control letters. Given that it appeared in the "fine print", we believe it was likely unnoticed by taxpayers and had a negligible impact on their response to the letters.

Figure I.1: Sample of Control and Treatment Messages: Poor Households, 2024 Experimental Design

(a) Control Letter Message




(b) Treatment Letter Message




*Note*: English translation (from Spanish) of the main pieces of information from the mailers. This text was contained on the second page of the mailer.

## I.2 Own-Rate Effects in the 2024 Reform

In this section, we replicated the regression discontinuity design analysis conducted for the 2023 tax reform to evaluate the effects of the 2024 reform. The 2024 analysis used the same approach, estimating first- and reduced-form effects of the tax changes at the valuation thresholds to assess behavioral responses in terms of tax compliance. This replication serves to validate the robustness of the findings across years and examine any differences in the effects due to contextual variations.

The 2024 reduced-form compliance effects, shown in Figure I.2, closely resemble those from 2023, highlighting the consistency of taxpayer behavioral responses. However, among rich households, the coefficient is similar in magnitude to its 2023 counterpart but is borderline statistically insignificant (p-value = 0.109). Additionally, the first-stage estimates of tax rate changes are slightly larger in 2024, reflecting stronger discontinuities at the thresholds. Figure I.3 presents the placebo tests for 2024, which perform well overall, confirming that the observed effects are not driven by spurious factors. Furthermore, the density estimates in Figure I.4 closely resemble those from 2023.

Finally, we replicated the elasticities estimated in equations (1) and (2) using data from 2024. Let $\beta_j^{own}$ represent the reduced-form effect on compliance for group $j$, where $j \in \{poor, rich\}$, obtained from the RDD. Let $\alpha_j$ denote the first-stage effect of the reform on tax rates for group $j$, also obtained from the RDD analysis. Let $C_j^{own}$ and $\tau_j$ denote the baseline compliance levels and tax rates, captured by the levels around the relevant thresholds. Both elasticities are expressed in terms of two-month to two-month changes (given dataset limitations in 2024), comparing periods before and after the reform. For each elasticity shown below, we report 90% confidence intervals in brackets, calculated with 5,000 bootstrap iterations:

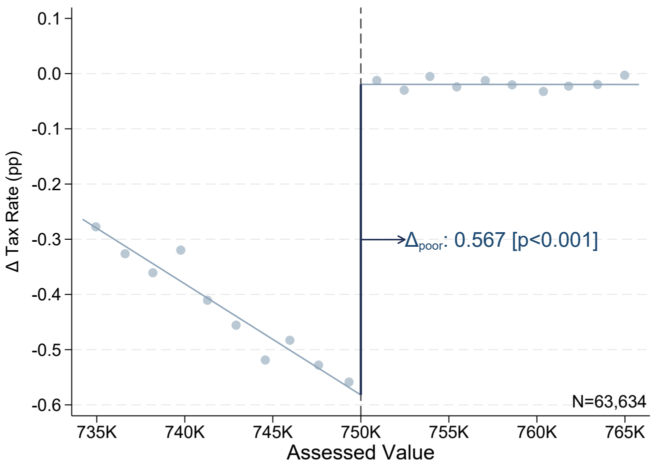
$$\varepsilon_{poor}^{own} = \frac{\frac{\beta_{poor}^{own}}{C_{poor}^{own}}}{\frac{\alpha_{poor}}{\tau_{poor}}} = \frac{\frac{-2.559}{35.3}}{\frac{0.567}{2.252}} = \underset{[-0.529,-0.087]}{\text{-0.288}} \tag{I.1}$$

$$\varepsilon_{rich}^{own} = \frac{\frac{\beta_{rich}^{own}}{C_{rich}^{own}}}{\frac{\alpha_{rich}}{\tau_{rich}}} = \frac{\frac{-2.605}{35.6}}{\frac{0.294}{1.894}} = \underset{[-1.083,-0.038]}{\text{-0.471}} \tag{I.2}$$

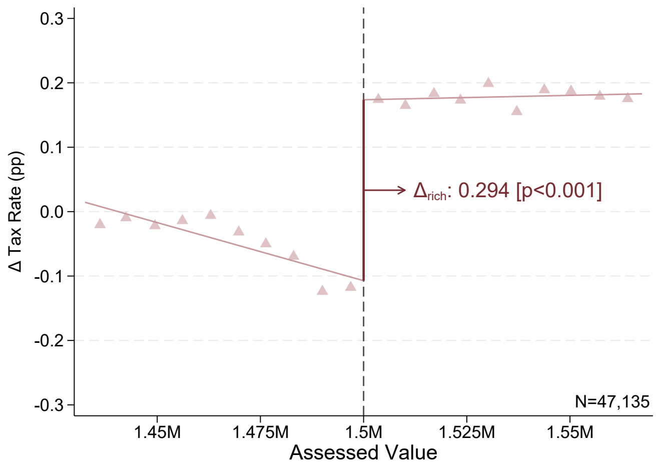
We estimated a significant own-rate elasticity for poor households of -0.288, indicating that for every 1% decrease in their tax rate, compliance increased by 0.288%. Similarly, the own-rate elasticity for rich households is -0.471, implying that for every 1% increase in their tax rate, compliance decreased by 0.471%. To test whether the difference in elasticities between the two groups is statistically significant, we conducted 5,000 bootstrap iterations, which revealed no significant difference (p-value = 0.678).

Figure I.2: 2024 RDD: Own-Rate Effects in Tax Compliance

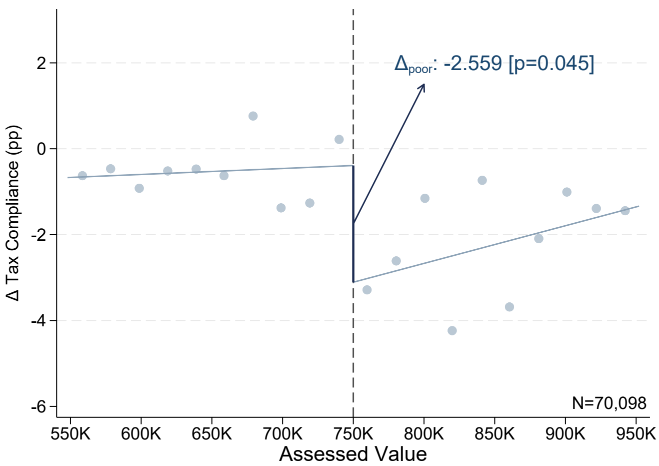
(a) First-Stage: Poor Households



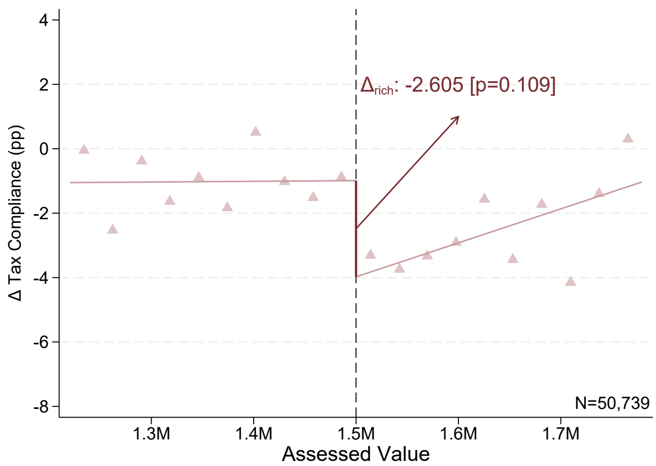

(b) First-Stage: Rich Households




(c) Reduced-Form: Poor Households

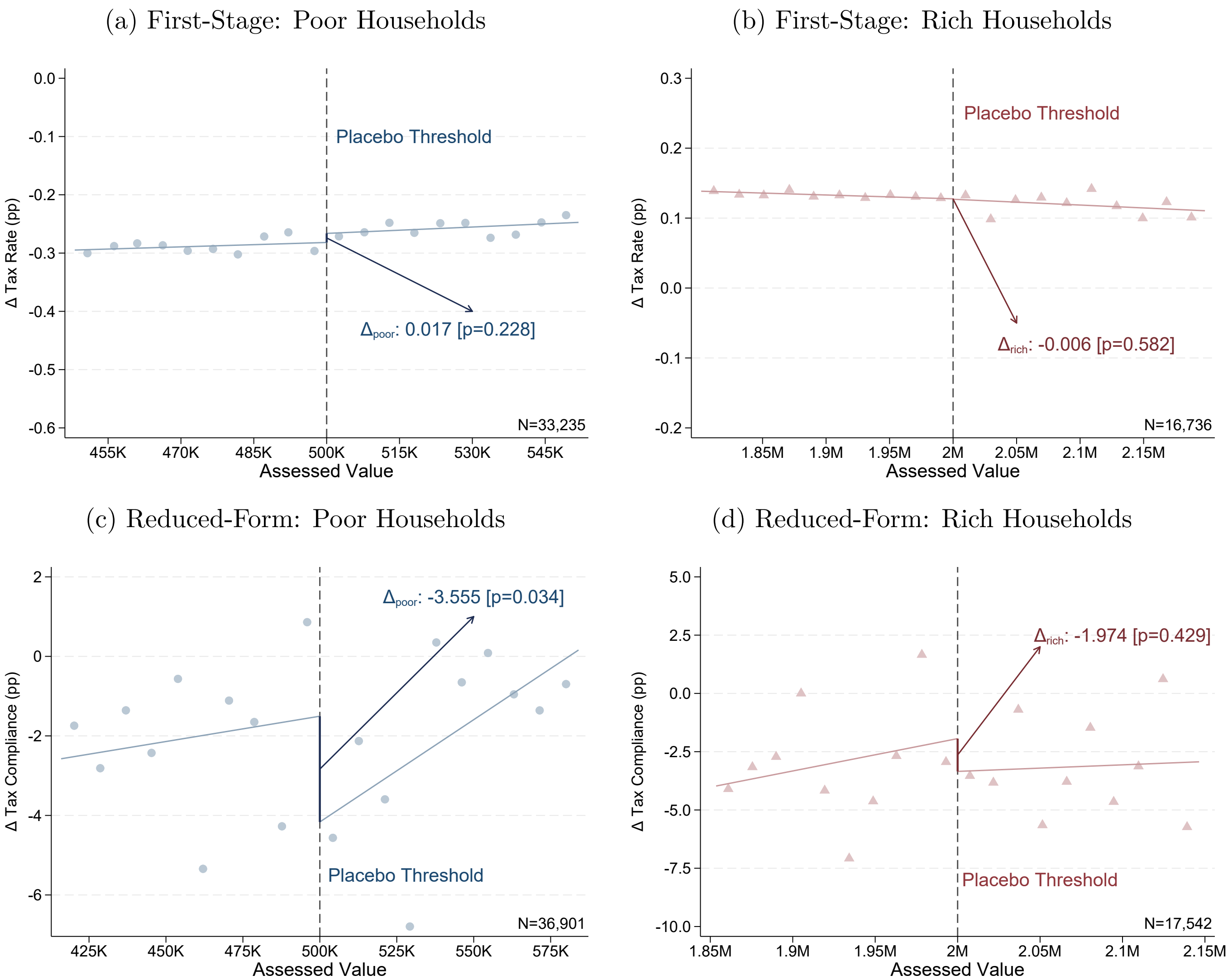



(d) Reduced-Form: Rich Households




*Note*: Panels (a) and (b) represent the first-stage for poor and rich, in which we evaluate the increase in the amounts owed between the last bimester of 2023 and the fist bimester of 2024, in relation to the value of the property, in the cut-off of AR$ 750K and AR$ 1.5M, respectively. Panels (c) and (d) represent the reduced-form for poor and rich, in which we evaluate the change in tax compliance for both groups between the last bimester of 2023 and the fist bimester of 2024. In panels (a), (b), (c) and (d), the x-axis corresponds to the tax valuation of the properties at 2024 values. Each figure indicates the estimated effect around the cut-off point, where the p-value, taken from a robust bias-corrected inference, is indicated in parentheses.

Figure I.3: 2024 Falsification Test: Placebo Thresholds in RDD Analysis

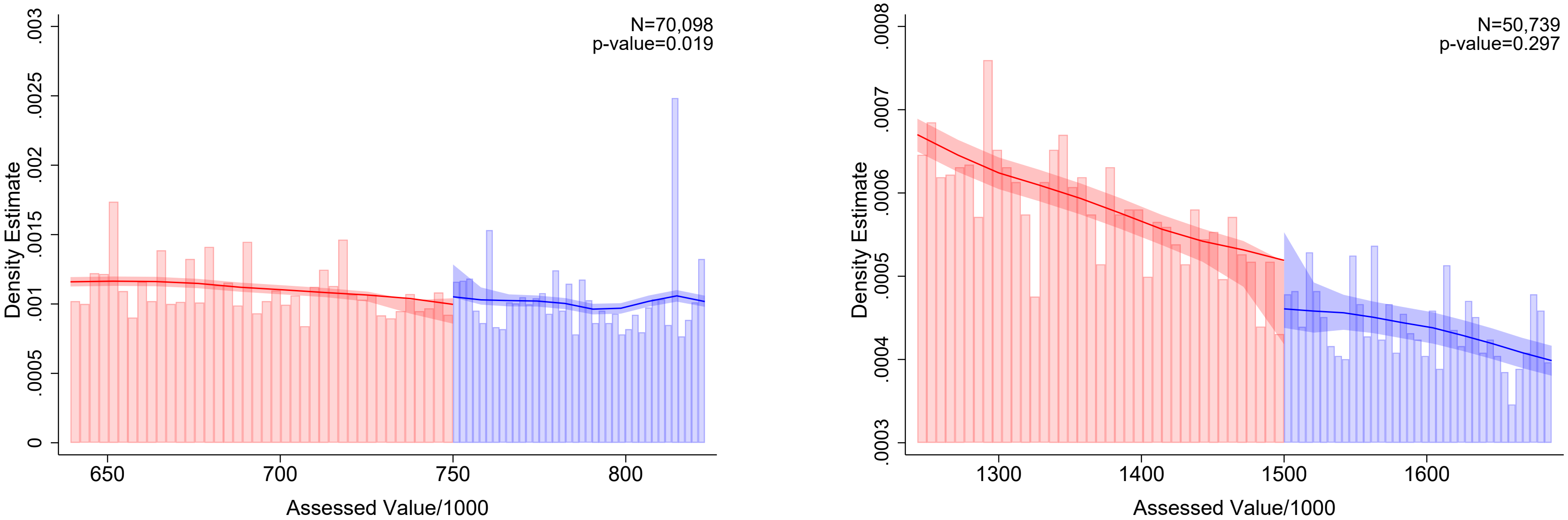



*Note*: Panels (a) and (b) represent the first-stage RDD for poor and rich, in which we evaluate the increase in the amounts owed between the last bimester of 2023 and the fist bimester of 2024, in relation to the value of the property, in the cut-off of AR$ 500K and AR$ 2M, respectively. Panels (c) and (d) represent the reduced-form RDD for poor and rich, in which we evaluate the change in tax compliance for both groups between the last bimester of 2023 and the fist bimester of 2024. The x-axis corresponds to the tax valuation of the properties at 2024 values. Each figure indicates the estimated effect around the cut-off point, where the p-value, taken from a robust bias-corrected inference, is indicated in brackets.

Figure I.4: Manipulation Test of Assessed Values around the Discontinuities: 2024

(a) Poor Cutoff - 2024

(b) Rich Cutoff - 2024




*Note*: Panels (a) and (b) show the McCrary (2008) manipulation test (with their corresponding p-value) of the property assessed values at the low-value threshold (poor) and the high-value threshold (rich) in 2024. The shaded areas represent the 90% confidence intervals. The x-axis represents assessed property value in AR$ thousands.

## I.3 Cross-Rate Effects in the 2024 Field Experiment

This section presents estimates of the cross-rate effects by comparing treatment and control groups in terms of tax compliance behavior for the 2024 replication. As Figure I.5 shows, during the months preceding the intervention, compliance rates between the treatment and control groups showed no statistically significant differences, indicating no preexisting differences in payment behavior. After the letters were sent, however, compliance among lower-value properties in the treatment group increased, aligning with the 2023 findings. Conversely, compliance among high and middle households in the treatment group showed no statistically significant changes.

Figure I.6 provides a detailed analysis of compliance effects, combining a month-by-month evaluation with pooled pre- and post-treatment data. Panel (a) shows that pretreatment estimates are statistically insignificant across all valuation brackets, confirming the robustness of the experimental design. Post-treatment, poor households exhibit increased compliance in the two months following the intervention, significant at the 10% level, while rich properties show slightly negative but statistically insignificant effects. Panel (b) shows the pooled analysis, analyzing two months before and after the intervention, and similarly revealed no significant pre-treatment differences but does reveal a compliance increase among lower-value properties that received information on the reform's progressivity, with no statistically significant changes for rich properties, consistent with the findings from 2023.

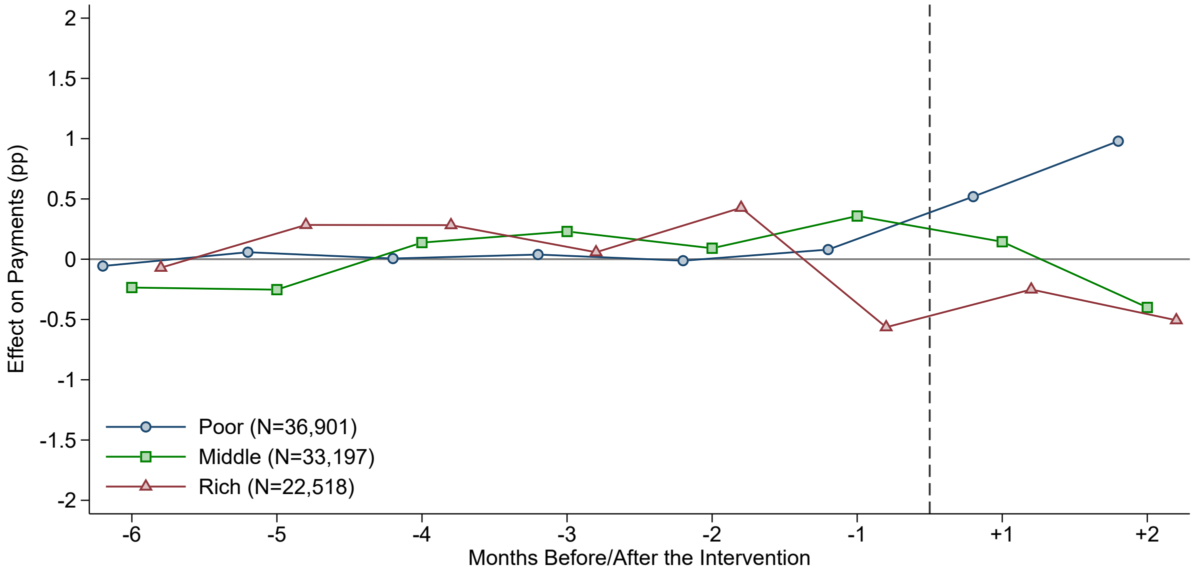
Let $\beta_j^{cross}$ represent the treatment effect on compliance for group $j$ as reported in the experiment, where $j \in \{poor, rich\}$. Let $C_j^{cross}$ denote the baseline compliance of each control group in 2024, and let $\frac{\Delta\tau_{-j}}{\tau_{-j}}$ represent the tax rate change for the opposite group. That is, for a poor household, the relevant change in the opposite group's tax rate corresponds to the tax hike on the rich; for a rich household, it corresponds to the tax cut for the poor. For simplicity, we characterize the tax changes as a 30% decrease in the tax rate for poor households and a 15% increase for rich households, matching the information conveyed in the letters regarding the tax liability. The results are presented below, with 90% confidence intervals in brackets, calculated using 5,000 bootstrap iterations:

$$\varepsilon_{poor}^{cross} = \frac{\frac{\beta_{poor}^{cross}}{C_{poor}^{cross}}}{\frac{\Delta\tau_{rich}}{\tau_{rich}}} = \frac{\frac{0.721}{38.4}}{0.15} = \underset{[0.025, 0.229]}{0.125} \tag{I.3}$$

$$\varepsilon_{rich}^{cross} = \frac{\frac{\beta_{rich}^{cross}}{C_{rich}^{cross}}}{\frac{\Delta\tau_{poor}}{\tau_{poor}}} = \frac{\frac{-0.409}{32.1}}{\text{-0.3}} = \underset{[-0.024, 0.110]}{0.042} \tag{I.4}$$

We estimated a significant cross-rate elasticity for poor households of 0.125, indicating that for each 1% increase in the tax rate of the rich households, poor households increased their compliance rate by 0.125% when they become aware of the change in the other group. We also estimated a cross-rate elasticity for rich households of 0.042, although this is not statistically significant at conventional levels. These results are not scaled-up for non-uptake and thus should be interpreted as ITT elasticities. Lastly, we tested whether the difference in elasticities between the two groups is statistically significant, as we did in 2023, but found no evidence of significant differences for this year (p-val=0.271).

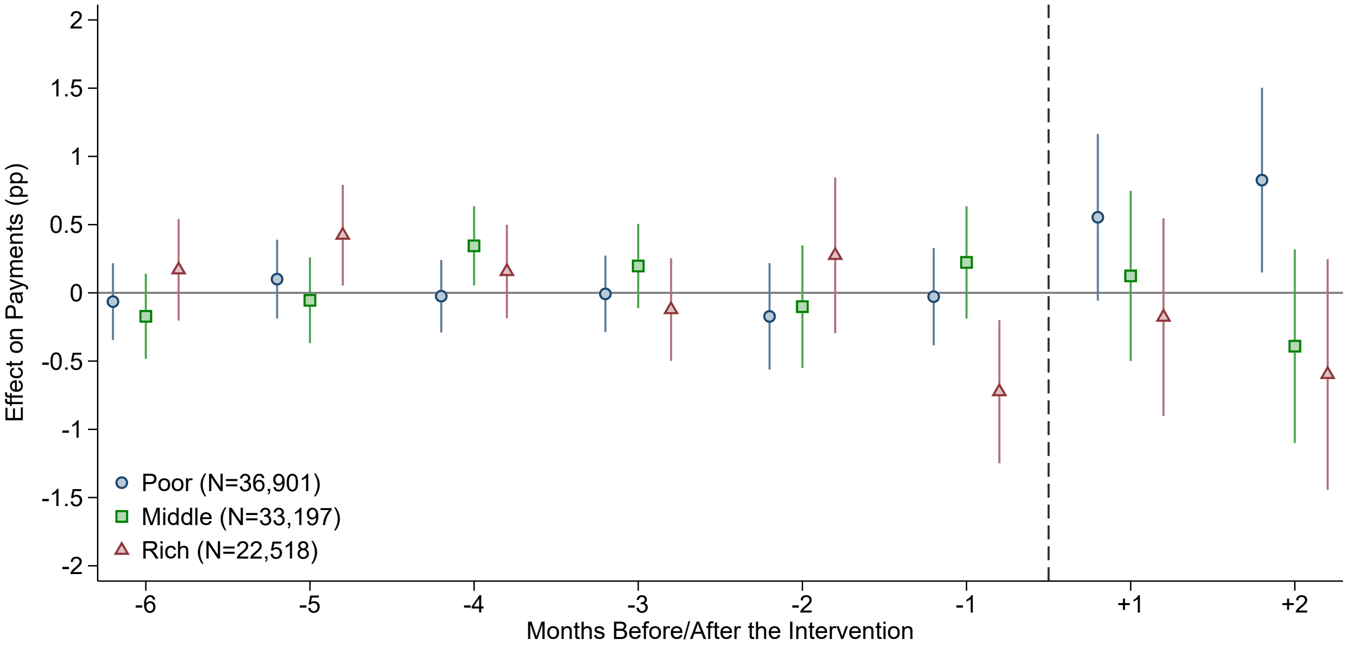
Figure I.5: Difference in Payment Rates between Control and Treatment Letters: 2024



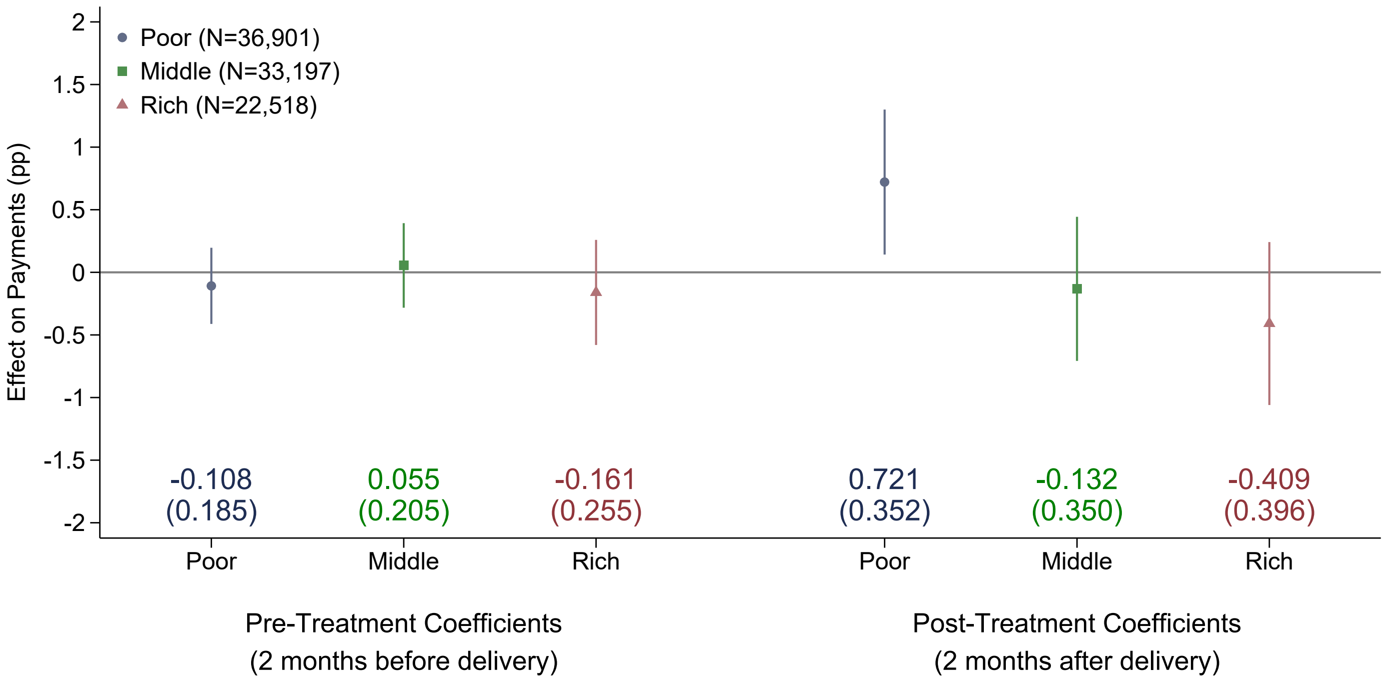

*Note*: This figure compares the likelihood of paying the monthly bill between treatment and control, normalized by the average pre-treatment difference. We estimated the coefficient for each monthly bill in separate regressions. The sample includes residential and non-residential properties and excludes units that made payments for 2024 before we sent the letters.

Figure I.6: Cross-rate Effects of a Progressive Tax Reform on Own Tax Compliance: 2024

(a) Event-Study Analysis (Monthly Level)




(b) Pre- and Post-Treatment Effects (Bimester level)




*Note*: This figure compares the likelihood of paying the monthly property tax bill between treatment and control for the 2024 experiment. The control group received a letter with information solely about how the reform changed the household's own tax rate. The treatment group letter included additional information about how the reform affected the tax rates of other households (i.e., informing the progressive nature of the reform). See Figure I.1 for an example. Panel (a) shows the dynamic effect of the treatment, while panel (b) shows the two months pooled effects of the treatment (pre- and post-treatment). 'Poor' (in blue) denotes households with properties valued at AR$ 750K or less, 'Middle' (in green) denotes properties valued between AR$ 750K and AR$ 1.5M, and 'Rich' (in red) denotes properties valued more than AR$ 1.5M. We estimated the coefficient for each monthly bill in separate regressions, including the 90% confidence interval for each group. Clustered standard errors for the pooled estimates in parenthesis. The sample includes residential and non-residential properties and excludes units that made payments for 2024 before we sent the letters. We controlled for the past 12 months payments in each pre-treatment regression, and for the 2023 monthly payments in each post-treatment regression. Additionally, we included controls for property valuation (in logs), a residential dummy, dummies indicating whether the property was an "always payer" or "never payer" in the previous year, and a dummy for properties that made an annual payment in the previous year. We included dummies equal to one for missing controls to retain all observations in the regressions. Standard errors clustered at the individual taxpayer level.

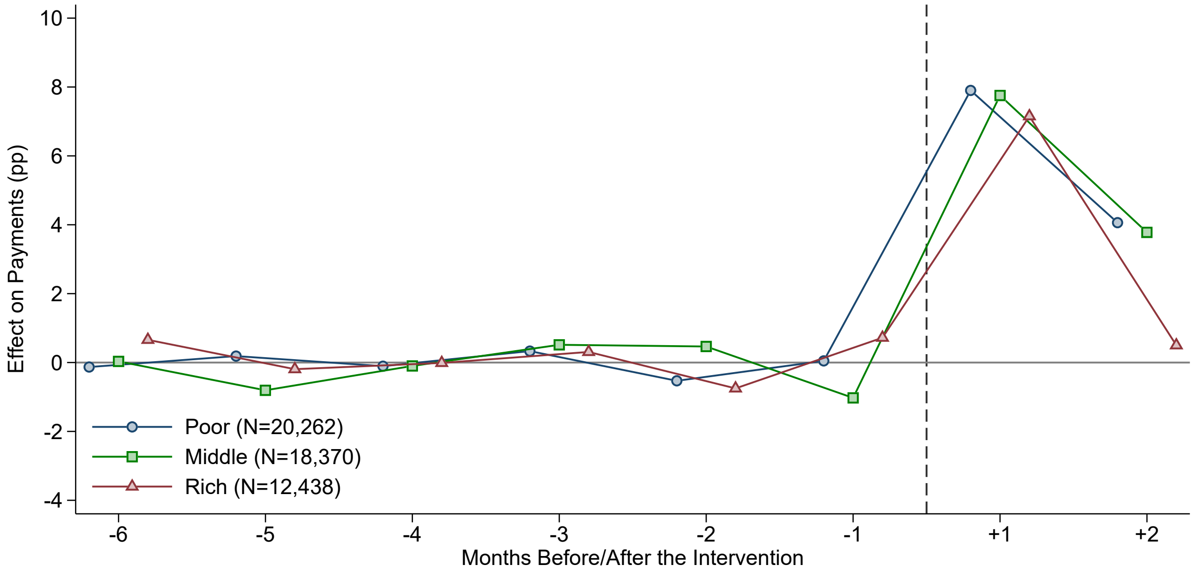
## I.4 Control Letter vs No Letter

The analysis of this section replicates the structure and findings of the 2023 analysis, which compared compliance rates between groups receiving control letters and those receiving no letters in Appendix F.2. The results for 2024, illustrated in Figure I.7, show that pre-treatment payment patterns were statistically indistinguishable between these groups, aligning with Figure F.2 in the 2023 analysis. This establishes a valid baseline for assessing the treatment effects.

Post-treatment, properties that received a control letter exhibited higher compliance rates than properties receiving no letter, as seen in panel (a) of Figure I.8. This mirrors the 2023 results, where panel (a) of Figure F.3 demonstrated a clear increase in payments within the first two months after the intervention. For 2024, our data showed compliance increased by approximately 8 percentage points in the first month, with a substantially smaller effect in the second month. Panel (b) of Figure I.8 confirms these patterns when analyzing pooled data, in line with the findings shown in panel (b) of Figure F.3.

In conclusion, the results of this section validated and replicated the findings from the 2023 analysis.

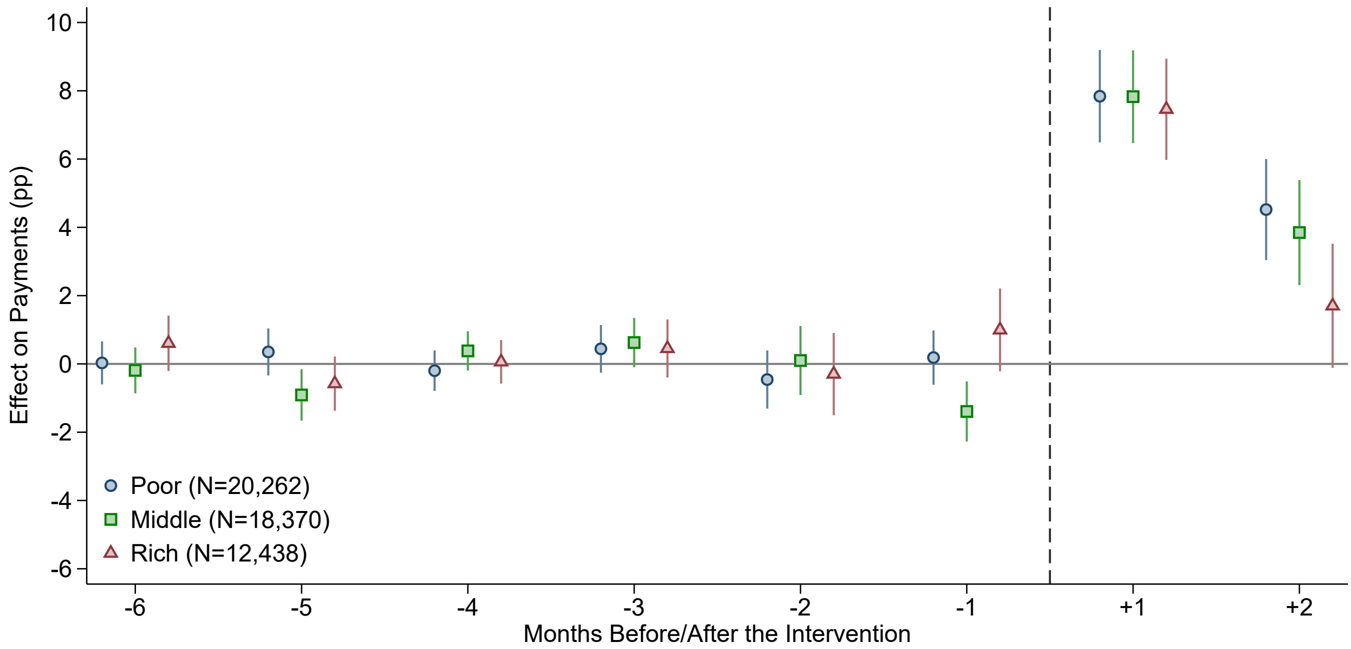
Figure I.7: Difference in Payment Rates between Control and No-Letter: 2024



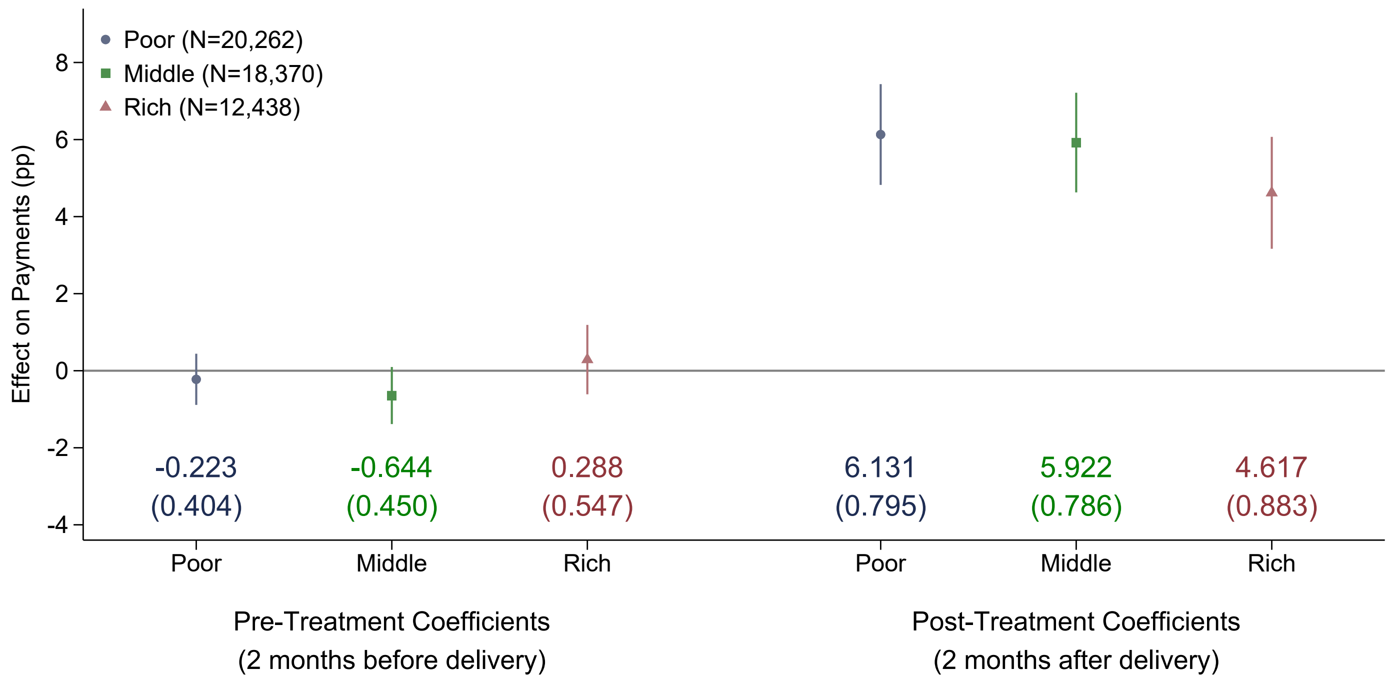

*Note*: This figure compares the likelihood of paying the monthly bill between treatment and control, normalized by the average pre-treatment difference. We estimated the coefficient for each monthly bill in separate regressions. The sample includes residential and non-residential properties and excludes units that made payments for 2024 before we sent the letters.

Figure I.8: The Effect of Receiving a Letter on Own Tax Compliance: 2024

(a) Event-Study Analysis (Monthly Level)




(b) Pre- and Post-Treatment Effects (Bimester level)




*Note*: This figure compares the likelihood of paying the monthly property tax bill between control and the no letter group. The control group received a letter with information solely about how the reform changed the household's own tax rate, while the no letter group received none. Panel (a) shows the dynamic effect of the letters, while panel (b) shows the two months pooled effects of the letter (pre- and post-treatment). 'Poor' (in blue) denotes households with properties valued at AR$ 750K or less, 'Middle' (in green) denotes properties valued between AR$ 750K and AR$ 1.5M, and 'Rich' (in red) denotes properties valued more than AR$ 1.5M. We estimated the coefficient for each monthly bill in separate regressions, including the 90% confidence interval for each group. Clustered standard errors for the pooled estimates in parenthesis. The sample includes residential and non-residential properties and excludes units that made payments for 2024 before we sent the letters. We controled for the past 12 months payments in each pre-treatment regression, and for the 2023 monthly payments in each post-treatment regression. Additionally, we included controls for property valuation (in logs), a residential dummy, dummies indicating whether the property was an "always payer" or "never payer" in the previous year, and an indicator for properties that made an annual payment in the previous year. We included dummies equal to one for missing controls to retain all observations in the regressions. Standard errors clustered at the individual taxpayer level.

# J Counterfactual Analysis

## J.1 Additional Details of the Counterfactual Analysis

This section explains the structure and interpretation of the detailed results presented in Table J.1, which summarizes key outcomes of the counterfactual analysis under different scenarios. Each panel represents a specific policy context: the pre-reform scenario, the post-reform scenario without behavioral responses, and the post-reform scenario with behavioral responses that vary by group.

The table is structured along three dimensions: columns, rows, and panels. The columns report key variables. $L_j^s$ represents the average tax liability for each group $j$ in scenario $s$, while $C_j^s$ denotes the proportion of properties that comply with tax payments on time in the given scenario. The total effective taxes paid by each group, $T_j^s$, is computed as the product of the average tax liability and the compliance rate.

The rows classify households into three groups based on property valuation: *poor* (low-valuation properties), *middle* (middle-valuation properties), and *rich* (high-valuation properties). Additionally, the table includes a weighted sum, which averages the values across groups based on population shares, and a measure of the rich-poor gap, which captures the difference between the rich and the poor in each scenario. Specifically, the rich-poor gap in taxes paid is computed by normalizing the effective taxes paid by the average valuation of each group.

The panels distinguish between different policy contexts. Panel (a) presents the pre-reform scenario, reflecting the status quo before any policy changes. The weighted sum in this panel represents total per-capita tax revenue, while the rich-poor gap provides a baseline measure of inequality. Panel (b) considers the post-reform scenario assuming no behavioral responses, isolating the mechanical effect of tax rate changes without accounting for adjustments in compliance. In contrast, panels (c), (d), and (e) incorporate behavioral responses, differentiating cases where adjustments in compliance occur only among the poor, only among the rich, or in both groups. These estimates include confidence intervals (e.g., [49.9%, 52.3%]), derived from 5,000 bootstrap iterations.

For our analysis, the most relevant elements of the table are the rich-poor gap and the weighted sum of taxes paid. The rich-poor gap quantifies the difference in taxes paid between the rich and the poor, serving as a key measure of inequality before and after the reform. Meanwhile, the weighted sum represents the average tax revenue per person, which is crucial for understanding the reform's impact on overall tax collection. By comparing these values across the pre-reform and post-reform scenarios, with and without behavioral responses, we can clearly see how changes in compliance and tax rates affect both revenue and inequality.

Table J.1: Counterfactual Analysis: No Behavioral Responses vs Behavioral Responses

| | (1) | (2) | (3) |
|---|---|---|---|
| | $L_j^0$ | $C_j^0$ | $T_j^0$ |
| Panel (a): Pre-Reform | | | |
| $j = \text{Poor}^{(i)}$ | $14,161$ | $46.6\%$ | $6,597$ |
| $j = \text{Middle}^{(ii)}$ | $22,159$ | $49.3\%$ | $10,929$ |
| $j = \text{Rich}^{(iii)}$ | $48,071$ | $50.3\%$ | $24,156$ |
| Weighted $\text{Sum}^{[(i)+(ii)+(iii)]}$ | $26,090$ | $48.6\%$ | $12,843$ |
| Rich-Poor $\text{Gap}^{[(iii)-(i)]}$ | $33,910$ | $3.7\%$ | $-0.313$ |
| | $L_j^C$ | $C_j^C$ | $T_j^C$ |
| Panel (b): Post-Reform without BR | | | |
| $j = \text{Poor}^{(i)}$ | $7,718$ | $46.6\%$ | $3,595$ |
| $j = \text{Middle}^{(ii)}$ | $22,159$ | $49.3\%$ | $10,929$ |
| $j = \text{Rich}^{(iii)}$ | $56,397$ | $50.3\%$ | $28,340$ |
| Weighted $\text{Sum}^{[(i)+(ii)+(iii)]}$ | $25,915$ | $48.6\%$ | $12,843$ |
| Rich-Poor $\text{Gap}^{[(iii)-(i)]}$ | $48,679$ | $3.7\%$ | $0.5$ |
| | $L_j^{AP}$ | $C_j^{AP}$ | $T_j^{AP}$ |
| Panel (c): Post-Reform, BR from *poor* | | | |
| $j = \text{Poor}^{(i)}$ | $7,718$ | $53.6\%$ | $4,137$ |
| | $-$ | $[50.3\%, 56.6\%]$ | $[3,880, 4,367]$ |
| $j = \text{Middle}^{(ii)}$ | $22,159$ | $49.3\%$ | $10,929$ |
| | $-$ | $-$ | $-$ |
| $j = \text{Rich}^{(iii)}$ | $56,397$ | $50.3\%$ | $28,340$ |
| | $-$ | $-$ | $-$ |
| Weighted $\text{Sum}^{[(i)+(ii)+(iii)]}$ | $25,915$ | $51.2\%$ | $13,044$ |
| | $-$ | $[49.9\%, 52.3\%]$ | $[12,949, 13,129]$ |
| Rich-Poor $\text{Gap}^{[(iii)-(i)]}$ | $48,679$ | $-3.3\%$ | $0.387$ |
| | $-$ | $[-6.3\%, 0.0\%]$ | $[0.339, 0.441]$ |
| | $L_j^{AR}$ | $C_j^{AR}$ | $T_j^{AR}$ |
| Panel (d): Post-Reform, BR from *rich* | | | |
| $j = \text{Poor}^{(i)}$ | $7,718$ | $46.6\%$ | $3,595$ |
| | $-$ | $-$ | $-$ |
| $j = \text{Middle}^{(ii)}$ | $22,159$ | $49.3\%$ | $10,929$ |
| | $-$ | $-$ | $-$ |
| $j = \text{Rich}^{(iii)}$ | $56,397$ | $45.2\%$ | $25,473$ |
| | $-$ | $[41.0\%, 49.4\%]$ | $[23,149, 27,862]$ |
| Weighted $\text{Sum}^{[(i)+(ii)+(iii)]}$ | $25,915$ | $47.2\%$ | $12,079$ |
| | $-$ | $[46.1\%, 48.3\%]$ | $[11,461, 12,715]$ |
| Rich-Poor $\text{Gap}^{[(iii)-(i)]}$ | $48,679$ | $-1.4\%$ | $0.373$ |
| | $-$ | $[-5.5\%, 2.8\%]$ | $[0.271, 0.479]$ |
| | $L_j^{A}$ | $C_j^{A}$ | $T_j^{A}$ |
| Panel (e): Post-Reform, BR from *poor* and *rich* | | | |
| $j = \text{Poor}^{(i)}$ | $7,718$ | $53.6\%$ | $4,137$ |
| | $-$ | $[50.3\%, 56.6\%]$ | $[3,880, 4,367]$ |
| $j = \text{Middle}^{(ii)}$ | $22,159$ | $49.3\%$ | $10,929$ |
| | $-$ | $-$ | $-$ |
| $j = \text{Rich}^{(iii)}$ | $56,397$ | $45.2\%$ | $25,473$ |
| | $-$ | $[41.0\%, 49.4\%]$ | $[23,149, 27,862]$ |
| Weighted $\text{Sum}^{[(i)+(ii)+(iii)]}$ | $25,915$ | $49.8\%$ | $12,280$ |
| | $-$ | $[48.1\%, 51.4\%]$ | $[11,652, 12,918]$ |
| Rich-Poor $\text{Gap}^{[(iii)-(i)]}$ | $48,679$ | $-8.4\%$ | $0.260$ |
| | $-$ | $[-13.5\%, -2.9\%]$ | $[0.147, 0.379]$ |

*Note*: Estimates made using Administrative Tax Data from Tres de Febrero. Panel (a) shows the pre-reform estimates. Panel (b) shows the post-reform estimates without behavioral responses. Panel (c) shows the post-reform estimates with behavioral responses for poor households only. Panel (d) shows the post-reform estimates with behavioral responses for rich households only. Panel (e) shows the post-reform estimates with behavioral responses for both groups. Additionally, to compute the rich-poor gap of the paid taxes we divided the effective taxes paid by each group for the average valuation of each group. 90% confidence intervals obtained from a 5,000 repetitions bootstrap.

## J.2 Relative Contributions of Own-Rate and Cross-Rate Effects to the Total Behavioral Response

Our framework provides a way to assess the relative contributions of the cross-rate and own-rate effects to the overall behavioral response to the tax reform. The public finance literature typically only accounts for the own-rate effects (Saez and Zucman, 2023), and this analysis offers a measure of the quantitative importance of these additional effects and the consequences of ignoring them. An advantage of our setup is that it allows us to measure and compare own and social effects within a unified framework and with the same metric.

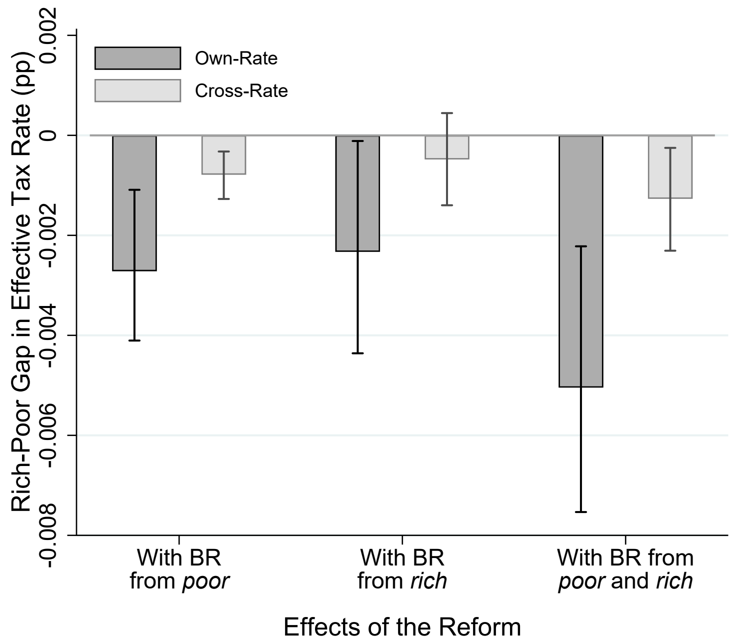
We now present the methodology used to separate the effects of changes in taxpayers' own rates (own-rate effects) from changes in others' rates (cross-rate effects) and their contribution to compliance and revenue outcomes.

Figure J.1 illustrates how these effects drive divergences in these outcomes. Panel (a) of Figure J.1 shows that effective progressivity—adjusted for compliance changes—is significantly lower. This reduction is primarily driven by own-rate effects, with cross-rate effects contributing to a smaller degree. The direction and magnitude of both the own-rate and cross-rate effects are similar for the poor (low property value bracket) and the rich (high property value bracket), with both exhibiting a negative impact on compliance for these groups. The combined result shows a significant decline in the progressivity of the reform.

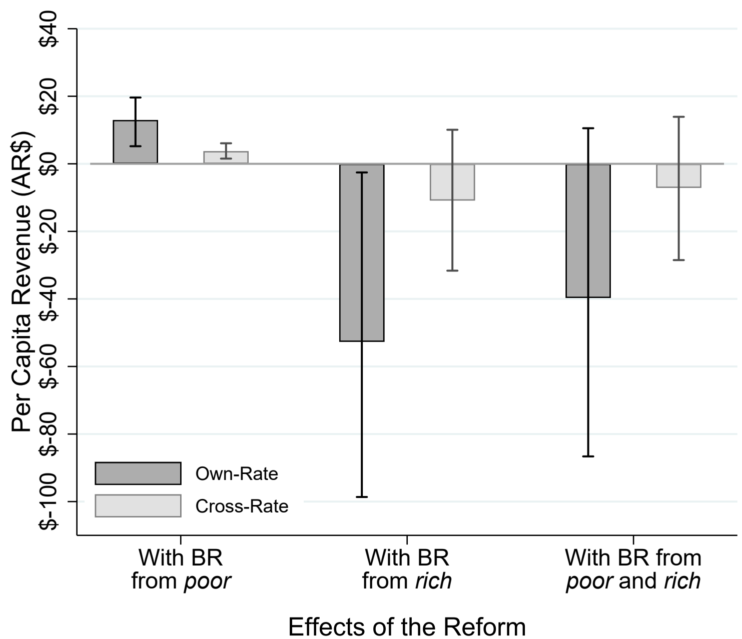
In contrast, panel (b) of Figure J.1 highlights that revenue increases for poor households through both the own-rate and cross-rate channels. Conversely, revenue declines for rich households through these same channels, though the cross-rate effect for rich properties is not statistically significant. When combining the effects across both groups, the negative impact of the own-rate and cross-rate effects tend to dominate. Nevertheless, the combined result lacks statistical precision at conventional significance levels.

Figure J.1: Own-Rate vs Cross-Rate: Rich-Poor Gap and Per Capita Revenue

(a) Rich-Poor Gap: Decomposition




(b) Per Capita Revenue: Decomposition




*Note*: Estimates made using Administrative Tax Data from Tres de Febrero. Panel (a) shows the decomposition by own-rate and cross-rate of the rich-poor gap, assuming different scenarios of behavioral responses. Panel (b) shows the decomposition by own-rate and cross-rate of the per capita revenue effects, assuming once again different scenarios of behavioral responses. 90% confidence intervals obtained from a 5,000 repetitions bootstrap.

# K Forecast Survey

This section presents the findings from a forecast survey conducted with 39 experts to evaluate their predictions about the effects of the progressive tax reform on tax compliance. The survey provided participants with comprehensive information about the experimental design, including details about the tax reform, randomized messaging interventions, and the primary outcome measure: changes in tax compliance across valuation brackets. Experts were asked to estimate the expected effects and indicate their confidence in these predictions (the full questionnaire can be found in Appendix M.3).

The sample of experts consisted primarily of senior academics, with 27 Professors, 8 Assistant Professors, and 4 researchers and postdoctoral fellows, as shown in Figure K.1, panel (a). Most participants had backgrounds in Economics, with a smaller number coming from Business (panel (b)). Notably, approximately 85% of the experts reported previous research experience in taxation, while 40% conducted research on preferences for redistribution, providing a strong foundation for making informed predictions (panels (c) and (d)).

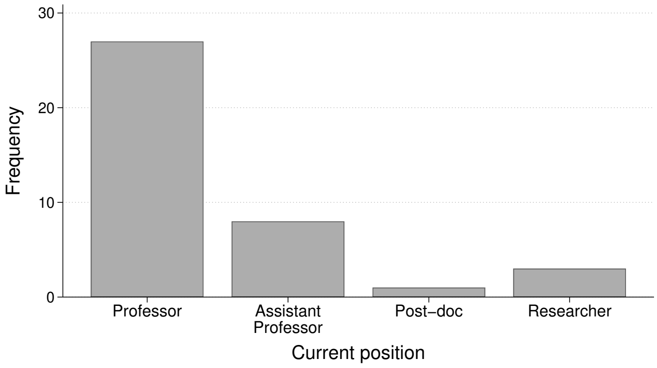
Figure K.2 shows that experts' predictions revealed significant variability across valuation groups. For poor properties, the majority expected a positive compliance response, attributing this to increased perceptions of fairness arising from the reform. However, for middle properties, predictions were more mixed, with some experts expecting no change and others anticipating increases in compliance. For rich properties, opinions varied significantly: while some predicted slight improvements in compliance, others forecasted decreases or no changes in behavior, reflecting uncertainty about the role of fairness perceptions in this group.

The survey revealed substantial dispersion in the predicted treatment effects, both in terms of magnitude and direction. Only a small number of experts predicted outcomes that aligned closely with the actual experimental findings. Furthermore, the majority of experts expressed low confidence in their predictions, describing themselves as only "slightly confident" or "somewhat confident" across all valuation brackets.

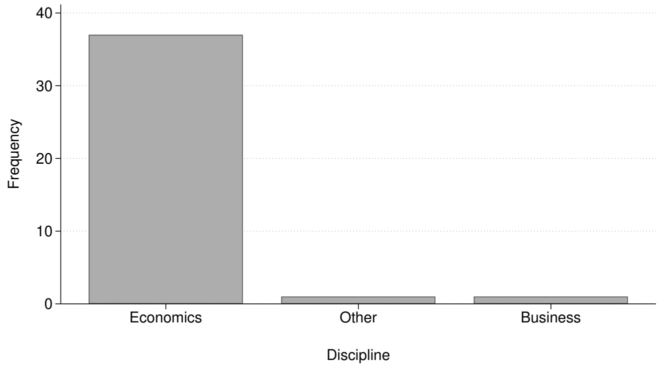
Overall, the survey results underscore the distinctive nature of the experimental findings. Although experts accurately anticipated positive compliance effects among poor properties, they significantly overestimated the strength of the treatment's impact. In contrast, the responses for medium- and high-value properties were less predictable, with expert forecasts diverging from the observed results. These findings highlight the importance of empirical evidence in capturing the effects of informing taxpayers about progressivity reforms.

Figure K.1: Descriptive Statistics: Forecast Survey

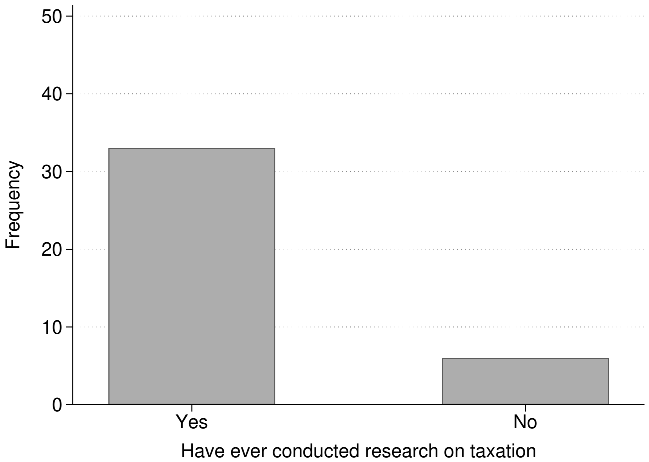
(a) Current Positions of Participants



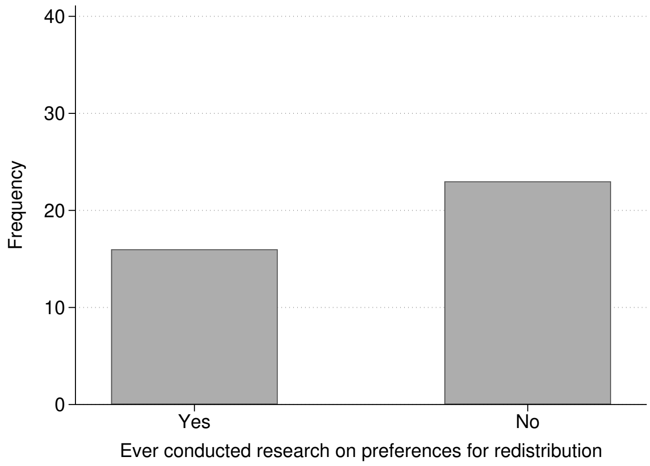

(b) Disciplines of Participants




(c) Research on Taxation



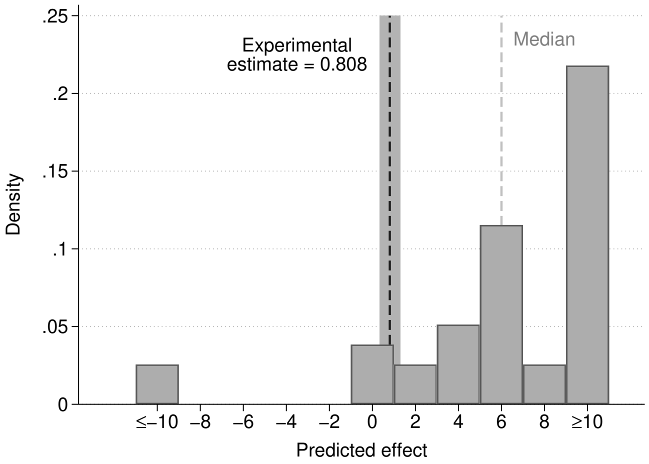

(d) Research on Redistribution



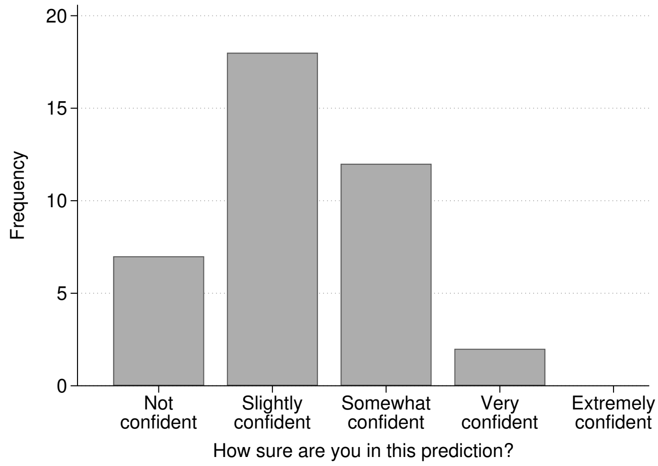

*Note*: These figures summarize the descriptive statistics of the expert sample from the forecast survey. Panel (a) shows the distribution of participants' current positions, while panel (b) presents the disciplines they specialize in. Panels (c) and (d) display the proportion of participants who have conducted research on taxation and preferences for redistribution, respectively.

Figure K.2: Predictions and Confidence Levels: Forecast Survey

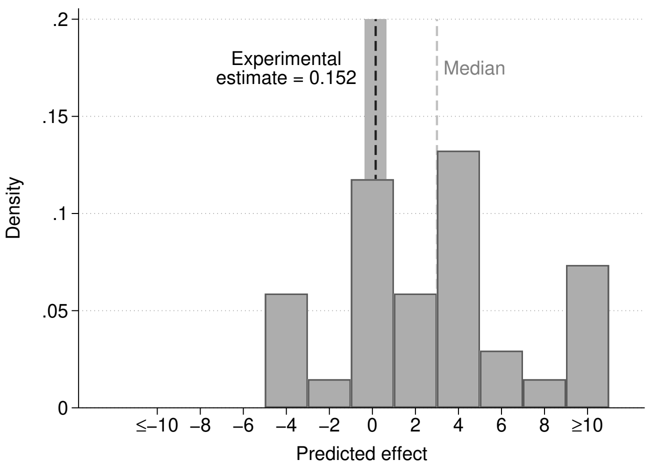
(a) Predictions: Poor-Households



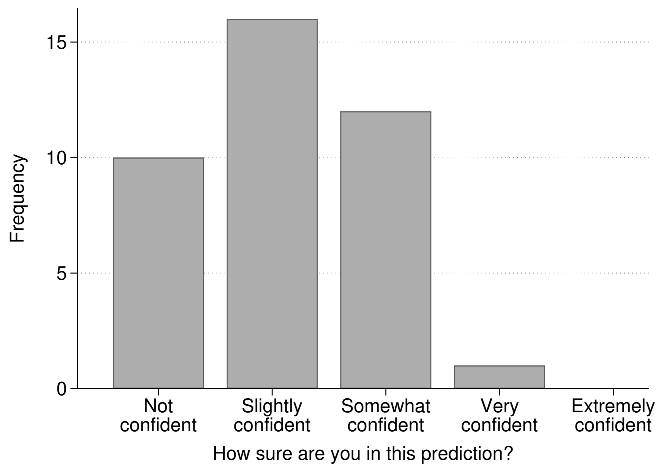

(b) Confidence: Poor-Households



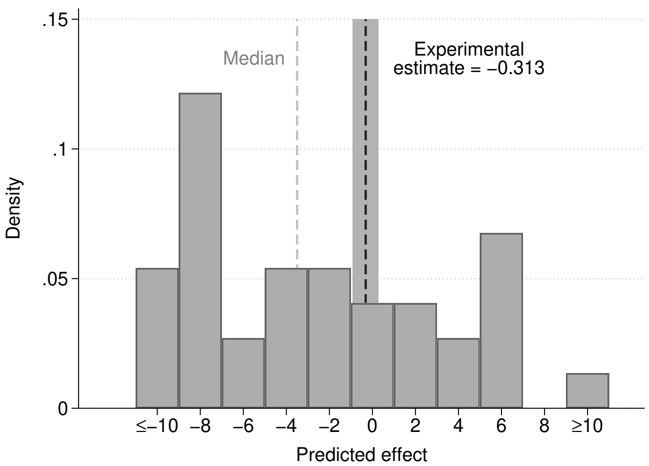

(c) Predictions: Middle-Households



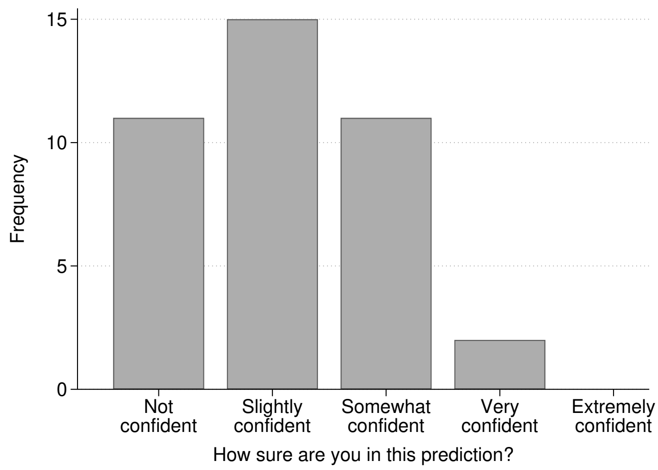

(d) Confidence: Middle-Households




(e) Predictions: Rich-Households




(f) Confidence: Rich-Households




*Note*: Panels (a), (c), and (e) show the distribution of predictions for poor, middle, and rich households, respectively. Panels (b), (d), and (f) illustrate the levels of confidence experts had in their predictions for each valuation group. The results reflect the expectations and uncertainty regarding the effects of the tax reform on compliance across property valuations. Shaded areas around the experimental estimate correspond to the 90% confidence intervals.

# L Sample Letters

## L.1 Sample Letters: 2023 Field Experiment (in English)

This section provides an English translation of the sample letters used in the 2023 field experiment, detailing the messaging strategies and content included in the communications. Both letters contained procedural information to facilitate taxpayer understanding and compliance, which included a detailed breakdown of monthly tax amounts, specific payment deadlines, and contact information for a helpline and email support. Importantly, our intervention through letters was designed to be subtle enough not to generate experimenter demand effects.

On the one hand, the control letter offered a straightforward summary of the tax changes, focusing exclusively on the recipient's situation. Crucially, the control letter avoided references to the reform's broader redistributive goals, maintaining a neutral tone that limited the information to immediate personal implications.

On the other hand, the treatment letter included the same personalized details as the control letter, but expanded its scope to emphasize the progressive nature of the tax reform. It explicitly framed the changes as part of a broader initiative to improve the progressivity of the tax system. The letter detailed not only how the reform affected the recipient, but also highlighted its impact on other taxpayer groups. This was designed to affect perceptions of fairness and encourage compliance. To enhance clarity and engagement, the treatment letter incorporated graphical elements that visually depicted changes in tax rates across different property valuation brackets.

The differences in information between the control and treatment letters were central to the experimental design. By comparing the responses to these letters, the study aimed to isolate the impact of emphasizing fairness and redistribution on taxpayer behavior.

**Municipality of Tres de Febrero**

# General Services Tax

**MR/MRS:** S[redacted]O ALBERTO J.

**ADDRESS:**
[redacted]L 399 - JOSÉ LEÓN SUÁREZ 1655

**LOCATION:**
[redacted] 985

**ACCOUNT:** [redacted]

**CADASTRAL NOMENCLATURE**
Cir: 4 [redacted]1

**RATE:** 0.0121 **CATEGORY:** 1

**VALUATION:** $682.124

## Monthly calculation

| | | |
|---|---|---|
| GENERAL SERVICES | $ | 1.100,50 |
| PROGRESSIVITY CORRECTION | $ | -330,15 |
| ADD. SECURITY | $ | 400,00 |
| ADD. HEALTH | $ | 320,00 |
| ADD. CEDEM MAINTENANCE | $ | 232,00 |
| ADD. VOLUNTEER FIREFIGHTERS | $ | 23,20 |
| PUBLIC LIGHTING ANTICIPATION | $ | -877,00 |
| **TOTAL:** | **$** | 868,55 |

## ANNUAL PAYMENT

TWO MONTHS DISCOUNT

**DUE DATE** 15/02/2023
**AMOUNT** $ 8.685,50

Scan the QR and pay through Mercado Pago

2000001[redacted]30215

## FIRST THREE MONTHS PAYMENTS

If you made the annual payment, waive the following fees

### JANUARY

DUE DATE 15/02/2023 **AMOUNT** $ 868,55

2023001[redacted]30215

### FEBRUARY

DUE DATE 15/02/2023 **AMOUNT** $ 868,55

2023002[redacted]30215

### MARCH

DUE DATE 15/03/2023 **AMOUNT** $ 868,55

2023003[redacted]30315

This is the annual invoice for the TSG and for the first three months of 2023. It is the only time in the year that you will receive it printed at home. Keep in mind that you can download it and pay for it from **mi3f .tresdefebrero.gov .ar.**

Remember that if instead of paying the annual fee you do it month by month and you have no debt, you will receive a bonus for July's fee.

**Municipality of Tres de Febrero**

## LESS TAXES, MORE WORK

At Tres de Febrero we are on the side of those who produce and generate employment, that is why we take measures that benefit those who invest.

### FREE ABILITATIONS

For any sector or size.

Shops

SMEs and industries

Commercial vehicles

### NO OBSTRUCTIONS OR BUREAUCRACY

We eliminated more than 400 fees and simplified 30 thousand procedures per year.

### WE HELP THOSE WHO GIVE WORK

Credits and tax benefits for those who hire Tres de Febrero workers.

## SIMPLER FOR YOU

All your experience with the Municipality in one place

**With your user you can:**

- Load urban reports from your neighborhood
- Pay your municipal taxes
- Carry out procedures online

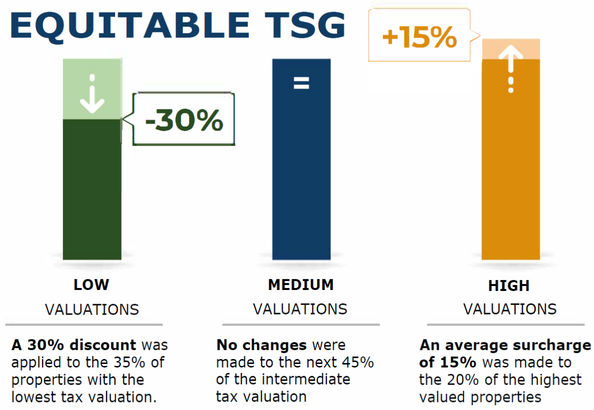
Register for one time only at **mi3f.tresdefebrero.gov.ar**

## FAIRER AND MORE EQUITABLE TSG

The TSG **increased below inflation** as in recent years and also now it is **fairer and more equitable**.

We applied a 30% discount on the variable component of the TSG for properties with the lowest tax valuation, and an average increase of 15% for those with the highest value.

**Based on your tax valuation, your TSG increased in relation to the rest.**




### TELL US WHAT YOU THINK ABOUT OUR TAX POLICY

Knowing what you thought helps us improve.
Scan the QR or go to opinion.tresdefebrero.gov.ar/tasas2023

Figure L.1: Sample of Control and Treatment Messages: Middle Households

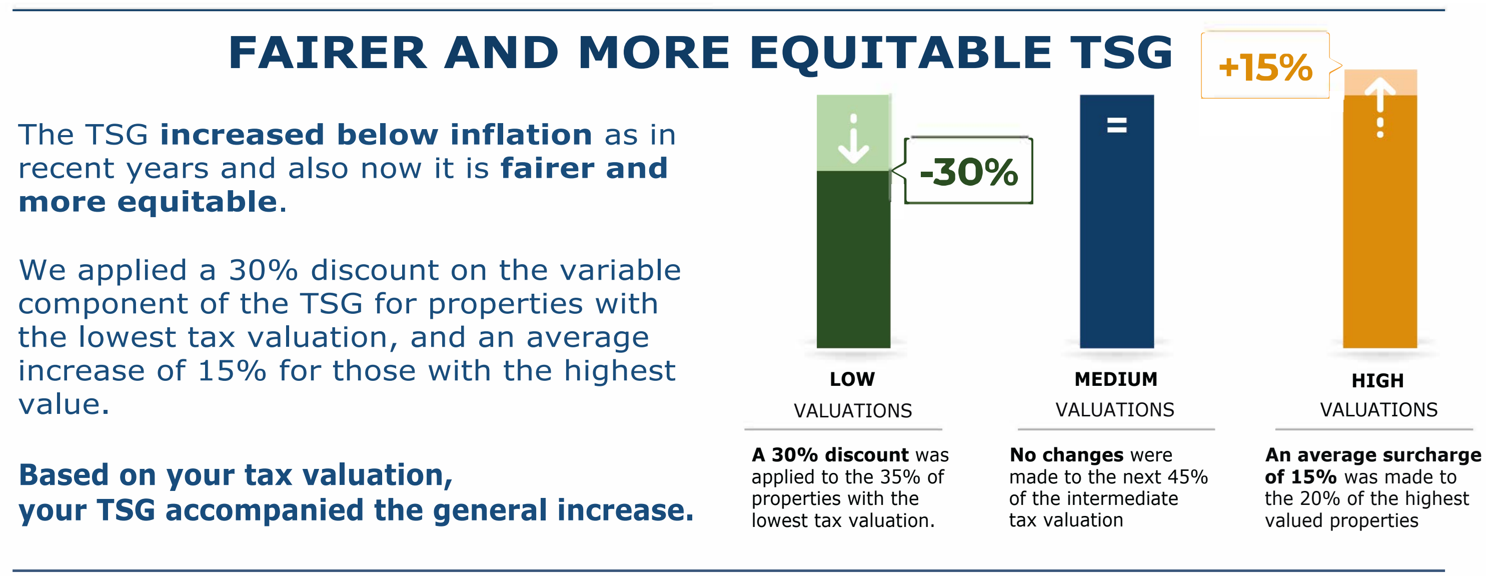
(a) Control Letter Message


FAIRER AND MORE EQUITABLE TSG

The TSG **increased below inflation** as in recent years and is also now **fairer and more equitable**.

**Based on your tax valuation, your TSG accompanied the general increase.**

(b) Treatment Letter Message




*Note*: English translation (from Spanish) of the main pieces of information from the mailers. This text was contained on the second page of the mailer.

Figure L.2: Sample of Control and Treatment Messages: Rich Households

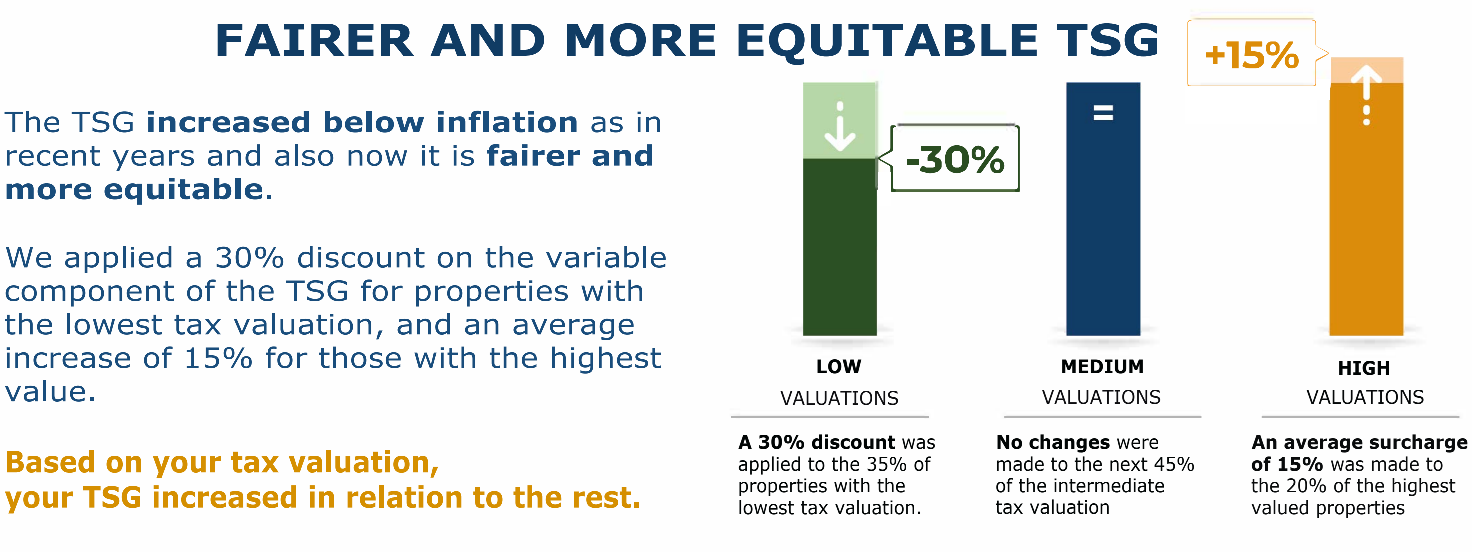
(a) Control Letter Message

FAIRER AND MORE EQUITABLE TSG

The TSG **increased below inflation** as in recent years and is also now **fairer and more equitable**.

**Based on your tax valuation, your TSG increased in relation to the rest.**

(b) Treatment Letter Message




*Note*: English translation (from Spanish) of the main pieces of information from the mailers. This text was contained on the second page of the mailer.

## L.2 Sample Letters: 2024 Field Experiment Replication (in English)

This section provides an English translation of the sample letters used in the 2024 field experiment, which closely resembled those from 2023 with only minor refinements. The treatment letters for both years were substantively identical, emphasizing the progressive nature of the tax reform and its broader redistributive goals. These letters detailed not only the recipient's tax changes but also the reform's impact on other taxpayer groups, accompanied by graphical elements that visually represented the adjustments across property valuation brackets.

However, the control letter differed between 2023 and 2024. In 2023, the control letter informed recipients whether their tax rate had decreased, remained unchanged, or increased, but without specifying the magnitude of the change. In 2024, the control letter provided additional detail by explicitly stating the exact percentage change in the recipient's tax rate. Additionally, the 2024 control letter included a small graphical representation illustrating how the reform affected tax rates for different taxpayer brackets (poor, middle, and rich properties).

This modification of the 2024 control letter provided an opportunity to examine whether detailed quantitative information and visual representations influenced taxpayer behavior differently compared to the categorical information provided in 2023. By maintaining identical treatment letters while enhancing the control letter's content, the 2024 experiment enabled us to isolate the effects of different presentations of the progressive reform on tax compliance through social preference channels.

Municipality of
Tres de Febrero

# Tasa por Servicios Generales

**ACCOUNT:**

**MR/MRS:**
**IN CHARGE:**
**LOCATION:**
**ADDRESS:**
**CADASTRAL NOMENCLATURE**
Cir: Sec: Cod Mz: Mz: Pc: UF:
**CATEGORY:**
**MODULE VALUE:** **RATE:**
**VALUATION:**

## Monthly calculation

**MONTHLY TOTAL: $**

## Amount owed to date

PREVIOUS PERIODS:
CURRENT *MORATORIAS*:

**TOTAL: $**

The debt mentioned is for informational purposes. Go to **tresdefebrero.gov.ar/tsg** to find out the payment facilities available.

## Payment methods

Or come from 9 a.m. to 3 p.m. at:
- **Municipality** (CAV hall): Alberdi 4840 - Caseros
- **Ciudadela Municipal Headquarters** : Av. Rivadavia y San Martín
- **Neighbor care center**

Remember that you can also pay at **tasas.tresdefebrero.gov.ar** and join the automatic debit through **mi3f.tresdefebrero.gov.ar**

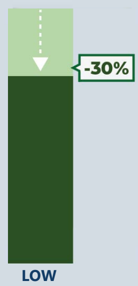
Due Date AMOUNT $

**Municipality of Tres de Febrero**

## TSG EVEN FAIRER AND EQUITABLE

From January 2024, the General Services Rate will be even more fair and equitable compared to last year.

**Due to your tax valuation, your TSG decreased in relation to the rest.**




A **30% discount** to the variable component of the General Service Rate was applied to the 35% of properties with the lowest tax valuation.

## SIMPLER FOR YOU

All your experience with the Municipality in one place

### With your user you can:

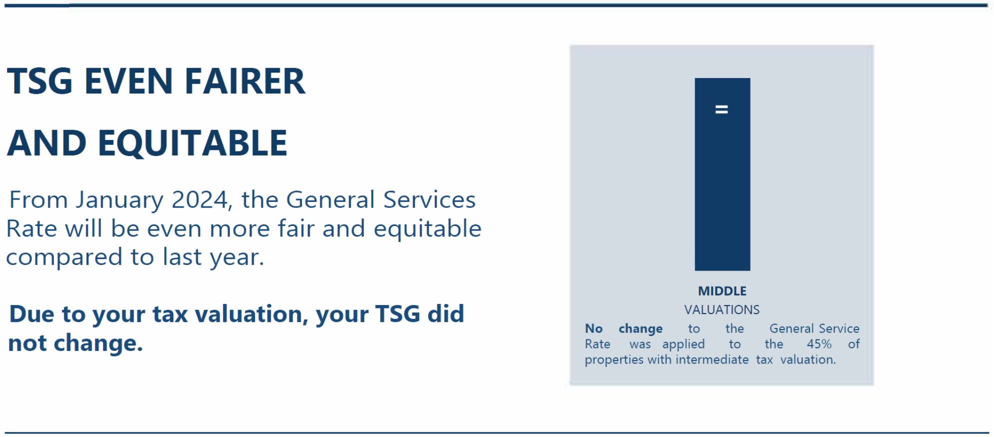
- **Pay your municipal taxes**
- **Carry out online procedures and registrations**
- **Load urban reports from your neighborhood**

Scan the **QR** to link your **TSG** account.

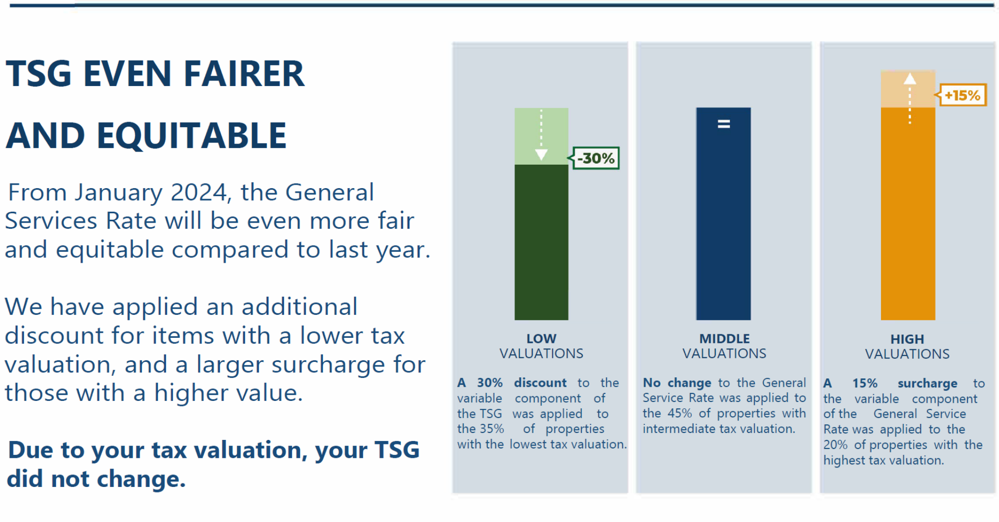
Remember that you can view all your **invoices** and join the **automatic debit** through **mi3f.tresdefebrero.gov.ar**

Figure L.3: Sample of Control and Treatment Messages: Middle Households, 2024 Experimental Design

(a) Control Letter Message



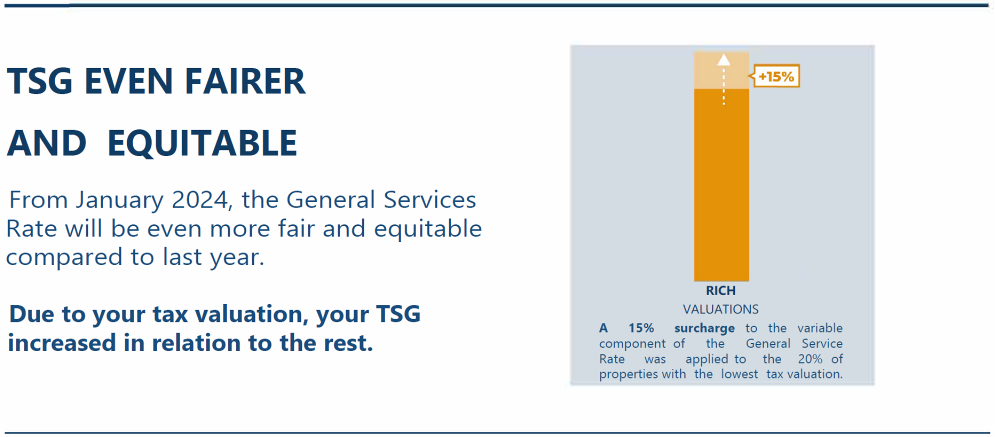

(b) Treatment Letter Message



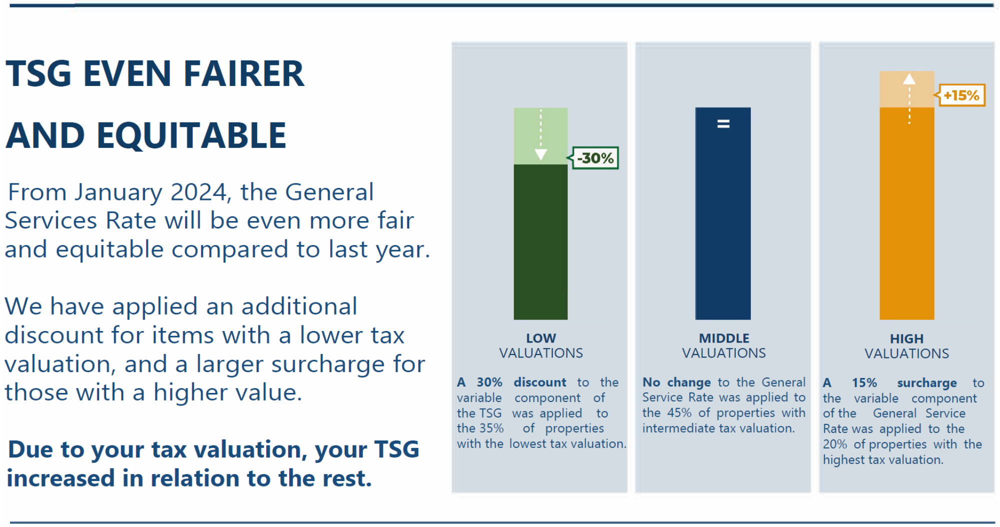

*Note*: English translation (from Spanish) of the main pieces of information from the mailers. This text was contained on the second page of the mailer.

Figure L.4: Sample of Control and Treatment Messages: Rich Households, 2024 Experimental Design

(a) Control Letter Message




(b) Treatment Letter Message




*Note*: English translation (from Spanish) of the main pieces of information from the mailers. This text was contained on the second page of the mailer.

# M Survey Questionnaires

## M.1 Email Invitation to Taxpayer Survey (English Translation)

Gmail

**What do you think about rates and taxes in Tres de Febrero?**
1 message

**Tres de Febrero** <ANONYMIZED> To:
ANONYMIZED

MUNICIPALIDAD

JOIN THE MUNICIPALITY WHATSAPP

**Hello ANONYMIZED!**

In Tres de Febrero, we are working to make taxes and rates fairer and more equitable. That's why we want to hear your opinion on the measures we have implemented in the municipality.

We only ask for three minutes to complete a short survey and share your thoughts.

**CLICK HERE TO COMPLETE**

Remember, you can pay your taxes, submit reports, or complete procedures online by registering on Mi3F.

**Municipality of Tres de Febrero**

**LESS TAXES, MORE WORK**

At Tres de Febrero we are on the side of those who produce and generate employment, that is why we take measures that benefit those who invest.

**FREE ABILITATIONS**

For any sector or size.

Shops — SMEs and industries — Commercial vehicles

**NO OBSTRUCTIONS OR BUREAUCRACY**

We eliminated more than 400 fees and simplified 30 thousand procedures per year.

**WE HELP THOSE WHO GIVE WORK**

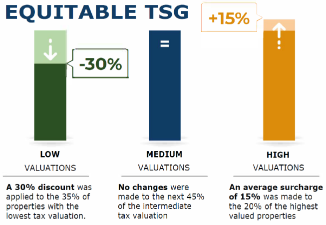
Credits and tax benefits for those who hire Tres de Febrero workers.

**FAIRER AND MORE EQUITABLE TSG**

The TSG **increased below inflation** as in recent years and also now it is **fairer and more equitable**.

We applied a 30% discount on the variable component of the TSG for properties with the lowest tax valuation, and an average increase of 15% for those with the highest value.

**Based on your tax valuation, your TSG increased in relation to the rest.**




**CLICK HERE AND SHARE THE INFORMATION**

Unsubscribe Unsubscribe link

## M.2 Complementary Taxpayer Survey (English Translation)

3F MUNICIPALIDAD

Welcome!

Thank you very much for participating in this brief survey, which will take less than 3 minutes and will help us make 3F the place we all want.

We want to know your opinion on the administration of municipal rates and taxes and about some measures applied during the last year.

## Additional Information - Control Poor

### FAIRER AND MORE EQUITABLE TSG

The TSG **increased below inflation** as in recent years and is also now **fairer and more equitable**.

**Based on your tax valuation, your TSG decreased in relation to the rest.**

## Additional Information - Control Middle

### FAIRER AND MORE EQUITABLE TSG

The TSG **increased below inflation** as in recent years and is also now **fairer and more equitable**.

**Based on your tax valuation, your TSG accompanied the general increase.**

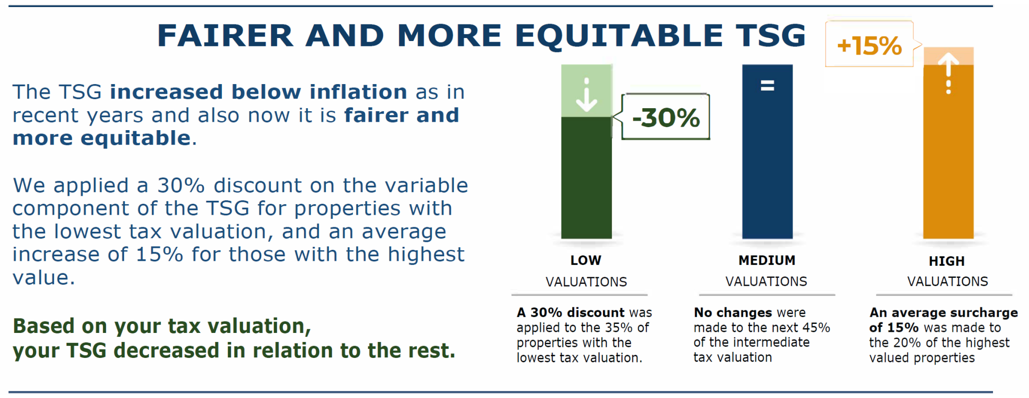
## Additional Information- Control Rich

### FAIRER AND MORE EQUITABLE TSG

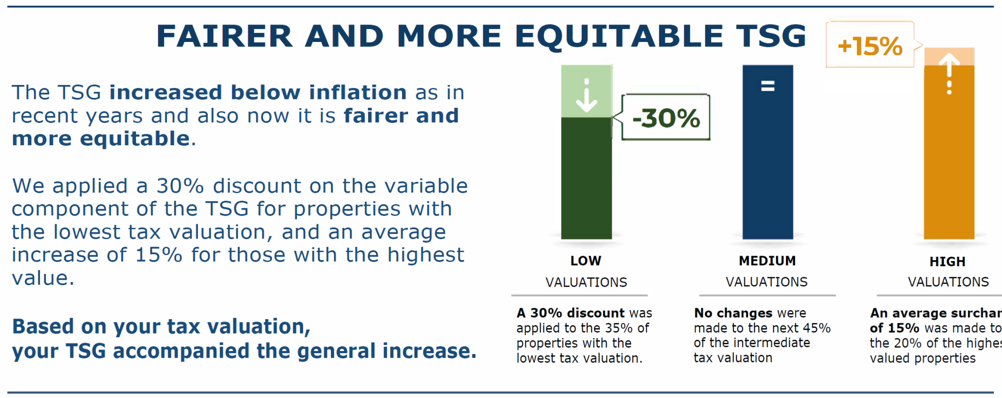
The TSG **increased below inflation** as in recent years and is also now **fairer and more equitable**.

**Based on your tax valuation, your TSG increased in relation to the rest.**

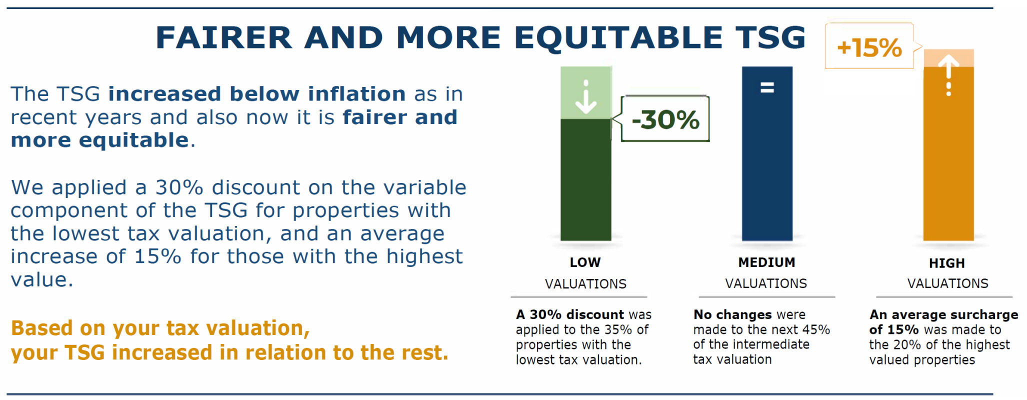
## Additional Information - Treatment Poor




## Additional Information - Treatment Middle




## Additional Information - Treatment Rich



Do you live in Tres de Febrero?

- ○ Yes
- ○ No

---

Did you know about any of the following measures recently taken in Tres de Febrero?

- ☐ Free and express commercial permits
- ☐ Credits to businesses for hiring employees and making improvements
- ☐ Elimination of certain rates, taxes, and administrative procedures
- ☐ TSG: discounts for low-valuation properties and additional charges for high-valuation properties
- ☐ No, I haven't heard of any

---

Were you benefited by any of these policies?

- ☐ Yes, my household has benefited
- ☐ Yes, my business/industry has benefited
- ☐ No, I have not seen any benefit from these measures

---

Are you satisfied with the payment options available for municipal rates and taxes?

- ◯ Yes
- ◯ No, I would like other options

---

What other payment options would you like for your municipal taxes?

---

Some people think the government at its various levels (national, provincial, municipal) should not be concerned with the differences between the rich and the poor. Others think the government should do everything it can to reduce inequality. What do you think about this issue?

It should not be concerned about inequality — It should do everything to resolve inequality

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| ◯ | ◯ | ◯ | ◯ | ◯ | ◯ | ◯ | ◯ | ◯ | ◯ | ◯ |

How fair do you think the current distribution of municipal tax rates is between people with more and fewer resources—that is, how much each of these groups pays?

Very unfair Very fair

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ |

Please tell us how much you agree or disagree with the following statements about the General Services Tax (TSG):

Lower the TSG for the poor (the 35% of households with the lowest valuations). There would be less money to provide services to the community.

Strongly disagree Strongly agree

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ |

Raise the TSG for the rich (the 20% of households with the highest valuations). There would be more money to provide services to the community.

Strongly disagree — Strongly agree

0 1 2 3 4 5 6 7 8 9 10

Lower the TSG for the poor (the 35% of households with the lowest valuations) and at the same time raise the TSG for the rich (the 20% of households with the highest valuations). The money used to provide services to the community will remain constant.

Strongly disagree — Strongly agree

0 1 2 3 4 5 6 7 8 9 10

---

Are you generally satisfied with the services provided by the municipality?

Very dissatisfied — Very satisfied

0 1 2 3 4 5 6 7 8 9 10

---

What could the municipality improve regarding the collection of rates and taxes?

What is your overall opinion of the municipal administration and what aspects do you think should be improved?

## M.3 Expert Forecast Survey

*This survey involves no more than minimal risk to participants (i.e., the level of risk encountered in daily life). Participation typically takes between 5 and 10 minutes and is strictly confidential. Many individuals find participation in this survey enjoyable, and no adverse reactions have been reported thus far. Participation is voluntary, and participants may withdraw from the survey at any time.*

I Agree

→

---

**Introduction:**

We conducted a field experiment about property taxes in a municipality from Buenos Aires, Argentina. Property taxes fund various services such as hospitals, parks, and roads. The taxpayers' residential properties are appraised each year, and the property taxes are computed as a function of the home's appraised value. For the sake of brevity, we assign each household to one of three groups:

- Low-income households: those with home values in the bottom 35% of the distribution.
- Middle-income households: those with home values in the middle 43% of the distribution.
- High-income households: those with home values in the top 22% of the distribution.

← →

---

The local government implemented a progressive tax reform in the property tax. Our study seeks to measure the effects of the reform on tax compliance. The tax rate is comprised of a fixed component and a component that is proportional to the assessed home value.

The tax reform affected households in the following way:

- Low-income households: the proportional component of their tax rate was reduced by 30%.
- Middle-income households: the proportional component of their tax rate did not change.
- High-income households: the proportional component of their tax rate increased by 15%.

← →

**Subject Pool:**

Households in the municipality receive a property tax bill by mail outlining their monthly obligations at the beginning of the year. In January 2023, we included a message in the paper bill mailed to the universe of about 100,000 residential households. We randomized some of the information contained in this message.

**Outcome of Interest:**

Property taxes are paid on a monthly basis and are due by the 15th of the corresponding billing period. Our main outcome of interest is whether the household paid at least one bill in the 3-month period after our intervention (i.e., between January and March 2023).

← →

**Subject Pool:**

Households in the municipality receive a property tax bill by mail outlining their monthly obligations at the beginning of the year. In January 2023, we included a message in the paper bill mailed to the universe of about 100,000 residential households. We randomized some of the information contained in this message.

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Property taxes are paid on a monthly basis and are due by the 15th of the corresponding billing period. Our main outcome of interest is whether the household paid at least one bill in the 3-month period after our intervention (i.e., between January and March 2023).

← →

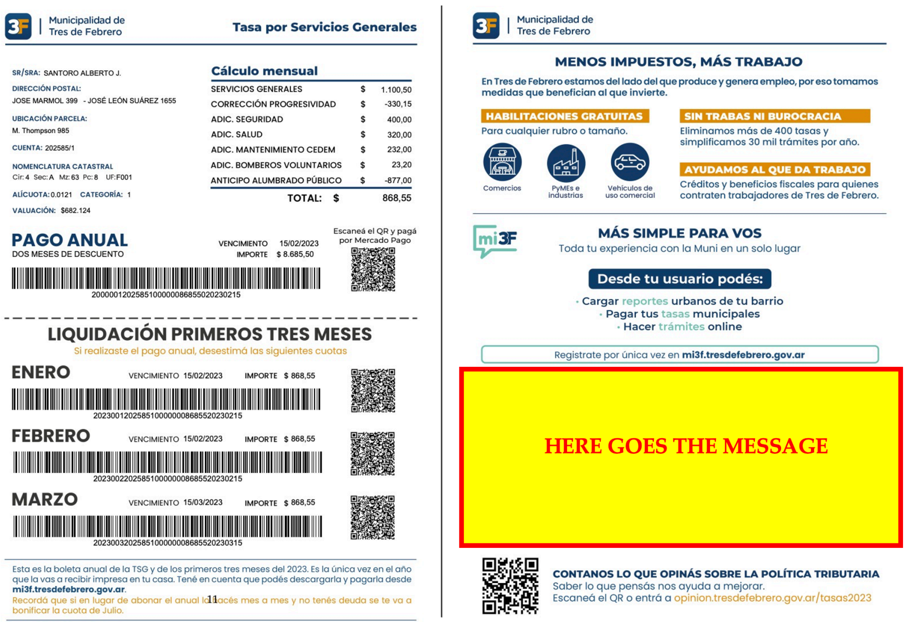
### Experimental Design

We included a message printed on the flip side of the bill, in the area depicted by the yellow rectangle below:

**Each household is randomized with equal chance to one of the following two messages:**

(1) Control message: contains summary information about the change (or lack thereof) in the recipient's own tax rates (i.e., reduced, no change or increased).

(2) Treatment message: in addition to the basic information about the recipient's own taxes, the message included an infographic with detailed information about the progressive nature of the reform – i.e., on how it affected the taxes of the three separate groups (low, middle and high property valuations).




11

**Prediction 1. Low-income group.**

Low-income households were randomized to one of the two following messages:

(1) Control message: due to the tax reform, the recipient's own tax rate decreased.
(2) Treatment message: due to the tax reform, the recipient's own tax rates decreased (as well as the tax rates of all low-income households). Additionally, tax rates increased for high-income households and remained unchanged for middle-income households (information conveyed by means of an infographic).

In the 3-months after the intervention, 44% of the low-income households ***with the control message*** paid at least one bill. What do you think is the corresponding share for low-income households ***with the treatment message***?

0 10 20 30 40 50 60 70 80 90 100

Share for Treatment Households (Low-income)

How sure are you in this prediction?

- Not confident at all
- Slightly confident
- Somewhat confident
- Very confident
- Extremely confident

Could you please provide an explanation to justify your prediction? (optional)

←  →

**Prediction 2. Middle-income group.**

Middle-income households were randomized to one of the two following messages:

(1) Control message: due to the tax reform, the recipient's own tax rate did not change.

(2) Treatment message: due to the tax reform, the recipient's own tax rates did not change. Additionally, tax rates increased for high-income households, remained unchanged for middle-income households, and decreased for low-income households (information conveyed by means of an infographic).

In the 3-months after the intervention, 46.5% of the middle-income households ***with the control message*** paid at least one bill. What do you think is the corresponding share for middle-income households ***with the treatment message***?

0 10 20 30 40 50 60 70 80 90 100

Share for Treatment Households (Middle-income)

How sure are you in this prediction?

- Not confident at all
- Slightly confident
- Somewhat confident
- Very confident
- Extremely confident

Could you please provide an explanation to justify your prediction? (optional)

←

→

**Prediction 3. High-income group.**

High-income households were randomized to one of the two following messages:

(1) Control message: due to the tax reform, the recipient's own tax rate increased.

(2) Treatment message: due to the tax reform, the recipient's own tax rates increased (as well as the tax rates of all high-income households). Additionally, tax rates remained unchanged for middle-income households, and decreased for low-income households (information conveyed by means of an infographic).

In the 3-months after the intervention, 48.5% of the high-income households ***with the control message*** paid at least one bill. What do you think is the corresponding share for high-income households ***with the treatment message***?

0 10 20 30 40 50 60 70 80 90 100

Share for Treatment Households (High-income)

How sure are you in this prediction?

- Not confident at all
- Slightly confident
- Somewhat confident
- Very confident
- Extremely confident

Could you please provide an explanation to justify your prediction? (optional)

←  →

This is the last section of the survey. We would appreciate if you could share some information about yourself.

←  →

Are you currently one of the following: faculty, graduate student (either Master level or PhD level), post-doc, researcher in an international organization or non-academic researcher?

- Yes
- No

←  →

Which of the following describes your current position?

- Professor
- Assistant Professor
- Post-doc
- Researcher
- PhD Student
- Master Student

← →

Please select your discipline

- Economics
- Business (management, accounting, finance, etc.)
- Political Science
- Psychology
- Sociology
- Other

← →

Have you ever conducted research on preferences for redistribution?

Yes

No

Have you ever conducted research on taxation?

Yes

No

← →

---

This is the end of the survey.

If you click next, you will submit your responses. We thank you for taking the time to provide your forecasts!

← Submit